AP CALCULUS AB · MASTER PRACTICE REPOSITORY

Eight-Unit Mixed Examination

250 Questions · Full Course Review

- 200 Multiple Choice Questions
- 50 Free Response Questions
- Units 1–8 · Star priorities · Tables and graphs
- Answer key and worked solutions at the end

Original AP-aligned practice · Not an official College Board examination

Structure & Priority Guide

Questions 1–200 are MCQs; FRQs begin at Question 201 and end at Question 250. Answers are placed after the entire question paper.

Priority Ratings

★★★★★	**Very Important**	Core concepts and multi-step AP-style reasoning.
★★★	**Mandatory**	Essential procedures and standard applications.
★★	**Important**	Foundational checks and direct applications.

- **Unit 1:** Unit 1 · Limits & Continuity — 25 MCQs, 6 FRQs.
- **Unit 2:** Unit 2 · Differentiation: Fundamental Properties — 25 MCQs, 6 FRQs.
- **Unit 3:** Unit 3 · Composite, Implicit & Inverse Functions — 25 MCQs, 6 FRQs.
- **Unit 4:** Unit 4 · Contextual Applications of Differentiation — 25 MCQs, 6 FRQs.
- **Unit 5:** Unit 5 · Analytical Applications of Differentiation — 25 MCQs, 6 FRQs.
- **Unit 6:** Unit 6 · Integration and Accumulation of Change — 25 MCQs, 6 FRQs.
- **Unit 7:** Unit 7 · Differential Equations — 25 MCQs, 7 FRQs.
- **Unit 8:** Unit 8 · Applications of Integration — 25 MCQs, 7 FRQs.
- **Scoring for this practice set:** 200 MCQs × 1 point and 50 FRQs × 9 points = 650 raw points. This is not an official AP score conversion.
- **Provenance:** Original AP-aligned practice; not copied College Board exam items.

> **Directions:** Attempt all questions before consulting the answer section. For FRQs, show the setup and justify each conclusion.

QUESTION 001

UNIT 1 · LIMITS & CONTINUITY · EASY

★★★ MANDATORY

$\lim_{x\to 3} (x^2 - 9)/(x - 3) =$

A. 6

B. 3

C. The limit does not exist

D. 0

QUESTION 002

UNIT 1 · LIMITS & CONTINUITY · EASY

★★ IMPORTANT

$\lim_{x\to 2} (x^2 + 3x - 1) =$

A. 13

B. 7

C. 9

D. 11

QUESTION 003

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} \sin(4x)/x =$

A. 1

B. 0

C. The limit does not exist

D. 4

QUESTION 004

UNIT 1 · LIMITS & CONTINUITY · EASY

★★★ MANDATORY

$\lim_{x\to\infty} (3x^2 + 2)/(5x^2 - 4) =$

A 1

B ∞

C 3/5

D 0

QUESTION 005

UNIT 1 · LIMITS & CONTINUITY · EASY

★★ IMPORTANT

$\lim_{x\to\infty} (2x + 1)/(x^2 + 3) =$

A ∞

B 1/3

C 2

D 0

QUESTION 006

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (\sqrt{(x+9)} - 3)/x =$

A 1/6

B 3

C 0

D 1/3

QUESTION 007

UNIT 1 · LIMITS & CONTINUITY · MODERATE ★★★ MANDATORY

Let $f(x) = \{x+1 \text{ for } x<2;\ 5 \text{ for } x=2;\ x^2 \text{ for } x>2\}$. Then $\lim_{x\to 2} f(x) =$

A. The limit does not exist
B. 3
C. 5
D. 4

QUESTION 008

UNIT 1 · LIMITS & CONTINUITY · MODERATE ★★★ MANDATORY

$\lim_{x\to 1} (x^3 - 1)/(x - 1) =$

A. 0
B. The limit does not exist
C. 3
D. 1

QUESTION 009

UNIT 1 · LIMITS & CONTINUITY · MODERATE ★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (1 - \cos x)/x =$

A. 1
B. 1/2
C. 0
D. The limit does not exist

QUESTION 010

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★★★ VERY IMPORTANT

The vertical asymptote(s) of $f(x) = (x+2)/(x^2 - x - 6)$ is/are

A. $x = 3$ only

B. $x = -2$ only

C. $x = 3$ and $x = -2$

D. There are no vertical asymptotes

QUESTION 011

UNIT 1 · LIMITS & CONTINUITY · EASY

★★★ MANDATORY

The horizontal asymptote of $f(x) = (2x^2 - 1)/(3x^2 + x)$ is

A. $y = 1$

B. $y = 2/3$

C. $y = 0$

D. $y = 3/2$

QUESTION 012

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★ MANDATORY

$\lim_{x \to 0} x^2 \sin(1/x) =$

A. 0

B. 1

C. The limit does not exist

D. -1

QUESTION 013

UNIT 1 · LIMITS & CONTINUITY · MODERATE ★★★★★ VERY IMPORTANT

Let $f(x) = \{(x^2 - 4)/(x - 2)$ for $x \neq 2$; k for $x = 2\}$. For f to be continuous at $x = 2$, $k =$

A 0

B 4

C No value of k works

D 2

QUESTION 014

UNIT 1 · LIMITS & CONTINUITY · MODERATE ★★★ MANDATORY

Let $f(x) = \{x^2 + k$ for $x \leq 1$; $2x + 3$ for $x > 1\}$. For f to be continuous at $x = 1$, $k =$

A 5

B 4

C 3

D 1

QUESTION 015

UNIT 1 · LIMITS & CONTINUITY · HARD ★★★ MANDATORY

$\lim_{x\to\infty} \sqrt{(9x^2 + x)}/(3x + 1) =$

A 0

B 1

C 3

D ∞

QUESTION 016

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★ IMPORTANT

$\lim_{x \to 2^-} 1/(x - 2) =$

A. $1/2$

B. $-\infty$

C. 0

D. $+\infty$

QUESTION 017

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★ IMPORTANT

$\lim_{x \to 0} |x|/x =$

A. 1

B. 0

C. The limit does not exist

D. -1

QUESTION 018

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★★★ VERY IMPORTANT

A function f is continuous on $[1, 3]$ with $f(1) = -2$ and $f(3) = 4$. The IVT guarantees that

A. $f(2) = 1$

B. the maximum of f on $[1, 3]$ is 4

C. there exists c in $(1, 3)$ with $f(c) = 6$

D. there exists c in $(1, 3)$ with $f(c) = 0$

QUESTION 019

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★ MANDATORY

$\lim_{x\to\pi} \sin(x)/(x-\pi) =$

A. 1
B. −1
C. 0
D. The limit does not exist

QUESTION 020

UNIT 1 · LIMITS & CONTINUITY · HARD

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (1-\cos(2x))/x^2 =$

A. 0
B. 1
C. 2
D. 4

QUESTION 021

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★ IMPORTANT

$\lim_{x\to-\infty} (x^3+1)/(x^2+2) =$

A. 0
B. $+\infty$
C. 1
D. $-\infty$

QUESTION 022

UNIT 1 · LIMITS & CONTINUITY · HARD

★★★ MANDATORY

$\lim_{x\to 0^+} x \ln x =$

A. 1
B. $+\infty$
C. $-\infty$
D. 0

QUESTION 023

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★★ MANDATORY

If $\lim_{x\to 4^-} f(x) = 3$, $\lim_{x\to 4^+} f(x) = 3$, and $f(4) = 1$, then f has

A. a removable discontinuity at $x = 4$
B. an infinite discontinuity at $x = 4$
C. continuity at $x = 4$
D. a jump discontinuity at $x = 4$

QUESTION 024

UNIT 1 · LIMITS & CONTINUITY · MODERATE

★★ IMPORTANT

$\lim_{x\to\infty} (\sin x)/x =$

A. 0
B. ∞
C. The limit does not exist
D. 1

QUESTION 025

UNIT 1 · LIMITS & CONTINUITY · EASY

★★★ MANDATORY

A table gives f values: x = 1.9, 1.99, 1.999, 2.001, 2.01, 2.1 with f(x) = 4.85, 4.98, 4.998, 5.002, 5.02, 5.2. The best estimate of $\lim_{x\to 2} f(x)$ is

A. 4.85
B. The limit does not exist
C. 2
D. 5

QUESTION 026

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

$\lim_{h\to 0} [(2+h)^3 - 8] / h =$

A. The limit does not exist
B. 12
C. 0
D. 4

QUESTION 027

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · EASY

★★★ MANDATORY

$\lim_{x\to 3} (x^2 - 9)/(x - 3)$ represents the derivative of which function at x = 3?

A. $f(x) = 2x$
B. $f(x) = x^2$
C. $f(x) = x^2 - 9$
D. $f(x) = x - 3$

QUESTION 028

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · EASY

★★★★★ VERY IMPORTANT

If a function f is differentiable at $x = c$, which of the following statements MUST be true?

A. f is continuous at $x = c$

B. $f''(c)$ exists

C. $f'(c) = 0$

D. f has a local extremum at $x = c$

QUESTION 029

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · EASY

★★★ MANDATORY

Which of the following functions is continuous at $x = 0$ but NOT differentiable at $x = 0$?

A. $f(x) = 1/x$

B. $f(x) = |x|$

C. $f(x) = x^2$

D. $f(x) = \sin x$

QUESTION 030

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

If $f(x) = 4x^{(-1/2)}$, then $f'(x) =$

A. $-2x^{(1/2)}$

B. $-2x^{(-3/2)}$

C. $-8x^{(-3/2)}$

D. $2x^{(-3/2)}$

QUESTION 031

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · EASY

★★ IMPORTANT

$d/dx\,[\,3/x^4\,] =$

A $12/x^5$

B $-3/(4x^3)$

C $-12/x^5$

D $-12/x^3$

QUESTION 032

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

The equation of the line tangent to $y = x^2 - 3x$ at $x = 2$ is

A $y = -x$

B $y = x - 4$

C $y = 4x - 10$

D $y = x - 2$

QUESTION 033

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

The equation of the line normal to $y = \sqrt{x}$ at $x = 4$ is

A $y - 2 = -(1/4)(x - 4)$

B $y - 2 = -4(x - 4)$

C $y - 2 = 4(x - 4)$

D $y - 2 = (1/4)(x - 4)$

QUESTION 034

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

For which value of x does the graph of $y = x^3 - 3x^2 + 2$ have a horizontal tangent line?

A. $x = 0$ and $x = 2$

B. $x = 0$ only

C. $x = 2$ only

D. $x = 1$ and $x = -1$

QUESTION 035

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

The graph of $y = x^{(1/3)}$ has which feature at $x = 0$?

A. A vertical tangent line

B. A jump discontinuity

C. A sharp corner with two distinct slopes

D. A horizontal tangent line

QUESTION 036

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

$\frac{d}{dx}[\, x^2 e^x \,] =$

A. $2x\, e^x$

B. $x^2 e^x + 2x\, e^x$

C. $x^2 e^x$

D. $2x\, e^x + x\, e^x$

QUESTION 037

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

If $f(x) = (2x - 1)\sin x$, then $f'(0) =$

A. 2
B. 0
C. 1
D. -1

QUESTION 038

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

$\frac{d}{dx}\left[\frac{x}{(x+1)}\right] =$

A. $(2x+1)/(x+1)^2$
B. $-1/(x+1)^2$
C. 1
D. $1/(x+1)^2$

QUESTION 039

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

If $g(x) = e^x / x$, then $g'(1) =$

A. e
B. 0
C. $e - 1$
D. $2e$

QUESTION 040

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · HARD

★★★ MANDATORY

If $f(x) = 3\sin x - 2\cos x$, then $f'(\pi/3) =$

A. $(3 - 2\sqrt{3})/2$

B. $(3\sqrt{3} + 2)/2$

C. $(2\sqrt{3} - 3)/2$

D. $(3 + 2\sqrt{3})/2$

QUESTION 041

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

$d/dx\,[\,2^x\,] =$

A. 2^x

B. $x \cdot 2^{x-1}$

C. $2^x / \ln 2$

D. $2^x \ln 2$

QUESTION 042

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

Let $f(x) = \{x^2$ for $x \le 1$; $2x - 1$ for $x > 1\}$. Is f differentiable at $x = 1$?

A. No, because the left derivative is 1 and right derivative is 2

B. No, because f is discontinuous at $x = 1$

C. Yes, because f is continuous and left/right derivatives both equal 2

D. No, because the derivative is undefined at all piecewise boundaries

QUESTION 043

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · EASY

★★★ MANDATORY

The table gives values of a differentiable function f.

x: 1, 3, 5 → f(x): 4, 10, 18.

The best estimate of $f'(3)$ using the average rate of change over $[1, 5]$ is

A 3

B 4

C 3.5

D 7

QUESTION 044

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

If $f(x) = \ln x$, then $f''(x) =$

A $-1/x^2$

B $1/x^2$

C $-2/x^3$

D $1/x$

QUESTION 045

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · EASY

★★ IMPORTANT

If $f(x) = x^4$, then the 4th derivative $f^{(4)}(x) =$

A 24

B 4x

C 0

D 24x

QUESTION 046

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE ★★★ MANDATORY

$\frac{d}{dx}\left[(x^2+1)(x-2)\right] =$

A $3x^2 - 4x + 1$

B $3x^2 + 4x - 1$

C $2x$

D $2x(x-2)$

QUESTION 047

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE ★★★ MANDATORY

Functions f and g satisfy $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, $g'(2) = 5$. If $h(x) = 3f(x) - 2g(x)$, then $h'(2) =$

A 7

B -13

C 13

D -7

QUESTION 048

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE ★★★★★ VERY IMPORTANT

Using the same table: $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, $g'(2) = 5$. If $P(x) = f(x)\cdot g(x)$, then $P'(2) =$

A 19

B 11

C -20

D -5

QUESTION 049

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★★★ VERY IMPORTANT

Using the same table: $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, $g'(2) = 5$. If $Q(x) = f(x)/g(x)$, then $Q'(2) =$

A $11/16$

B $-19/16$

C $19/16$

D $-11/16$

QUESTION 050

UNIT 2 · DIFFERENTIATION: FUNDAMENTAL PROPERTIES · MODERATE

★★★ MANDATORY

At what point on the curve $y = x^3$ does the tangent line have a slope of 12?

A $(4, 64)$

B $(2, 8)$ and $(-2, -8)$

C $(-2, -8)$ only

D $(2, 8)$ only

QUESTION 051

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · EASY

★★★★★ VERY IMPORTANT

$d/dx\,[\,\sin(x^3)\,] =$

A $3x^2\cos(x^3)$

B $-3x^2\cos(x^3)$

C $\cos(x^3)$

D $3x^2\sin(x^3-1)$

QUESTION 052

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · EASY

★★★ MANDATORY

$\frac{d}{dx}\left[e^{(-2x)} \right] =$

A. $e^{(-2x)}$

B. $-2e^{(-2x)}$

C. $2e^{(-2x)}$

D. $-2xe^{(-2x)}$

QUESTION 053

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

$\frac{d}{dx}\left[\ln(x^2 + 4) \right] =$

A. $2 / (x^2 + 4)$

B. $2x / (x^2 + 4)$

C. $2x \ln(x^2 + 4)$

D. $1 / (x^2 + 4)$

QUESTION 054

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★ MANDATORY

If $f(x) = (3x - 2)^5$, then $f'(1) =$

A. 75

B. 15

C. 5

D. 45

QUESTION 055

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

$\frac{d}{dx}\left[\sqrt{(1-x^2)}\right] =$

A $\frac{1}{(2\sqrt{(1-x^2)})}$

B $\frac{-x}{\sqrt{(1-x^2)}}$

C $\frac{-2x}{\sqrt{(1-x^2)}}$

D $\frac{x}{\sqrt{(1-x^2)}}$

QUESTION 056

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

$\frac{d}{dx}\left[\cos^3(2x)\right] =$

A $-3\cos^2(2x)\sin(2x)$

B $-6\cos(2x)\sin(2x)$

C $-6\cos^2(2x)\sin(2x)$

D $6\cos^2(2x)\sin(2x)$

QUESTION 057

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

The table gives values of differentiable functions f and g.

$x = 2$: $f(2) = 3$, $f'(2) = 4$, $g(2) = 2$, $g'(2) = 5$.

$x = 3$: $f(3) = 1$, $f'(3) = -2$.

If $h(x) = f(g(x))$, then $h'(2) =$

A 20

B -10

C 4

D -2

QUESTION 058

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · EASY

★★★★★ VERY IMPORTANT

If $x^2 + y^2 = 25$, then $dy/dx =$

A y/x

B $-y/x$

C x/y

D $-x/y$

QUESTION 059

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★ MANDATORY

For the curve $xy = 4$, the slope of the tangent line at the point $(2, 2)$ is

A -1

B 1

C 2

D -2

QUESTION 060

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

If $x^2 + 2xy - y^2 = 7$, then at the point $(3, 2)$, $dy/dx =$

A -5

B 5

C $-8/2$

D $-5/2$

QUESTION 061

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

If $x^2 + y^2 = 1$, then $d^2y/dx^2 =$

A $-1/y^2$

B $-1/y^3$

C $-x/y^2$

D $1/y^3$

QUESTION 062

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

The curve $x^2 - xy + y^2 = 3$ has a horizontal tangent line when

A $x = 0$

B $y = 0$

C $2x - y = 0$

D $y = 2x$ and $x \neq 0$

QUESTION 063

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

If f is a differentiable, invertible function with $f(3) = 5$ and $f'(3) = 2$, then $(f^{-1})'(5) =$

A $1/5$

B 2

C 5

D $1/2$

QUESTION 064

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

Let $f(x) = x^3 + x - 2$. What is the value of $(f^{-1})'(0)$?

A) 4

B) 1/4

C) 0

D) 1/2

QUESTION 065

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

The table shows values of f and f′.

$x = 1$: $f(1) = 2$, $f'(1) = 3$.

$x = 2$: $f(2) = 4$, $f'(2) = 5$.

If $g = f^{-1}$, then $g'(2) =$

A) 5

B) 1/5

C) 1/3

D) 3

QUESTION 066

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

$\frac{d}{dx}[\arctan(3x)] =$

A) $3 / \sqrt{(1 - 9x^2)}$

B) $3 / (1 + 9x^2)$

C) $1 / (1 + 9x^2)$

D) $3 / (1 + 3x^2)$

QUESTION 067

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

$\frac{d}{dx}\left[\arcsin\left(\frac{x}{2}\right)\right] =$

A $\frac{2}{\sqrt{4 - x^2}}$

B $\frac{1}{\sqrt{4 - x^2}}$

C $\frac{1}{\sqrt{1 - x^2/4}}$

D $\frac{1}{2\sqrt{1 - x^2}}$

QUESTION 068

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · EASY

★★★ MANDATORY

$\frac{d}{dx}\left[\arccos(x)\right]$ at $x = 0$ is

A 1

B Undefined

C 0

D -1

QUESTION 069

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★ MANDATORY

$\frac{d}{dx}\left[x \cdot \arctan x\right] =$

A $\arctan x + \frac{1}{1 + x^2}$

B $\arctan x + \frac{x}{1 + x^2}$

C $\frac{1}{1 + x^2}$

D $\frac{x}{1 + x^2}$

QUESTION 070

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★ MANDATORY

$\frac{d}{dx}\left[\arcsin(e^x)\right] =$

A. $e^x / \sqrt{(1 - e^x)}$

B. $e^x / \sqrt{(1 - e^{(2x)})}$

C. $e^x / (1 + e^{(2x)})$

D. $1 / \sqrt{(1 - e^{(2x)})}$

QUESTION 071

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★★★ VERY IMPORTANT

$\frac{d}{dx}\left[\ln(\cos x)\right] =$

A. $\tan x$

B. $-\tan x$

C. $\cot x$

D. $-\cot x$

QUESTION 072

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★ MANDATORY

If $\sin y = x$ for $-\pi/2 < y < \pi/2$, then $\frac{dy}{dx} =$

A. $1 / \cos y$

B. $-1 / \cos y$

C. $1 / \sin y$

D. $\cos y$

QUESTION 073

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

If f(x) = e^(3x), what is $(f^{-1})'(e^3)$?

A $3e^3$

B $1 / (3e^3)$

C $e^3 / 3$

D $1 / e^3$

QUESTION 074

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · MODERATE

★★★ MANDATORY

The slope of the line normal to $x^2 + y^2 = 25$ at the point $(3, 4)$ is

A $-3/4$

B $3/4$

C $-4/3$

D $4/3$

QUESTION 075

UNIT 3 · COMPOSITE, IMPLICIT & INVERSE FUNCTIONS · HARD

★★★★★ VERY IMPORTANT

$d/dx [\arctan(1/x)]$ for $x > 0 =$

A $1 / (1 + x^2)$

B $1 / (x^2 + 1)$

C $-1 / (1 + x^2)$

D $-x^2 / (1 + x^2)$

QUESTION 076

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · EASY

★★★★★ VERY IMPORTANT

If $W(t)$ represents the amount of water in a tank in gallons at time t in minutes, what does $W'(5) = -3$ mean in context?

A. The average leak rate is 3 gallons per minute

B. At $t = 5$ min, the tank contains -3 gallons

C. At $t = 5$ min, water is leaking out at 3 gallons per minute

D. Over the first 5 minutes, 3 gallons leaked out

QUESTION 077

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$. The particle is at rest when

A. $t = 1$ and $t = 3$

B. $t = 1$ only

C. $t = 3$ only

D. $t = 0$ only

QUESTION 078

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

For the same particle $s(t) = t^3 - 6t^2 + 9t$, the particle is moving to the left when

A. $t > 1$

B. $t > 3$

C. $0 < t < 1$

D. $1 < t < 3$

QUESTION 079

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

If $v(t) < 0$ and $a(t) < 0$ at time $t = 2$, which statement is true about the particle at $t = 2$?

A. The particle is moving right and speeding up

B. The particle is moving left and slowing down

C. The particle is moving left and speeding up

D. The particle is at rest

QUESTION 080

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

If $v(t) = -4$ and $a(t) = 2$ at time $t = 3$, the particle is

A. Moving right and speeding up

B. Moving left and slowing down

C. Moving left and speeding up

D. Moving right and slowing down

QUESTION 081

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

A particle has position $s(t) = t^4 - 4t^3$. Its acceleration at $t = 2$ is

A. 0

B. 12

C. -16

D. 24

QUESTION 082

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

The radius of a circle increases at a rate of 3 cm/s. How fast is the area increasing when $r = 4$ cm?

A. 48π cm^2/s

B. 24π cm^2/s

C. 12π cm^2/s

D. 16π cm^2/s

QUESTION 083

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · HARD

★★★★★ VERY IMPORTANT

A 10-ft ladder leans against a vertical wall. If the bottom slides away at 3 ft/s, how fast is the top sliding down when the bottom is 6 ft from the wall?

A. 4 ft/s

B. 1.5 ft/s

C. 3 ft/s

D. 2.25 ft/s

QUESTION 084

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

A spherical balloon is inflated so that its volume increases at 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm? ($V = (4/3)\pi r^3$)

A. $25/\pi$ cm/s

B. $1/\pi$ cm/s

C. $4/\pi$ cm/s

D. $2/\pi$ cm/s

QUESTION 085

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · HARD

★★★★★ VERY IMPORTANT

Water pours into a conical tank (vertex down) of radius 4 ft and height 8 ft at 6 ft^3/min. Find dh/dt when h = 4 ft.

A. $3/(2\pi)$ ft/min

B. $6/\pi$ ft/min

C. $3/(8\pi)$ ft/min

D. $6/(4\pi)$ ft/min

QUESTION 086

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

Two cars leave an intersection at the same time: Car A travels north at 30 mph, Car B travels east at 40 mph. How fast is the distance z between them increasing after 2 hours?

A. 100 mph

B. 50 mph

C. 35 mph

D. 70 mph

QUESTION 087

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

The linear approximation of $f(x) = \sqrt{x}$ at x = 4 is L(x) =

A. $2 + (1/2)(x - 4)$

B. $(1/4)x + 2$

C. $2 + 4(x - 4)$

D. $2 + (1/4)(x - 4)$

QUESTION 088

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · EASY

★★★ MANDATORY

Using the tangent line approximation for $f(x) = \sqrt{x}$ at $x = 4$, $\sqrt{4.08} \approx$

A 2.04

B 2.01

C 2.02

D 2.08

QUESTION 089

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

Because $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$ ($f''(x) < 0$), the tangent line approximation 2.02 for $\sqrt{4.08}$ is

A Cannot be determined without a calculator

B An overestimate of the true value

C An underestimate of the true value

D An exact value

QUESTION 090

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

$\lim (x\to0) [e^(3x) - 1] / \sin(2x) =$

A 0

B 2/3

C 1

D 3/2

QUESTION 091

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} [1 - \cos(4x)] / x^2 =$

A 2

B 8

C 4

D 16

QUESTION 092

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

$\lim_{x\to \infty} (\ln x) / \sqrt{x} =$

A 0

B ∞

C 1/2

D 1

QUESTION 093

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · HARD

★★★★★ VERY IMPORTANT

$\lim_{x\to 0^+} x \ln x =$

A $+\infty$

B 0

C $-\infty$

D 1

QUESTION 094

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE ★★★ MANDATORY

A searchlight on the ground 20 ft from a wall tracks a man walking toward the wall. If angle θ changes, related rates uses which formula relating distance x and angle θ?

A. $\tan\theta = 20/x$

B. $\sin\theta = x/20$

C. $\tan\theta = x/20$

D. $\cos\theta = 20/x$

QUESTION 095

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · HARD ★★★★★ VERY IMPORTANT

The rate of change of the volume of a cube with side length s is related to the rate of change of its surface area $A = 6s^2$ by which relationship?

A. $dV/dt = s/4 \cdot dA/dt$

B. $dV/dt = 3s^2 \cdot dA/dt$

C. $dV/dt = s/2 \cdot dA/dt$

D. $dV/dt = 12s \cdot dA/dt$

QUESTION 096

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · HARD ★★★★★ VERY IMPORTANT

A particle has velocity $v(t) = t^2 - 2t - 3$ for $0 \le t \le 4$. Total distance traveled on $[0, 4]$ is

A. $34/3$

B. $-16/3$

C. 10

D. $16/3$

QUESTION 097

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · EASY

★★ IMPORTANT

Using linear approximation for $f(x) = e^x$ at $x = 0$, e^(0.05) ≈

A 1.5

B 0.95

C 1.005

D 1.05

QUESTION 098

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

Which of the following limits CANNOT be evaluated using L'Hôpital's Rule directly?

A $\lim_{x\to 0} (\cos x)/x$

B $\lim_{x\to 0} (\sin x)/x$

C $\lim_{x\to\infty} x / e^x$

D $\lim_{x\to 1} (x^2 - 1)/(x - 1)$

QUESTION 099

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · MODERATE

★★★ MANDATORY

If the edge of a square increases at 2 cm/s, how fast is the area increasing when the perimeter is 20 cm?

A 50 cm^2/s

B 40 cm^2/s

C 20 cm^2/s

D 10 cm^2/s

QUESTION 100

UNIT 4 · CONTEXTUAL APPLICATIONS OF DIFFERENTIATION · EASY

★★★ MANDATORY

The graph of velocity v(t) for a particle on [0, 6] is shown. When is the particle's acceleration equal to zero?

(Imagine graph: v rises from (0,0) to (2,4), horizontal from (2,4) to (4,4), falls from (4,4) to (6,0))

A Nowhere on (0, 6)

B At $t = 0$ and $t = 6$

C On the open interval (2, 4)

D At $t = 3$ only

QUESTION 101

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For $f(x)=x^2-4x$ on $[0,4]$, the value c guaranteed by the Mean Value Theorem is

A 2

B 0

C 1

D 4

QUESTION 102

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For $f(x)=x^3$ on $[0,2]$, the MVT guarantees a c satisfying $f'(c)=4$. The value is

A $\sqrt{3}$

B $1/\sqrt{3}$

C $4/3$

D $2/\sqrt{3}$

QUESTION 103

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For f(x)=sin x on [0,π], a value guaranteed by the MVT is

A. 3π/4

B. π/4

C. π

D. π/2

QUESTION 104

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For $f(x)=x^3$ on $[-1,1]$, the MVT equation $f'(c)=(f(1)-f(-1))/2$ has

A. two solutions $c=\pm 1/\sqrt{3}$

B. no solution

C. exactly one solution $c=0$

D. one solution $c=1/\sqrt{3}$ only

QUESTION 105

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

If $f'(x)=-(x+2)^2$, which statement is correct?

A. f has a local minimum at $x=-2$

B. f is strictly decreasing on the real line

C. f has a local maximum at $x=-2$

D. f is strictly increasing on the real line

QUESTION 106

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

Let $f'(x)=(x+1)(x-2)^2$. Which is true?

A. f has local minima at both points

B. f has a local maximum at $x=-1$ and a local minimum at $x=2$

C. f has a local minimum at $x=-1$ and no extremum at $x=2$

D. f has a local maximum at $x=2$

QUESTION 107

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

For a differentiable function, f' changes from positive to negative at c. Then

A. f has an inflection point at c

B. f is constant near c

C. f has a local maximum at c

D. f has a local minimum at c

QUESTION 108

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

If $f'(x)>0$ throughout an interval, then f is

A. increasing on that interval

B. constant on that interval

C. concave up on that interval

D. decreasing on that interval

QUESTION 109

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

If $f'(x)=-(x-3)^2$, then at $x=3$ the function

A changes from decreasing to increasing

B continues decreasing through a stationary point

C has a local maximum

D has a local minimum

QUESTION 110

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For $f(x)=x^2-2x$ on $[0,3]$, the absolute minimum value is

A 3

B -1

C 0

D 1

QUESTION 111

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

A continuous function on $[-3,5]$ is guaranteed to

A attain both an absolute maximum and an absolute minimum

B have exactly one local maximum

C have an interior critical number

D be differentiable everywhere

QUESTION 112

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For $f(x)=x^2-4x$ on $[0,5]$, the absolute maximum value is

A -4

B 0

C 4

D 5

QUESTION 113

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

For f(x)=x^(2/3) on $[-1,1]$, the absolute minimum is

A -1 at $x=0$

B 0 at $x=0$

C 1 at both endpoints

D no minimum

QUESTION 114

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

If $f''(x)=12(x-1)^2$, what can be concluded about concavity at $x=1$?

A There is a local maximum at 1

B There is an inflection point at 1

C There is no change in concavity at 1

D There is a local minimum at 1

QUESTION 115

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

At an interior critical number c, if $f'(c)=0$ and $f''(c)<0$, then f has

A no conclusion

B a local maximum

C a local minimum

D an inflection point

QUESTION 116

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

A point c is an inflection point of f when

A $f''(c)=0$

B $f'(c)=0$

C f has a local extremum at c

D the concavity of f changes at c and the point lies on the graph

QUESTION 117

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

For $f(x)=x^4$, $f''(0)=0$. Which statement is true?

A x=0 is not an inflection point because concavity is upward on both sides

B x=0 must be an inflection point

C f is decreasing on both sides

D f has a local maximum at 0

QUESTION 118

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

An inflection point can occur where f″ is undefined if

A. f′ is positive on both sides

B. f″ is zero there

C. f is discontinuous there

D. f is continuous and its concavity changes across the point

QUESTION 119

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

If $f'(x)>0$ and $f''(x)<0$ on an interval, f is

A. decreasing and concave up

B. decreasing and concave down

C. increasing and concave down

D. increasing and concave up

QUESTION 120

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

If $f'(x)<0$ and $f''(x)>0$ throughout an interval, then f is

A. increasing and concave up

B. increasing and concave down

C. decreasing and concave up

D. decreasing and concave down

QUESTION 121

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

A rectangle has perimeter 40 units. Its maximum possible area is

A 80

B 400

C 90

D 100

QUESTION 122

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

A symmetric rectangle is inscribed under $y=9-x^2$ and above the x-axis. If x is its positive half-width, its maximum area is

A 18

B 9

C $12\sqrt{3}$

D $9\sqrt{3}$

QUESTION 123

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

A rectangle has area 64 square units. Its minimum possible perimeter is

A 32

B 16

C 24

D 64

QUESTION 124

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★ MANDATORY

A company's profit is $P(x)=-x^2+40x-100$ for $0\le x\le 50$. The production level giving maximum profit is

A. 10

B. 40

C. 50

D. 20

QUESTION 125

UNIT 5 · ANALYTICAL APPLICATIONS OF DIFFERENTIATION · PRACTICE

★★★★★ VERY IMPORTANT

A continuous function on $[-2,4]$ has f' positive on $(-2,0)$, negative on $(0,3)$, and positive on $(3,4)$. Its absolute maximum must occur at

A. $x=3$ only

B. $x=-2$ only

C. $x=4$ only

D. $x=0$ or an endpoint, so values must be compared

QUESTION 126

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★★ MANDATORY

If $f'(t) = 3$ for all t and $f(0) = 10$, then $f(4) =$

A. 18

B. 30

C. 12

D. 22

QUESTION 127

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★★ MANDATORY

A tank drains at a constant 3 L/min. The net change in volume over 4 minutes, $\int_0^4 (-3)\,dt$, equals

A. 7 L
B. 12 L
C. −12 L
D. −7 L

QUESTION 128

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★ IMPORTANT

$\int_0^3 0\,dt =$

A. Undefined
B. 1
C. 0
D. 3

QUESTION 129

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★★★ VERY IMPORTANT

The midpoint Riemann sum for $f(x) = x^2$ on $[0, 4]$ with $n = 2$ is

A. 20
B. 28
C. 10
D. 24

QUESTION 130

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

The midpoint Riemann sum for $f(x) = x$ on $[0, 6]$ with $n = 3$ is

A 21

B 9

C 18

D 27

QUESTION 131

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

The right Riemann sum for $f(x) = \sqrt{x}$ on $[0, 4]$ with $n = 4$ is approximately

A 6.15

B 5.00

C 7.50

D 4.00

QUESTION 132

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★ IMPORTANT

$\Sigma(i = 1 \text{ to } 4)\ i =$

A 16

B 10

C 4

D 24

QUESTION 133

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★ IMPORTANT

$\Sigma(i = 1 \text{ to } 4)\ 3 =$

A 3

B 12

C 7

D 24

QUESTION 134

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

$\int_{-2}^{3} 2x\ dx =$

A 10

B 5

C 0

D 13

QUESTION 135

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★ IMPORTANT

$\int_{0}^{2} (x + 1)\ dx =$

A 6

B 3

C 4

D 2

QUESTION 136

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★ IMPORTANT

$\int 5\,dx =$

A. $5x + C$

B. $5 + C$

C. $5x$

D. $x^5 + C$

QUESTION 137

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

$\int x^{-2}\,dx =$

A. $x^{-1} + C$

B. $-x^{-1} + C$

C. $-2x^{-3} + C$

D. $\ln|x| + C$

QUESTION 138

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★★★★ VERY IMPORTANT

$\int e^x\,dx =$

A. $e^{x+1} + C$

B. $\ln x + C$

C. $e^x + C$

D. $xe^x + C$

QUESTION 139

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

$\int (x^2 - 4x + 1)\,dx =$

A. $3x - 4 + C$

B. $x^3 - 4x^2 + x + C$

C. $x^3/3 - 2x^2 + x + C$

D. $x^3/3 - 4x^2 + x + C$

QUESTION 140

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★★★ VERY IMPORTANT

$\int_0^{\pi} \sin x\,dx =$

A. 2

B. 1

C. π

D. 0

QUESTION 141

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

$\int_{-1}^{1} (x^2 + 1)\,dx =$

A. 4

B. 4/3

C. 2

D. 8/3

QUESTION 142

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

$\int_0^3 (2x - 4)\,dx =$

A 0

B -6

C 3

D -3

QUESTION 143

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★★★★ VERY IMPORTANT

Reversing the limits of a definite integral gives

A the reciprocal

B the negative of the value

C zero

D the same value

QUESTION 144

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★ MANDATORY

$\int_{-3}^3 x^2\,dx$ equals

A 27

B 9

C 0

D 18

QUESTION 145

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · HARD

★★★★★ VERY IMPORTANT

If $H(x) = \int_0^{(x^2)} \sin t \, dt$, then $H'(x) =$

A. $x^2 \cos(x^2)$

B. $2x \sin(x^2)$

C. $\sin(x^2)$

D. $\sin(x^2) \cdot 2x + C$

QUESTION 146

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★★★ VERY IMPORTANT

If $A(x) = \int_1^x (1/t) \, dt$, then $A(e) =$

A. $1/e$

B. 0

C. 1

D. e

QUESTION 147

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · HARD

★★★★★ VERY IMPORTANT

$d/dx \left[\int_0^{(x^3)} e^t \, dt \right] =$

A. $e^{(x^3)}$

B. $e^{(3x^2)}$

C. $e^{(x^3)} \cdot x^3$

D. $3x^2 e^{(x^3)}$

QUESTION 148

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★★★ VERY IMPORTANT

The average value of $f(x) = \sin x$ on $[0, \pi]$ is

A. 2
B. $2/\pi$
C. 0
D. $1/\pi$

QUESTION 149

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · MODERATE

★★★★★ VERY IMPORTANT

If $v(t) = 3t^2$, the displacement on $[0, 2]$ is

A. 4
B. 12
C. 8
D. 6

QUESTION 150

UNIT 6 · INTEGRATION AND ACCUMULATION OF CHANGE · EASY

★★★ MANDATORY

If $v(t) = 4$ on $[1, 5]$, the distance traveled is

A. 20
B. 4
C. 8
D. 16

QUESTION 151

UNIT 7 · DIFFERENTIAL EQUATIONS · EASY

★★★★★ VERY IMPORTANT

A population P grows continuously at 4% per year. Which differential equation models this?

A $P'=0.04$

B $P=0.04t$

C $P'=0.04P$

D $P'=4P$

QUESTION 152

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

A quantity A receives a constant input of 12 units per hour and is removed at a rate proportional to A, with proportionality constant 0.2. The model is

A $A'=0.2A-12$

B $A'=12-0.2A$

C $A'=12+0.2A$

D $A'=12A-0.2$

QUESTION 153

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

For $P'=0.2P(1-P/50)$, if $P=60$ then P' is

A positive

B negative

C undefined

D zero

QUESTION 154

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

Which function solves $y' = 2y$?

A $y = Ce^{x}$

B $y = 2Ce^{x}$

C $y = C + 2x$

D $y = Ce^{2x}$

QUESTION 155

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

Which constant function is a solution of $y' = y(5 - y)$?

A $y = 5/2$

B $y = -1$

C $y = 0$

D $y = 1$

QUESTION 156

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

Which function solves $y' = 2xy$?

A $y = Ce^{x2}$

B $y = Cx^{2}$

C $y = Ce^{2x}$

D $y = Ce^{x}$

QUESTION 157

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

For $y' = 2x - y$, the slope at $(1,3)$ is

A 1

B 3

C -1

D 5

QUESTION 158

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

For $y' = (x-1)(y+2)$, the slope is zero when

A $x = -1$ or $y = 2$

B $x = y$

C $x = 1$ or $y = -2$

D only at $(1,-2)$

QUESTION 159

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

For $y' = 1 + y^2$, which statement is true about the slope field?

A Slopes are zero on $y = 0$

B Slopes are positive at every point

C Slopes depend only on x

D Slopes are negative for $y < 0$

QUESTION 160

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

The equilibrium solutions of $y' = 0.5y(1 - y/20)$ are

A. $y = 10$ and $y = 20$

B. $y = 0$ and $y = 20$

C. $y = 0$ only

D. $y = 20$ only

QUESTION 161

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

For $y' = y(5 - y)$ with 0

A. decreases toward 0

B. increases toward 5

C. increases without bound

D. is constant

QUESTION 162

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

For $y' = y^2$ at $y = -2$, the slope is

A. -2

B. 0

C. -4

D. 4

QUESTION 163

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

For $y'=(y-2)(y+1)$, which equilibrium is stable?

A $y=2$

B both

C $y=-1$

D neither

QUESTION 164

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

On an interval not containing $x=0$, the general solution of $y'=y/x$ is

A $y=Cx$

B $y=C/x$

C $y=Ce^{x}$

D $y=Cx^{2}$

QUESTION 165

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

A separated solution of $y'=x^{2}/y^{2}$ satisfies

A $y^{3}=x^{3}+C$

B $y^{2}=x^{3}+C$

C $y^{3}=3x^{2}+C$

D $y=x^{3}+C$

QUESTION 166

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

For $y>0$, the separated general solution of $y'=(x+1)/\sqrt{y}$ can be written

A. $y^{(3/2)}=x^2+x+C$

B. $2\sqrt{y}=x^2/2+x+C$

C. $(2/3)y^{(1/2)}=x^2/2+x+C$

D. $(2/3)y^{(3/2)}=x^2/2+x+C$

QUESTION 167

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

The solution of $y'=y(1-y/4)$, $y(0)=1$, is

A. $4/(1+3e^{-x})$

B. $4/(1+e^{-x})$

C. $1/(1+3e^{-x})$

D. $4/(1-3e^{-x})$

QUESTION 168

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

Solve $y'=-0.5y$, $y(0)=80$.

A. $80-0.5t$

B. $40e^{-t}$

C. $80e^{-0.5t}$

D. $80e^{0.5t}$

QUESTION 169

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

Solve $y' = -2y$ with $y(1) = 5$.

A $5e^{2(x-1)}$

B $5e^{-2x}$

C $5e^{-2(x-1)}$

D $5 - 2(x-1)$

QUESTION 170

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

An exponential model has half-life 6 hours and satisfies $A' = -kA$. The value of k is

A $-\ln 2/6$

B $\ln 2/6$

C $6/\ln 2$

D $\ln 6/2$

QUESTION 171

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

For $A' = -0.04A$, the half-life is

A $0.04/\ln 2$

B $\ln 2/0.04$

C $\ln(0.04)/2$

D 25 hours

QUESTION 172

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★ MANDATORY

If $N'=0.05N$ and $N=400$ at an instant, then N' at that instant is

A 0.05

B 20

C 400

D 8

QUESTION 173

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

For $A'=-0.2A$, the time for A to fall to one quarter of its initial value is

A $4/0.2$

B $\ln 4/0.2$

C $0.2 \ln 4$

D $\ln 2/0.2$

QUESTION 174

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

For $P'=0.3P(1-P/800)$, the carrying capacity is

A 800

B 400

C 0.3

D 0

QUESTION 175

UNIT 7 · DIFFERENTIAL EQUATIONS · MODERATE

★★★★★ VERY IMPORTANT

The general positive, non-equilibrium solution of $P' = rP(1 - P/K)$, $r > 0$, has the form

A $P = K(1 + Ae^{-rt})$

B $P = Ae^{rt}$

C $P = K/(1 - Ae^{rt})$

D $P = K/(1 + Ae^{-rt})$

QUESTION 176

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

The average value of $f(x) = x^2$ on $[0,3]$ is

A 6

B 9

C 3

D 2

QUESTION 177

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

The average value of $f(x) = 3x + 2$ on $[1,4]$ is

A 8

B 19/2

C 10

D 57/2

QUESTION 178

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

The average value of sin x on $[0,\pi]$ is

A $1/\pi$

B 1

C $2/\pi$

D 2

QUESTION 179

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

For $f(x)=x^2$ on $[0,2]$, the value c in $[0,2]$ satisfying $f(c)$=the average value of f is

A $1/\sqrt{3}$

B $2/\sqrt{3}$

C $4/3$

D $\sqrt{3}$

QUESTION 180

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

A particle has velocity $v(t)=t-1$ on $[0,3]$. Its displacement is

A 1

B 3/2

C $-3/2$

D 5/2

QUESTION 181

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

For $v(t)=t-1$ on $[0,3]$, the total distance traveled is

A) 2

B) 5/2

C) 7/2

D) 3/2

QUESTION 182

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

A particle has velocity $v(t)=(t-1)(t-3)$. On which interval is it moving in the negative direction?

A) $t>3$

B) All $t>0$

C) 0

D) 1

QUESTION 183

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

A particle's velocity is linear between the table values below.

t	0	1	2	3
v(t)	0	2	0	−2

Using the trapezoidal rule on each unit interval, the displacement on $[0,3]$ is

A) 0

B) 2

C) -1

D) 1

QUESTION 184

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

The area between $y=1$ and $y=x^2$ on $[-1,1]$ is

A 4/3

B 1

C 2

D 2/3

QUESTION 185

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

The area between $y=x$ and $y=x^2$ on $[-1,1]$ is

A 4/3

B 3/2

C 1

D 1/2

QUESTION 186

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

The area between $y=x+2$ and $y=x^2$ from $x=-1$ to $x=2$ is

A 6

B 9/2

C 3

D 4

QUESTION 187

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

Using horizontal slices, the area between $x=y^2$ and $x=2y$ for $0\le y\le 2$ is

A $2/3$

B 2

C $4/3$

D $8/3$

QUESTION 188

UNIT 8 · APPLICATIONS OF INTEGRATION · EASY

★★★ MANDATORY

Using horizontal slices, the area between $x=y$ and $x=2$ for $0\le y\le 2$ is

A 1

B 3

C 4

D 2

QUESTION 189

UNIT 8 · APPLICATIONS OF INTEGRATION · EASY

★★★ MANDATORY

The area of the upper semicircle $y=\sqrt{(4-x^2)}$, $-2\le x\le 2$, is

A 4π

B 8π

C 2π

D π

QUESTION 190

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

A solid has square cross sections perpendicular to the x-axis. The base extends vertically from $y=0$ to $y=1-x^2$ for $-1\le x\le 1$. The volume is

A. 8/15
B. 2
C. 4/3
D. 16/15

QUESTION 191

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

Rectangular cross sections have base $1-x$ and height twice the base for $0\le x\le 1$. Their volume is

A. 1
B. 4/3
C. 1/3
D. 2/3

QUESTION 192

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

Semicircular cross sections have diameter $\sqrt{(1-x^2)}$ for $-1\le x\le 1$. The volume is

A. $\pi/6$
B. $\pi/3$
C. $2\pi/3$
D. $\pi/12$

QUESTION 193

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

Equilateral-triangle cross sections have side $\sqrt{(4-x^2)}$ for $-2\le x\le 2$. The volume is

A $8\sqrt{3}$

B $4\sqrt{3}/3$

C $8\sqrt{3}/3$

D $16\sqrt{3}/3$

QUESTION 194

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

The region under $y=\sqrt{(4-x^2)}$ on $[-2,2]$ is revolved about the x-axis. The volume is

A $32\pi/3$

B 16π

C 8π

D $16\pi/3$

QUESTION 195

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

The region under $y=x$ on $0\le x\le 3$ is revolved about the x-axis. The volume is

A 9π

B 27π

C 3π

D 6π

QUESTION 196

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

The region between $x=0$ and $x=\sqrt{y}$ for $0\le y\le 4$ is revolved about the y-axis. The volume is

A 4π

B 8π

C 16π

D 6π

QUESTION 197

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

The region between $y=2$ and $y=x$ on $0\le x\le 2$ is revolved about the x-axis. The washer volume is

A $8\pi/3$

B 16π

C $16\pi/3$

D 8π

QUESTION 198

UNIT 8 · APPLICATIONS OF INTEGRATION · HARD

★★★★★ VERY IMPORTANT

The region between $y=2x$ and $y=x^2$ on $0\le x\le 2$ is revolved about the x-axis. The volume is

A $32\pi/15$

B $16\pi/3$

C $8\pi/3$

D $64\pi/15$

QUESTION 199

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★★★ VERY IMPORTANT

The region under $y=x$ on $[0,2]$ is revolved about $y=3$. The volume is

A $16\pi/3$

B $32\pi/3$

C $20\pi/3$

D $28\pi/3$

QUESTION 200

UNIT 8 · APPLICATIONS OF INTEGRATION · MODERATE

★★★ MANDATORY

The region under $y=3-x$ on $0\le x\le 3$ is revolved about the x-axis. The volume is

A 6π

B 18π

C 9π

D 3π

Question 201: Piecewise Continuity with Two Parameters

★★★★★ Very Important | 9 points

Let f be defined by $f(x) = \{x^2 + a \text{ for } x < 1;\ 4 \text{ for } x = 1;\ bx + 2 \text{ for } x > 1\}$, where a and b are constants.

(a) [2 points] Find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$ in terms of a and b.

(b) [4 points] Determine a and b so that f is continuous at x = 1, justifying with the three conditions of continuity.

(c) [3 points] If a = 2 and b = 5, classify the discontinuity at x = 1 with justification.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 202: Zeros of a Continuous Function from a Table

★★★★★ Very Important | 9 points

The function f is continuous on [0, 4]. Selected values are given.

x: 0, 1, 2, 3, 4 → f(x): 3, −1, 2, −2, 5

(a) [2 points] Explain why the Intermediate Value Theorem applies to f on [0, 4].

(b) [4 points] Find the minimum number of values of x in (0, 4) for which f(x) = 0. Justify your answer.

(c) [3 points] Must there be a value c in (0, 1) with f(c) = 1? Justify.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 203: Asymptotes of a Rational Function

★★★★★ Very Important · 9 points

Let $f(x) = (x^2 - 1)/(x^2 - 4)$.

(a) [2 points] Factor the numerator and the denominator completely.

(b) [4 points] Find all vertical asymptotes and evaluate $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$.

(c) [3 points] Find $\lim_{x \to \infty} f(x)$ and state the horizontal asymptote.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 204: The Squeeze Theorem

★★★★★ Very Important · 9 points

It is given that $4x - x^2 \le g(x) \le x^2 + 2$ for all x in an open interval containing $x = 1$.

(a) [2 points] Evaluate $\lim_{x \to 1}(4x - x^2)$ and $\lim_{x \to 1}(x^2 + 2)$.

(b) [3 points] Use the Squeeze Theorem to find $\lim_{x \to 1} g(x)$, stating the required conditions.

(c) [4 points] Evaluate $\lim_{x \to 0} x^3 \cos(1/x)$ using the Squeeze Theorem.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 205: Limits at Infinity with a Radical

★★★★★ Very Important · 9 points

Let $g(x) = 3x/\sqrt{(x^2 + 1)}$.

(a) [3 points] Evaluate $\lim_{x\to\infty} g(x)$.

(b) [4 points] Evaluate $\lim_{x\to-\infty} g(x)$, carefully justifying the sign.

(c) [2 points] Does g have any vertical asymptotes? Justify.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 206: Applied Model: Cooling Coffee

★★★★★ Very Important | 9 points

The temperature of coffee, in °C, t minutes after pouring is modeled by $T(t) = 20 + 70e^{(-0.05t)}$.

(a) [2 points] Find T(0).

(b) [3 points] Evaluate lim (t→∞) T(t) and interpret the value in context.

(c) [4 points] Given e^(−1.5) ≈ 0.223, use the IVT to show the coffee reaches 50°C at some time t in (0, 30).

Show your reasoning. State the theorem or method used and include units when relevant.

Question 207: Definition of Derivative and Tangent Line

★★★★★ Very Important | 9 points

Let f be the function defined by $f(x) = 1/(x + 1)$ for $x \neq -1$.

(a) [4 points] Use the limit definition of the derivative to show that $f'(x) = -1/(x + 1)^2$.

(b) [3 points] Write an equation for the line tangent to the graph of f at $x = 1$.

(c) [2 points] Find all points on the graph of f where the tangent line is parallel to the line $x + 4y = 8$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 208: Piecewise Continuity and Differentiability

★★★★★ Very Important | 9 points

Let g be the function defined by $g(x) = \{ax^2 + 1 \text{ for } x \le 2;\ bx - 3 \text{ for } x > 2\}$, where a and b are constants.

(a) [3 points] Find a relationship between a and b so that g is continuous at x = 2.

(b) [4 points] Find the values of a and b for which g is differentiable at x = 2.

(c) [2 points] Using the values from part (b), write an equation for the tangent line to g at x = 2.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 209: Analyzing Differentiability from a Graph

★★★★★ Very Important | 9 points

The graph of f consists of a line segment from $(-3, 0)$ to $(0, 3)$, a quarter circle of radius 3 from $(0, 3)$ to $(3, 0)$ centered at $(0, 0)$, and a line segment from $(3, 0)$ to $(5, -4)$.

(a) [2 points] Find $f'(-1)$ and explain your reasoning.

(b) [3 points] Is f differentiable at $x = 0$? Justify your answer.

(c) [4 points] State all values of x in $(-3, 5)$ where f is not differentiable, and classify each reason.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 210: Tabular Data: Average Rates and Tangent Approximation

★★★★★ Very Important 9 points

The table shows values of a twice-differentiable function h.
t (sec): 0, 2, 5, 8 → h(t) (meters): 1, 7, 19, 37.

(a) [2 points] Find the average velocity of h on [0, 8]. Include units.

(b) [3 points] Estimate $h'(5)$ using values from the table. Include units.

(c) [4 points] Use the tangent line to h at $t = 2$ to approximate $h(2.3)$, given $h'(2) = 3.5$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 211: Derivatives of Products and Quotients with Trig

★★★★★ Very Important | 9 points

Let $f(x) = (x^2 + 2)\cos x$.

(a) [3 points] Find $f'(x)$.

(b) [2 points] Find the slope of the tangent line to the graph of f at $x = 0$.

(c) [4 points] Find $f''(x)$ and evaluate $f''(0)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 212: Horizontal Tangents and Normal Lines to a Rational Function

★★★★★ Very Important 9 points

Let $R(x) = (x - 1)/(x^2 + 3)$.

(a) [3 points] Find $R'(x)$.

(b) [3 points] Find all values of x where R has a horizontal tangent line.

(c) [3 points] Write an equation for the normal line to the graph of R at $x = 1$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 213: Implicit Curve: Tangents and Asymptotes

★★★★★ Very Important | 9 points

Consider the curve defined by $x^3 + y^3 = 6xy$.

(a) [3 points] Show that $dy/dx = (2y - x^2) / (y^2 - 2x)$.

(b) [3 points] Find the equation of the tangent line to the curve at the point (3, 3).

(c) [3 points] Find the coordinates of the point in the first quadrant (other than the origin) where the tangent line is horizontal.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 214: Derivative of an Inverse Function with Table

★★★★★ Very Important | 9 points

Let f be a twice-differentiable, strictly increasing function. The table gives values of f and f′.

x: 1, 2, 4 → f(x): 2, 4, 9 → f′(x): 3, 5, 8.

Let $g = f^{-1}$ be the inverse function of f.

(a) [3 points] Find g(4) and g′(4). Show your work.

(b) [3 points] Write an equation for the line tangent to the graph of y = g(x) at x = 4.

(c) [3 points] Find g″(4), given f″(2) = 6.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 215: Chain Rule with Tables for Composites

★★★★★ Very Important | 9 points

The table shows values of differentiable functions f and g and their derivatives.

x: 1, 2, 3 → f: 3, 1, 2 → f′: −2, 4, 5 → g: 2, 3, 1 → g′: 6, −1, 3.

(a) [3 points] Let $h(x) = f(g(x))$. Find $h'(1)$.

(b) [3 points] Let $k(x) = g(f(x))$. Find $k'(2)$.

(c) [3 points] Let $p(x) = [f(x)]^2 \cdot g(x)$. Find $p'(3)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 216: Inverse Trigonometric Function and Tangent Lines

★★★★★ Very Important 9 points

Let $f(x) = \arctan(2x)$.

(a) [2 points] Find $f'(x)$.

(b) [4 points] Find the equation of the tangent line to the graph of f at $x = 1/2$.

(c) [3 points] Find all values of x where the slope of the tangent line to f equals 1.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 217: Higher-Order Implicit Differentiation

★★★★★ Very Important | 9 points

Consider the curve $x^2 + 4y^2 = 8$.

(a) [2 points] Find dy/dx in terms of x and y.

(b) [4 points] Show that $d^2y/dx^2 = -2 / y^3$.

(c) [3 points] Determine the concavity of the curve in the upper half-plane ($y > 0$).

Show your reasoning. State the theorem or method used and include units when relevant.

Question 218: Multi-Layer Chain Rule Applied to Motion

★★★★★ Very Important | 9 points

A particle moves along the x-axis so that its position is $x(t) = \ln(t^2 + 1)$ for $t \geq 0$.

(a) [2 points] Find the velocity $v(t)$.

(b) [3 points] Find the acceleration $a(t)$.

(c) [4 points] At what time $t > 0$ is the velocity maximized?

Show your reasoning. State the theorem or method used and include units when relevant.

Question 219: Particle Motion Analysis

★★★★★ Very Important | 9 points

A particle moves along a straight line with position function $s(t) = 2t^3 - 9t^2 + 12t$ for $t \geq 0$, where s is in meters and t in seconds.

(a) [2 points] Find the velocity $v(t)$ and acceleration $a(t)$ of the particle.

(b) [2 points] During what time intervals is the particle moving to the left? Justify your answer.

(c) [2 points] At $t = 1.5$, is the particle speeding up or slowing down? Explain your reasoning.

(d) [3 points] Find the total distance traveled by the particle over the time interval $[0, 3]$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 220: Related Rates: Draining Conical Tank

★★★★★ Very Important | 9 points

A water tank has the shape of an inverted circular cone with base radius 6 feet and height 12 feet. Water is leaking out of the bottom at a constant rate of 5 ft^3/min.

(a) [3 points] Write the volume V of water in the tank solely as a function of the water's depth h.

(b) [4 points] At what rate is the water level falling when the water is 4 feet deep?

(c) [2 points] Is the water level falling faster or slower as the depth h decreases? Justify your answer mathematically.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 221: Related Rates: Sliding Ladder with Angle and Area

★★★★★ Very Important | 9 points

A 25-foot ladder leans against a vertical wall. The bottom of the ladder slides away from the wall at 2 ft/s.

(a) [3 points] Find the rate at which the top of the ladder is sliding down when the bottom is 7 feet from the wall.

(b) [3 points] Find the rate of change of the area of the triangle formed by the ladder, wall, and ground at this instant.

(c) [3 points] Find the rate of change of the angle θ between the ladder and the ground at this instant.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 222: Linearization, Error and Concavity from a Table

★★★★★ Very Important | 9 points

Selected values of a twice-differentiable function f are shown in the table.

x: 3, 5, 7 → f(x): 8, 14, 18 → f′(x): 3.5, 2.5, 1.5.

It is known that $f''(x) < 0$ for all x in [3, 7].

(a) [2 points] Write the equation of the line tangent to the graph of f at x = 5.

(b) [2 points] Use this tangent line to approximate f(5.2).

(c) [2 points] Is your approximation in part (b) an overestimate or underestimate? Justify your answer using information given about f.

(d) [3 points] Estimate f′′(5) using values of f′ from the table.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 223: Velocity Graph: Acceleration, Direction and Distance

★★★★★ Very Important | 9 points

A particle moves along a horizontal axis. Its velocity $v(t)$, in m/s, for $0 \le t \le 8$ is given by line segments joining the points $(0, 3)$, $(2, 3)$, $(4, -3)$, $(6, -3)$, and $(8, 0)$.

(a) [2 points] Find the acceleration of the particle at $t = 3$ and at $t = 5$.

(b) [2 points] At what time t in $(0, 8)$ does the particle change direction? Justify.

(c) [3 points] During what time intervals is the particle's speed increasing? Justify.

(d) [2 points] Find the total distance traveled by the particle on $[0, 8]$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 224: L'Hôpital's Rule with Parameters and Limits

★★★★★ Very Important · 9 points

Consider the function $f(x) = (\cos(ax) - 1) / (b\,x^2)$, where a and b are nonzero constants.

(a) [4 points] Evaluate $\lim_{x\to 0} f(x)$ in terms of a and b using L'Hôpital's Rule.

(b) [2 points] Given that $\lim_{x\to 0} f(x) = -2$ and $a = 4$, find the value of b.

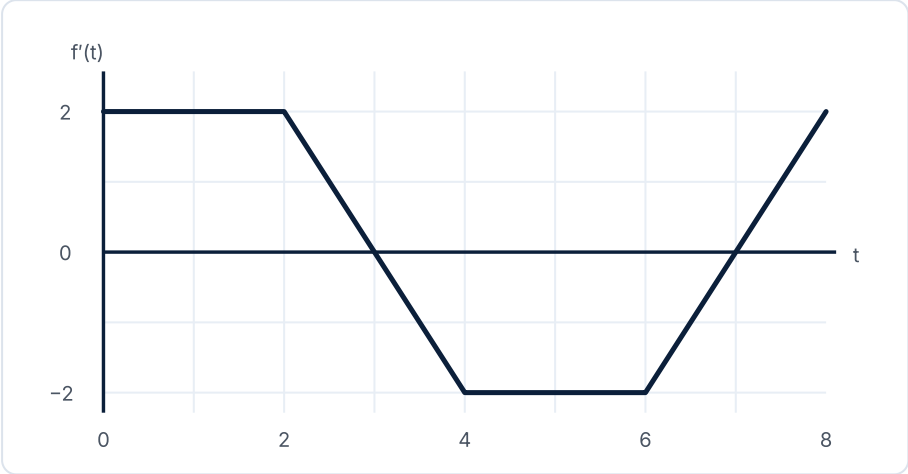
(c) [3 points] Evaluate $\lim_{x\to 0} [x - \sin x] / [x(1 - \cos x)]$. Show all steps.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 225: Analyzing f′ from a Graph

★★★★★ Very Important | 9 points

The graph shows $f'(t)$ for $0 \le t \le 8$. The straight segments connect the plotted points. Given $f(0)=0$.



Graph of f′ on $0 \le t \le 8$; straight segments connect the plotted points.

(a) [2 points] State the intervals on which f is increasing and decreasing.

(b) [2 points] Find and classify the relative extrema of f in (0,8).

(c) [3 points] Use signed areas under f′ to find f(8).

(d) [2 points] Find the absolute maximum and minimum values of f on [0,8].

Show your reasoning. State the theorem or method used and include units when relevant.

Question 226: Mean Value Theorem

★★★★★ Very Important | 9 points

Let $f(t)=t^2-4t+1$ on the interval $[0,4]$.

(a) [2 points] Explain why the Mean Value Theorem applies.

(b) [2 points] Find the average rate of change of f on [0,4].

(c) [3 points] Find all values c guaranteed by the theorem.

(d) [2 points] Explain why the tangent at t=c is parallel to the secant through the endpoints.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 227: Absolute Extrema and the Candidates Test

★★★★★ Very Important · 9 points

Let $g(x)=x^3-6x^2+9x$ on the closed interval $[0,5]$.

(a) [2 points] Find the interior critical numbers of g.

(b) [3 points] Evaluate g at the endpoints and at every interior critical number.

(c) [4 points] State the absolute maximum and minimum values and where they occur.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 228: Concavity and Points of Inflection

★★★★★ Very Important | 9 points

Let $f(x)=x^4/4-2x^3/3$.

(a) [2 points] Find $f'(x)$ and $f''(x)$, and factor $f''(x)$.

(b) [2 points] Determine the intervals where f is concave up and concave down.

(c) [3 points] Find the coordinates of all points of inflection.

(d) [2 points] Explain why checking $f''(x)=0$ alone is not enough to establish an inflection point.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 229: Second Derivative Test

★★★★★ Very Important | 9 points

Let $f(x)=x^4-4x^2$.

(a) [3 points] Find f', f'', and all critical numbers.

(b) [3 points] Use the Second Derivative Test to classify each critical number.

(c) [3 points] Find the coordinates of the relative extrema.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 230: Using a Sign Table for f′

★★★★★ Very Important | 9 points

The table describes a differentiable function f on $[-2,3]$.

x-interval	$(-2,-1)$	$(-1,1)$	$(1,3)$
sign of f′	+	−	+

x	−2	−1	1	3
f(x)	0	4	−2	1

(a) [3 points] State the intervals where f is increasing and decreasing.

(b) [2 points] Find and classify the relative extrema in $(-2,3)$.

(c) [4 points] Find the absolute maximum and minimum values on $[-2,3]$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 231: Riemann Sums from a Table

★★★★★ Very Important · 9 points

The function f is continuous and increasing on [0, 8]. Selected values are given in the table.

x	0	2	4	6	8
f(x)	3	5	4	2	1

Note: the table is consistent with a continuous curve that rises then falls, so use the listed values exactly as given.

(a) [3 points] Use a left Riemann sum with n = 4 to estimate $\int_0^8 f(x)\,dx$.

(b) [3 points] Use a right Riemann sum with n = 4 to estimate the same integral.

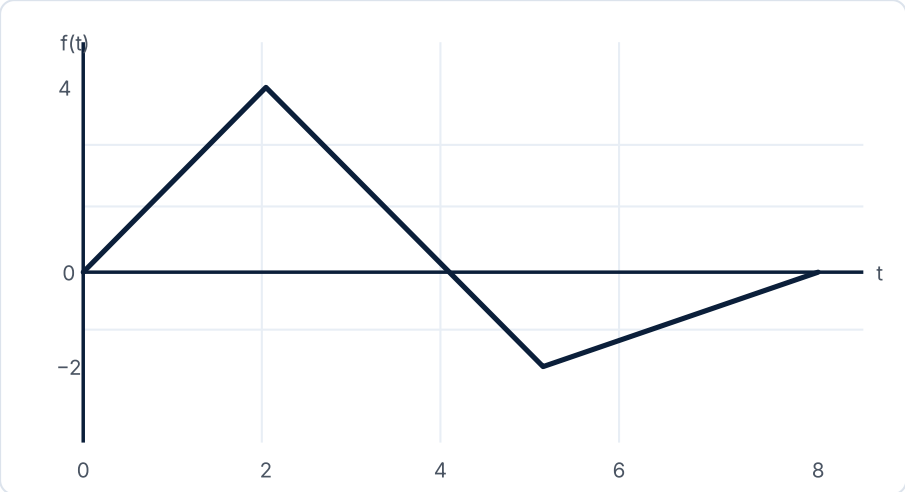
(c) [3 points] Use the trapezoid rule with n = 4 and explain why it is the average of your two earlier estimates.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 232: Definite Integrals from a Graph

★★★★★ Very Important | 9 points

The graph of f consists of straight line segments and is shown for $0 \le t \le 8$.



Graph of f on $0 \le t \le 8$, made of straight line segments.

(a) [3 points] Find $\int_0^8 f(t)\,dt$. Explain how the signs of the regions affect your answer.

(b) [3 points] Find the total area between the graph of f and the t-axis on [0, 8].

(c) [2 points] Find the average value of f on [0, 8].

(d) [1 points] Explain why your answers to (a) and (b) are different numbers.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 233: Analyzing an Accumulation Function

★★★★★ Very Important · 9 points

Let $F(x) = \int_0^x (t^2 - 4)\,dt$.

(a) [2 points] Find $F'(x)$ and state the theorem you used.

(b) [3 points] Find an explicit formula for $F(x)$.

(c) [2 points] Find the critical numbers of F and classify each as a local maximum or local minimum.

(d) [2 points] Evaluate $F(2)$ and $F(-2)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 234: FTC and Average Value

★★★★★ Very Important | 9 points

Let $f(x) = 3x^2 - 2x$.

(a) [3 points] Evaluate $\int_1^4 f(x)\,dx$.

(b) [2 points] Find the average value of f on [1, 4].

(c) [4 points] Find the value c in [1, 4] such that f(c) equals the average value from part (b).

Show your reasoning. State the theorem or method used and include units when relevant.

Question 235: Motion: Displacement and Distance

★★★★★ Very Important | 9 points

A particle moves along a line with velocity $v(t) = t^2 - 4t + 3$ for $0 \le t \le 5$, where v is in m/s.

(a) [3 points] Find the displacement over [0, 5].

(b) [4 points] Find the total distance traveled over [0, 5].

(c) [2 points] During which time intervals is the particle moving in the positive direction?

Show your reasoning. State the theorem or method used and include units when relevant.

Question 236: From a Riemann Sum to an Integral

★★★★★ Very Important · 9 points

Consider $\lim_{n\to\infty} \sum_{i=1}^{n} \left(2 + \frac{6i}{n}\right)^2 \cdot \frac{6}{n}$.

(a) [3 points] Identify Δx and the sample points x_i, and state the interval $[a, b]$.

(b) [3 points] Write the limit as a definite integral and identify the integrand $f(x)$.

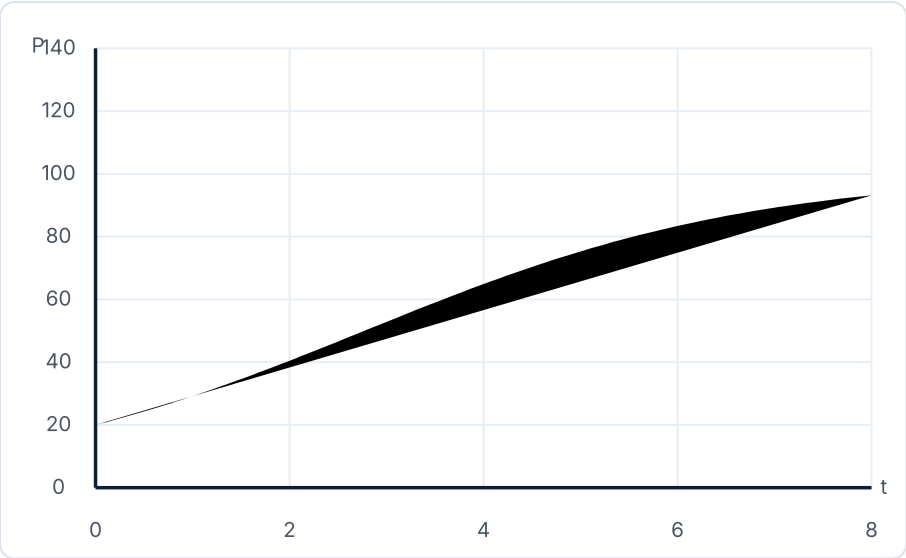
(c) [3 points] Evaluate the integral exactly.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 237: Logistic Slope Field and Long-Term Behavior

★★★★★ Very Important | 9 points

A population $P(t)$ satisfies $P' = 0.5P(1 - P/100)$. The slope field is shown below; the red curve is the solution with $P(0) = 20$.



Slope field for $P' = 0.5P(1 - P/100)$; red curve is the solution with $P(0) = 20$.

(a) [2 points] Find the equilibrium solutions of the differential equation.

(b) [2 points] Find the slope at the point $(0, 20)$, and interpret its sign.

(c) [2 points] Use the slope field to describe the long-term behavior of the solution starting at $P(0) = 20$.

(d) [3 points] Solve the initial-value problem and estimate $P(2)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 238: Verifying and Applying a Solution

★★★★★ Very Important | 9 points

Consider the differential equation $y' = 2xy$ and the proposed family $y = Ce^{x^2}$.

(a) [2 points] Verify by differentiation that $y = Ce^{x^2}$ satisfies the differential equation.

(b) [2 points] Use $y(0) = 3$ to determine C and write the particular solution.

(c) [2 points] Evaluate $y(1)$.

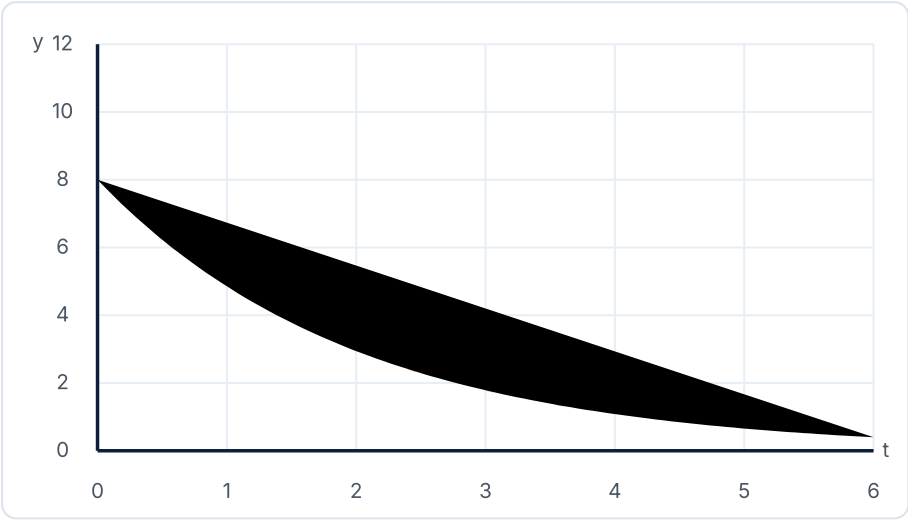
(d) [3 points] Show that $y = 0$ is also an equilibrium solution.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 239: Reasoning from an Exponential-Decay Slope Field

★★★★★ Very Important | 9 points

The slope field for $y' = -0.5y$ is shown. A solution passes through $(0,8)$.



Slope field for $y' = -0.5y$; red curve passes through $(0,8)$.

(a) [2 points] Find the slope at $(0,8)$.

(b) [3 points] Separate variables and solve the initial-value problem.

(c) [2 points] Find $y(2)$.

(d) [2 points] State the equilibrium solution and the limit of $y(t)$ as $t \to \infty$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 240: Separation of Variables with an Initial Condition

★★★★★ Very Important 9 points

A solution satisfies $y'=(x+1)/(y+1)$, $y(0)=1$, and $y>-1$.

(a) [2 points] Separate the variables.

(b) [3 points] Find the general implicit solution.

(c) [2 points] Use the initial condition to determine the constant.

(d) [2 points] Find $y(1)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 241: Exponential Population Model from a Table

★★★★★ Very Important | 9 points

A population follows $P' = kP$, where t is measured in hours. The table gives observed values.

t (hours)	0	2	4	6
P(t)	200	$200\sqrt{2}$	400	$400\sqrt{2}$

(a) [3 points] Use P(0) and P(4) to determine k.

(b) [2 points] Write the particular solution P(t).

(c) [2 points] Predict P(6).

(d) [2 points] Find $P'(4)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 242: Radioactive Decay and Half-Life

★★★★★ Very Important | 9 points

A radioactive sample has initial mass 640 grams and follows $A' = -kA$. Its half-life is 8 hours.

(a) [2 points] Determine k.

(b) [2 points] Write A(t).

(c) [2 points] Find the mass after 24 hours.

(d) [3 points] How long does it take for the mass to reach 40 grams?

Show your reasoning. State the theorem or method used and include units when relevant.

Question 243: Determining a Logistic Model from Rates

★★★★★ Very Important | 9 points

A population P is modeled by $P' = rP(1 - P/K)$. Measurements are shown below.

P	0	50	100	150	200
P′	0	30	40	30	0

(a) [2 points] Use the table to identify the carrying capacity K.

(b) [2 points] Determine r.

(c) [3 points] Find the maximum growth rate and the population at which it occurs.

(d) [2 points] Classify the equilibria P=0 and P=200 as stable or unstable.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 244: Average Value and the Integral Mean Value Theorem

★★★★★ Very Important 9 points

Let $f(x)=x^2$ on the interval $[0,4]$.

(a) [3 points] Find the average value of f on $[0,4]$.

(b) [3 points] Find $c \in [0,4]$ such that $f(c)$ equals its average value.

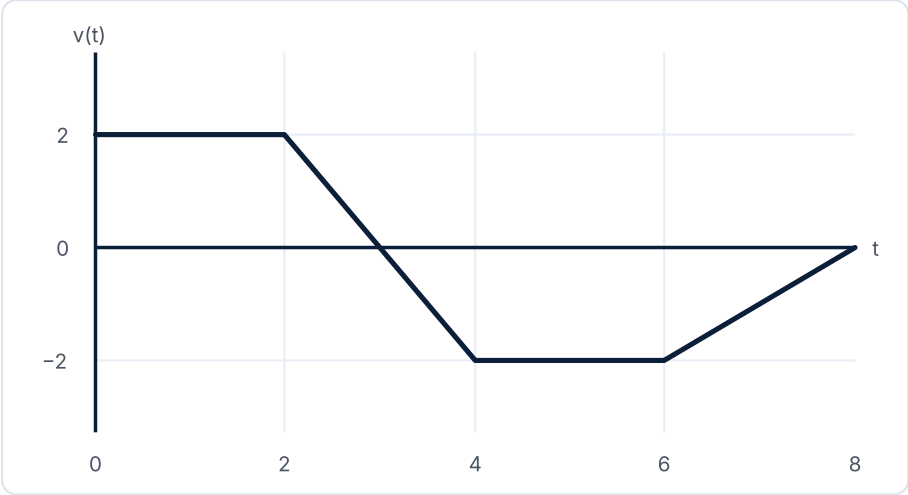
(c) [3 points] Explain why the Integral Mean Value Theorem guarantees at least one such c.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 245: Motion from a Velocity Graph

★★★★★ Very Important | 9 points

The graph shows $v(t)$, in meters per second, for a particle on $0 \le t \le 8$.



Velocity graph, in meters per second; straight segments join the listed vertices.

Given that the particle's initial position is $s(0) = 5$ m.

(a) [2 points] Find the acceleration on each open interval $(0,2)$, $(2,4)$, $(4,6)$, and $(6,8)$.

(b) [2 points] State when the particle is moving in the negative direction.

(c) [3 points] Find the displacement and total distance traveled on $[0,8]$.

(d) [2 points] Find the final position $s(8)$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 246: Area Between Curves and Average Vertical Distance

★★★★★ Very Important | 9 points

Consider the region bounded by $y=x+2$ and $y=x^2$ from $x=-1$ to $x=2$.

(a) [2 points] Find the intersection points of the two curves.

(b) [2 points] Determine which curve lies above the other between the intersections.

(c) [3 points] Set up and evaluate the area between the curves.

(d) [2 points] Find the average vertical distance between the curves on $[-1,2]$.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 247: Volume with Square Cross Sections

★★★★★ Very Important | 9 points

The base of a solid is the region under $y=4-x^2$ and above the x-axis for $-2\le x\le 2$. Cross sections perpendicular to the x-axis are squares.

(a) [2 points] State the side length of the square at a position x.

(b) [2 points] Write the cross-sectional area A(x).

(c) [3 points] Set up and evaluate the volume integral.

(d) [2 points] Explain why the area expression must be squared.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 248: Disk Method: A Semicircle Revolved

★★★★★ Very Important · 9 points

The region under $y=\sqrt{(4-x^2)}$, above the x-axis, and between $x=-2$ and $x=2$ is revolved about the x-axis.

(a) [2 points] Identify the disk radius at position x.

(b) [3 points] Set up the disk-method integral for the volume.

(c) [4 points] Evaluate the exact volume.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 249: Washer Method About the x-Axis

★★★★★ Very Important · 9 points

The region between $y=2$ and $y=x$ on $0\le x\le 2$ is revolved about the x-axis.

(a) [2 points] Identify the outer and inner radii.

(b) [3 points] Write the washer-method integral.

(c) [4 points] Evaluate the volume exactly.

Show your reasoning. State the theorem or method used and include units when relevant.

Question 250: Horizontal Slices and Rotation About the y-Axis

★★★★★ Very Important 9 points

Consider the region bounded by $x=y^2$ and $x=2y$ for $0\le y\le 2$.

(a) [2 points] Find the y-values where the bounding curves intersect.

(b) [3 points] Set up and evaluate the area using horizontal slices.

(c) [4 points] Revolve the region about the y-axis and find the volume using washers.

Show your reasoning. State the theorem or method used and include units when relevant.

Questions 1–250 Complete

The following pages contain the MCQ key, worked solutions, and FRQ model responses with scoring rubrics.

> **Important:** This practice set uses its own raw-point scheme; it does not convert to an official AP score.

Multiple-Choice Answer Key (Part 1)

Q001 **A** ★★★	Q002 **C** ★★	Q003 **D** ★★★★★	Q004 **C** ★★★	Q005 **D** ★★	Q006 **A** ★★★★★	Q007 **A** ★★★	Q008 **C** ★★★	Q009 **C** ★★★★★	Q010 **A** ★★★★★
Q011 **B** ★★★	Q012 **A** ★★★	Q013 **B** ★★★★★	Q014 **B** ★★★	Q015 **B** ★★★	Q016 **B** ★★	Q017 **C** ★★	Q018 **D** ★★★★★	Q019 **B** ★★★	Q020 **C** ★★★★★
Q021 **D** ★★	Q022 **D** ★★★	Q023 **A** ★★★	Q024 **A** ★★	Q025 **D** ★★★	Q026 **B** ★★★★★	Q027 **B** ★★★	Q028 **A** ★★★★★	Q029 **B** ★★★	Q030 **B** ★★★
Q031 **C** ★★	Q032 **B** ★★★	Q033 **B** ★★★★★	Q034 **A** ★★★	Q035 **A** ★★★★★	Q036 **B** ★★★★★	Q037 **D** ★★★	Q038 **D** ★★★	Q039 **B** ★★★	Q040 **D** ★★★
Q041 **D** ★★★★★	Q042 **C** ★★★★★	Q043 **C** ★★★	Q044 **A** ★★★	Q045 **A** ★★	Q046 **A** ★★★	Q047 **B** ★★★	Q048 **B** ★★★★★	Q049 **B** ★★★★★	Q050 **B** ★★★

Multiple-Choice Answer Key (Part 2)

Q051	Q052	Q053	Q054	Q055	Q056	Q057	Q058	Q059	Q060
A ★★★★★	B ★★★	B ★★★★★	B ★★★	B ★★★★★	C ★★★★★	A ★★★★★	D ★★★★★	A ★★★	A ★★★★★
Q061	**Q062**	**Q063**	**Q064**	**Q065**	**Q066**	**Q067**	**Q068**	**Q069**	**Q070**
B ★★★★★	C ★★★★★	D ★★★★★	B ★★★★★	C ★★★★★	B ★★★★★	B ★★★★★	D ★★★	B ★★★	B ★★★
Q071	**Q072**	**Q073**	**Q074**	**Q075**	**Q076**	**Q077**	**Q078**	**Q079**	**Q080**
B ★★★★★	A ★★★	B ★★★★★	D ★★★	C ★★★★★	C ★★★★★	A ★★★	D ★★★	C ★★★★★	B ★★★★★
Q081	**Q082**	**Q083**	**Q084**	**Q085**	**Q086**	**Q087**	**Q088**	**Q089**	**Q090**
A ★★★	B ★★★★★	D ★★★★★	B ★★★★★	A ★★★★★	B ★★★	D ★★★★★	C ★★★	B ★★★★★	D ★★★★★
Q091	**Q092**	**Q093**	**Q094**	**Q095**	**Q096**	**Q097**	**Q098**	**Q099**	**Q100**
B ★★★★★	A ★★★	B ★★★★★	C ★★★	A ★★★★★	A ★★★★★	D ★★	A ★★★	C ★★★	C ★★★

Multiple-Choice Answer Key (Part 3)

Q101 A ★★★	Q102 D ★★★	Q103 D ★★★	Q104 A ★★★	Q105 B ★★★	Q106 C ★★★★★	Q107 C ★★★★★	Q108 A ★★★	Q109 B ★★★	Q110 B ★★★
Q111 A ★★★★★	Q112 D ★★★	Q113 B ★★★	Q114 C ★★★★★	Q115 B ★★★★★	Q116 D ★★★★★	Q117 A ★★★★★	Q118 D ★★★★★	Q119 C ★★★★★	Q120 C ★★★
Q121 D ★★★★★	Q122 C ★★★★★	Q123 A ★★★	Q124 D ★★★	Q125 D ★★★★★	Q126 D ★★★	Q127 C ★★★	Q128 C ★★	Q129 A ★★★★★	Q130 C ★★★
Q131 A ★★★	Q132 B ★★	Q133 B ★★	Q134 B ★★★	Q135 C ★★	Q136 A ★★	Q137 B ★★★	Q138 C ★★★★★	Q139 C ★★★	Q140 A ★★★★★
Q141 D ★★★	Q142 D ★★★	Q143 B ★★★★★	Q144 D ★★★	Q145 B ★★★★★	Q146 C ★★★★★	Q147 D ★★★★★	Q148 B ★★★★★	Q149 C ★★★★★	Q150 D ★★★

Multiple-Choice Answer Key (Part 4)

Q151 C ★★★★★	Q152 B ★★★★★	Q153 B ★★★	Q154 D ★★★★★	Q155 C ★★★	Q156 A ★★★★★	Q157 C ★★★	Q158 C ★★★★★	Q159 B ★★★	Q160 B ★★★★★
Q161 B ★★★★★	Q162 D ★★★	Q163 C ★★★★★	Q164 A ★★★	Q165 A ★★★★★	Q166 D ★★★	Q167 A ★★★★★	Q168 C ★★★★★	Q169 C ★★★	Q170 B ★★★★★
Q171 B ★★★★★	Q172 B ★★★	Q173 B ★★★★★	Q174 A ★★★★★	Q175 D ★★★★★	Q176 C ★★★	Q177 B ★★★	Q178 C ★★★	Q179 B ★★★★★	Q180 B ★★★
Q181 B ★★★★★	Q182 D ★★★	Q183 D ★★★	Q184 A ★★★	Q185 C ★★★★★	Q186 B ★★★★★	Q187 C ★★★★★	Q188 D ★★★	Q189 C ★★★	Q190 D ★★★★★
Q191 D ★★★	Q192 A ★★★	Q193 C ★★★★★	Q194 A ★★★★★	Q195 A ★★★	Q196 B ★★★★★	Q197 C ★★★★★	Q198 D ★★★★★	Q199 D ★★★★★	Q200 C ★★★

Question 001 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (A) 6

STEP-BY-STEP SOLUTION

Factor the numerator: $(x-3)(x+3)/(x-3) = x+3$ for $x \neq 3$. Then $\lim (x\to 3)(x+3) = 6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Confuses the x-value 3 with the limit value.

✗ **(C)** A 0/0 form does not mean the limit fails to exist; factoring resolves it.

✗ **(D)** Substitutes $x = 3$ into the numerator only.

Question 002 · Unit 1 · Limits & Continuity

★★ Important

Correct answer: (C) 9

STEP-BY-STEP SOLUTION

Polynomials are continuous, so substitute directly: $4 + 6 - 1 = 9$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error.

✗ **(B)** Computes $4 + 3 = 7$, forgetting to multiply 3 by 2.

✗ **(D)** Sign error: $4 + 6 + 1$.

Question 003 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (D) 4

STEP-BY-STEP SOLUTION

Write $\sin(4x)/x = 4 \cdot \sin(4x)/(4x)$. As $x\to 0$, $\sin(4x)/(4x) \to 1$, so the limit is 4.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the coefficient 4.

✗ **(B)** Ignores that the denominator also approaches 0.

✗ **(C)** The standard sine limit is finite.

Question 004 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (C) 3/5

STEP-BY-STEP SOLUTION

Equal degrees (both 2): the limit is the ratio of leading coefficients, 3/5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Requires equal leading coefficients.

✗ **(B)** Requires the numerator degree to be larger.

✗ **(D)** Requires the denominator degree to be larger.

Question 005 · Unit 1 · Limits & Continuity

★★ Important

Correct answer: (D) 0

STEP-BY-STEP SOLUTION

Numerator degree (1) is less than denominator degree (2), so the limit is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Requires the numerator degree to be larger.

✗ **(B)** Uses the constant terms.

✗ **(C)** Uses the numerator's leading coefficient.

Question 006 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (A) 1/6

STEP-BY-STEP SOLUTION

Multiply by the conjugate: the numerator becomes x, leaving $1/(\sqrt{(x+9)}+3) \to 1/(3+3) = 1/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the a-value, not the limit.

✗ **(C)** Substitutes into the numerator only.

✗ **(D)** Uses $1/\sqrt{9}$, skipping the conjugate step.

Question 007 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (A) The limit does not exist

STEP-BY-STEP SOLUTION

Left limit: $2+1 = 3$. Right limit: $2^2 = 4$. Since $3 \neq 4$, the two-sided limit does not exist.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses only the left-hand piece.

✗ **(C)** This is f(2), which never affects the limit.

✗ **(D)** Uses only the right-hand piece.

Question 008 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (C) 3

STEP-BY-STEP SOLUTION

Factor: $(x-1)(x^2+x+1)/(x-1) = x^2+x+1 \to 1+1+1 = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Substitutes into the numerator only.

✗ **(B)** Factoring resolves the 0/0 form.

✗ **(D)** Confuses the x-value with the limit.

Question 009 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (C) 0

STEP-BY-STEP SOLUTION

Standard special limit: multiply by the conjugate to get $(\sin x/x)\cdot(\sin x/(1+\cos x)) \to 1\cdot(0/2) = 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the limit of $\sin x/x$.

✗ **(B)** This is the limit of $(1-\cos x)/x^2$.

✗ **(D)** The limit exists and equals 0.

Question 010 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (A) x = 3 only

STEP-BY-STEP SOLUTION

Factor the denominator: (x−3)(x+2). The (x+2) factor cancels with the numerator, leaving 1/(x−3). Only x = 3 is a vertical asymptote; x = −2 is a removable hole.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** x = −2 is a removable hole after cancellation.

✗ **(C)** x = −2 cancels with the numerator, so it is not an asymptote.

✗ **(D)** The denominator is zero at x = 3, so an asymptote exists.

Question 011 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (B) y = 2/3

STEP-BY-STEP SOLUTION

Equal degrees: the asymptote is the ratio of leading coefficients, y = 2/3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Requires equal leading coefficients.

✗ **(C)** Requires denominator dominance.

✗ **(D)** Inverts the ratio.

Question 012 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

Since $-1 \le \sin(1/x) \le 1$, we get $-x^2 \le x^2 \sin(1/x) \le x^2$. Both bounds $\to 0$, so by the Squeeze Theorem the limit is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the upper bound of the sine factor.

✗ **(C)** The x^2 factor damps the oscillation.

✗ **(D)** This is the lower bound of the sine factor.

Question 013 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (B) 4

STEP-BY-STEP SOLUTION

For $x \neq 2$, $f(x) = x + 2$, so the limit at 2 is 4. Continuity requires $k = 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the numerator value at $x = 2$.

✗ **(C)** The limit exists, so some value of k works.

✗ **(D)** Confuses the x-value with the limit.

Question 014 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (B) 4

STEP-BY-STEP SOLUTION

Left value: $1 + k$. Right value: $2(1) + 3 = 5$. Continuity needs $1 + k = 5$, so $k = 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sets k equal to the right-hand value directly.

✗ **(C)** Miscomputes the right-hand value as 3.

✗ **(D)** Sets $1 + k = 2$, an unjustified equation.

Question 015 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (B) 1

STEP-BY-STEP SOLUTION

For $x > 0$, $\sqrt{(9x^2+x)} = 3x\sqrt{(1 + 1/(9x))}$. Dividing gives $\sqrt{(1 + 1/(9x))}/(1 + 1/(3x)) \to 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the denominator dominates.

✗ **(C)** Ignores the linear denominator.

✗ **(D)** Numerator and denominator grow at the same rate.

Question 016 · Unit 1 · Limits & Continuity

★★ Important

Correct answer: (B) $-\infty$

STEP-BY-STEP SOLUTION

From the left, $x - 2$ is a small negative number, so $1/(\text{small negative}) \to -\infty$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is $1/x$ evaluated at $x = 2$.

✗ **(C)** The denominator shrinks rather than grows.

✗ **(D)** This is the right-hand limit.

Question 017 · Unit 1 · Limits & Continuity

★★ Important

Correct answer: (C) The limit does not exist

STEP-BY-STEP SOLUTION

For $x > 0$ the quotient is 1; for $x < 0$ it is -1. The one-sided limits differ, so the limit does not exist.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is only the right-hand limit.

✗ **(B)** The quotient is never 0.

✗ **(D)** This is only the left-hand limit.

Question 018 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (D) there exists c in (1, 3) with f(c) = 0

STEP-BY-STEP SOLUTION

0 lies between $f(1) = -2$ and $f(3) = 4$, so by the IVT some c in $(1, 3)$ satisfies $f(c) = 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The IVT does not identify the location of the zero.

✗ **(B)** The IVT says nothing about maxima.

✗ **(C)** 6 does not lie between -2 and 4.

Question 019 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (B) −1

STEP-BY-STEP SOLUTION

Let $h = x - \pi$. Then $\sin(\pi + h) = -\sin h$, so the expression becomes $-\sin(h)/h \to -1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Ignores the sign from the shift identity.

✗ **(C)** Uses $\sin(\pi) = 0$ in the numerator only.

✗ **(D)** The substitution $h = x - \pi$ gives a finite limit.

Question 020 · Unit 1 · Limits & Continuity

★★★★★ Very Important

Correct answer: (C) 2

STEP-BY-STEP SOLUTION

$1 - \cos(2x) = 2\sin^2 x$, so the expression is $2(\sin x/x)^2 \to 2(1) = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Ignores the quadratic denominator.

✗ **(B)** Forgets the factor of 2.

✗ **(D)** Squares the coefficient instead of the identity coefficient.

Question 021 · Unit 1 · Limits & Continuity

★★ Important

Correct answer: (D) $-\infty$

STEP-BY-STEP SOLUTION

Numerator degree (3) exceeds denominator degree (2). For $x \to -\infty$ the dominant term x^3 is negative while the denominator is positive, so the quotient $\to -\infty$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Requires the denominator degree to be larger.

✗ **(B)** The sign is negative for $x \to -\infty$.

✗ **(C)** Requires equal degrees.

Question 022 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (D) 0

STEP-BY-STEP SOLUTION

Write $x \ln x = \ln x / (1/x)$, an ∞/∞ form. By L'Hôpital: $(1/x)/(-1/x^2) = -x \to 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** No basis for this value.

✗ **(B)** The product tends to 0, not infinity.

✗ **(C)** $\ln x \to -\infty$, but the factor x drives the product to 0.

Question 023 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (A) a removable discontinuity at x = 4

STEP-BY-STEP SOLUTION

The limit exists and equals 3, but $f(4) = 1 \neq 3$. A limit that exists but differs from the function value is a removable discontinuity.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** No one-sided limit is infinite.

✗ **(C)** Continuity would require $f(4) = 3$.

✗ **(D)** A jump requires differing one-sided limits.

Question 024 · Unit 1 · Limits & Continuity

★★ Important

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

For $x > 0$, $-1/x \leq \sin x / x \leq 1/x$. Both bounds $\to 0$, so by the Squeeze Theorem the limit is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The numerator stays bounded.

✗ **(C)** The oscillation is damped by $1/x$.

✗ **(D)** This is the limit as $x \to 0$, not as $x \to \infty$.

Question 025 · Unit 1 · Limits & Continuity

★★★ Mandatory

Correct answer: (D) 5

STEP-BY-STEP SOLUTION

The values approach 5 from both sides (4.998 from the left, 5.002 from the right), so the best estimate is 5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is a single table entry, not the trend.

✗ **(B)** Both one-sided values approach the same number.

✗ **(C)** This is the x-value, not a function value.

Question 026 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (B) 12

STEP-BY-STEP SOLUTION

This is the definition of $f'(2)$ for $f(x) = x^3$. Since $f'(x) = 3x^2$, $f'(2) = 3(2)^2 = 12$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The difference quotient represents a derivative that exists.

✗ **(C)** Substitutes $h = 0$ into the difference quotient without evaluating the derivative.

✗ **(D)** Computes $2^2 = 4$ instead of $3(2)^2$.

Question 027 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) $f(x) = x^2$

STEP-BY-STEP SOLUTION

Using $f'(a) = \lim_{x \to a} [f(x) - f(a)]/(x - a)$ with $a = 3$ and $f(3) = 9$, this represents the derivative of $f(x) = x^2$ at $x = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $2x$ is the derivative, not the original function.

✗ **(C)** $f(3)$ would be 0, not giving $(x^2-9)/(x-3)$.

✗ **(D)** This is the denominator, not the function being differentiated.

Question 028 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (A) f is continuous at x = c

STEP-BY-STEP SOLUTION

Differentiability implies continuity. If f is differentiable at c, it is guaranteed to be continuous at c.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Existence of f′ does not guarantee existence of f″.

✗ **(C)** The derivative can be any real number, not just 0.

✗ **(D)** A differentiable function need not have an extremum at c.

Question 029 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) f(x) = |x|

STEP-BY-STEP SOLUTION

f(x) = |x| has lim (x→0) |x| = 0 = f(0) (continuous), but its left slope is −1 and right slope is 1, creating a sharp corner (not differentiable).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** 1/x is undefined and discontinuous at x = 0.

✗ **(C)** x^2 is differentiable everywhere with f′(0) = 0.

✗ **(D)** sin x is differentiable everywhere with f′(0) = 1.

Question 030 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) −2x^(−3/2)

STEP-BY-STEP SOLUTION

Power Rule: f′(x) = 4 · (−1/2) x^(−1/2 − 1) = −2 x^(−3/2).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds 1 to the exponent instead of subtracting 1.

✗ **(C)** Multiplies by −2 instead of −1/2.

✗ **(D)** Sign error: forgets the negative sign from the power −1/2.

Question 031 · Unit 2 · Differentiation: Fundamental Properties

★★ Important

Correct answer: (C) $-12/x^5$

STEP-BY-STEP SOLUTION

Rewrite as $3x^{-4}$. Differentiate: $3(-4x^{-5}) = -12x^{-5} = -12/x^5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign from differentiating a negative power.

✗ **(B)** Differentiates the denominator in place.

✗ **(D)** Subtracts −1 incorrectly to get −3 instead of −5.

Question 032 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) $y = x - 4$

STEP-BY-STEP SOLUTION

Point: $(2, 4 - 6) = (2, -2)$. Derivative: $y' = 2x - 3$. At $x = 2$, slope $m = 4 - 3 = 1$. Line: $y - (-2) = 1(x - 2) \Rightarrow y = x - 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses slope −1 instead of +1.

✗ **(C)** Uses slope $m = 4$ instead of 1.

✗ **(D)** Uses the wrong y-intercept from an arithmetic sign slip.

Question 033 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (B) $y - 2 = -4(x - 4)$

STEP-BY-STEP SOLUTION

Point: $(4, 2)$. $y' = 1/(2\sqrt{x})$, so at $x = 4$, tangent slope is $1/4$. The normal line slope is the negative reciprocal: $m = -4$. Equation: $y - 2 = -4(x - 4)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Takes the negative but forgets the reciprocal.

✗ **(C)** Takes the reciprocal but forgets the negative sign.

✗ **(D)** This is the tangent line equation, not the normal line.

Question 034 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (A) x = 0 and x = 2

STEP-BY-STEP SOLUTION

Horizontal tangent means $y' = 0$. $y' = 3x^2 - 6x = 3x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses the critical point x = 2.

✗ **(C)** Misses the critical point x = 0.

✗ **(D)** Roots of $3x^2 - 3$, not $3x^2 - 6x$.

Question 035 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (A) A vertical tangent line

STEP-BY-STEP SOLUTION

y′ = (1/3)x^(−2/3) = 1/(3 x^(2/3)). As x → 0, y′ → +∞ from both sides. The function is continuous, so it has a vertical tangent line at x = 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The function is continuous at x = 0.

✗ **(C)** The one-sided derivative limits both approach +∞; there is no corner.

✗ **(D)** y′ approaches infinity, not zero.

Question 036 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (B) $x^2 e^x + 2x e^x$

STEP-BY-STEP SOLUTION

Product Rule: $(x^2)' e^x + x^2 (e^x)' = 2x e^x + x^2 e^x = e^x(x^2 + 2x)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates only the x^2 term.

✗ **(C)** Omits the derivative of the first factor entirely.

✗ **(D)** Differentiates x^2 as x instead of 2x.

Question 037 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (D) −1

STEP-BY-STEP SOLUTION

Product Rule: $f'(x) = 2\sin x + (2x - 1)\cos x$. At $x = 0$: $f'(0) = 2(0) + (-1)(1) = -1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only the derivative of $2x - 1$.

✗ **(B)** Assumes the whole derivative vanishes because $\sin(0) = 0$.

✗ **(C)** Sign error on the $(2x - 1)$ factor at $x = 0$.

Question 038 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (D) $1 / (x + 1)^2$

STEP-BY-STEP SOLUTION

Quotient Rule: $[(1)(x + 1) - x(1)] / (x + 1)^2 = (x + 1 - x) / (x + 1)^2 = 1 / (x + 1)^2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds instead of subtracting in the quotient rule numerator.

✗ **(B)** Sign error in the numerator subtraction: $x - (x + 1) = -1$.

✗ **(C)** Differentiates numerator and denominator separately: $1/1 = 1$.

Question 039 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) 0

STEP-BY-STEP SOLUTION

Quotient Rule: $g'(x) = (e^x \cdot x - e^x \cdot 1) / x^2 = e^x(x - 1) / x^2$. At $x = 1$: $g'(1) = e(0)/1 = 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only the numerator's e^x term.

✗ **(C)** Subtracts 1 from e incorrectly.

✗ **(D)** Calculates $e(1 + 1)/1^2$ by adding instead of subtracting.

Question 040 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (D) (3 + 2√3)/2

STEP-BY-STEP SOLUTION

$f'(x) = 3\cos x + 2\sin x$. At $x = \pi/3$: $3\cos(\pi/3) + 2\sin(\pi/3) = 3(1/2) + 2(\sqrt{3}/2) = (3 + 2\sqrt{3})/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Subtracts the two terms instead of adding.

✗ **(B)** Swaps the values of $\cos(\pi/3)$ and $\sin(\pi/3)$.

✗ **(C)** Sign error: differentiates $\cos x$ as $+\sin x$, giving $-2\sin x$.

Question 041 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (D) $2^x \ln 2$

STEP-BY-STEP SOLUTION

General exponential rule: $d/dx(a^x) = a^x \ln a$. For $a = 2$, this is $2^x \ln 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This rule applies only to e^x where $\ln e = 1$.

✗ **(B)** Applies the power rule to an exponential function.

✗ **(C)** Divides by $\ln 2$ instead of multiplying.

Question 042 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (C) Yes, because f is continuous and left/right derivatives both equal 2

STEP-BY-STEP SOLUTION

Continuity: $\lim_{x\to1^-} x^2 = 1 = \lim_{x\to1^+} (2x - 1) = f(1) = 1$ (continuous). Derivatives: $d/dx(x^2)\big|_1 = 2(1) = 2$; $d/dx(2x - 1)\big|_1 = 2$. Left and right derivatives match, so $f'(1) = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Left derivative is $2x$ at 1, which equals 2, not 1.

✗ **(B)** f is continuous: $1^2 = 2(1) - 1 = 1$.

✗ **(D)** Differentiability can hold at piecewise boundaries if limits and derivatives match.

Question 043 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (C) 3.5

STEP-BY-STEP SOLUTION

Symmetric difference quotient: [f(5) − f(1)] / (5 − 1) = (18 − 4) / 4 = 14 / 4 = 3.5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses [f(3) − f(1)] / (3 − 1) = (10 − 4)/2 = 3 (forward/backward step, not symmetric over [1,5]).

✗ **(B)** Uses [f(5) − f(3)] / 2 = 8/2 = 4.

✗ **(D)** Computes (18 − 4)/2, forgetting the denominator is 4.

Question 044 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (A) $-1/x^2$

STEP-BY-STEP SOLUTION

$f'(x) = x^{-1}$. Differentiating again: $f''(x) = -1 \cdot x^{-2} = -1/x^2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Drops the negative sign from the power rule.

✗ **(C)** This is the third derivative $f'''(x)$.

✗ **(D)** This is the first derivative $f'(x)$.

Question 045 · Unit 2 · Differentiation: Fundamental Properties

★★ Important

Correct answer: (A) 24

STEP-BY-STEP SOLUTION

$f' = 4x^3$, $f'' = 12x^2$, $f''' = 24x$, $f^{(4)} = 24$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is related to the first derivative.

✗ **(C)** The 5th derivative is 0; the 4th is the constant 24.

✗ **(D)** This is the third derivative $f'''(x)$.

Question 046 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (A) $3x^2 - 4x + 1$

STEP-BY-STEP SOLUTION

Expand first: $(x^2 + 1)(x - 2) = x^3 - 2x^2 + x - 2$. Differentiate: $3x^2 - 4x + 1$. (Product rule yields the identical result.)

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign errors on the terms.

✗ **(C)** Differentiates only the x^2 term.

✗ **(D)** Differentiates only the first factor.

Question 047 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) −13

STEP-BY-STEP SOLUTION

$h'(x) = 3f'(x) - 2g'(x)$. At $x = 2$: $h'(2) = 3(-1) - 2(5) = -3 - 10 = -13$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $3(1) + 2(2)$ using wrong values.

✗ **(C)** Computes $3 + 10 = 13$.

✗ **(D)** Sign error on the first term: computes $+3 - 10 = -7$.

Question 048 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (B) 11

STEP-BY-STEP SOLUTION

Product Rule: $P'(2) = f'(2)g(2) + f(2)g'(2) = (-1)(4) + (3)(5) = -4 + 15 = 11$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds incorrectly: $4 + 15 = 19$.

✗ **(C)** Multiplies function values and derivatives incorrectly.

✗ **(D)** Multiplies derivatives directly: $(-1)(5) = -5$.

Question 049 · Unit 2 · Differentiation: Fundamental Properties

★★★★★ Very Important

Correct answer: (B) −19/16

STEP-BY-STEP SOLUTION

Quotient Rule: $Q'(2) = [f'(2)g(2) - f(2)g'(2)] / [g(2)]^2 = [(-1)(4) - (3)(5)] / 4^2 = (-4 - 15)/16 = -19/16$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error giving positive 11/16.

✗ **(C)** Drops the negative sign from −19.

✗ **(D)** Reverses the sign in the subtraction: −4 + 15 = 11/16.

Question 050 · Unit 2 · Differentiation: Fundamental Properties

★★★ Mandatory

Correct answer: (B) (2, 8) and (−2, −8)

STEP-BY-STEP SOLUTION

$y' = 3x^2 = 12 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$. At $x = 2$, $y = 8$; at $x = -2$, $y = -8$. Both points have tangent slope 12.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Solves $3x = 12$ instead of $3x^2 = 12$.

✗ **(C)** Misses the positive solution $x = 2$.

✗ **(D)** Misses the negative solution $x = -2$.

Question 051 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (A) $3x^2 \cos(x^3)$

STEP-BY-STEP SOLUTION

Chain Rule: outer derivative $\cos(x^3)$ times inner derivative $3x^2$ gives $3x^2 \cos(x^3)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Derivative of sine is positive cosine, not negative.

✗ **(C)** Forgets to multiply by the inner derivative $3x^2$.

✗ **(D)** Mixes the power rule into the sine argument.

Question 052 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (B) −2e^(−2x)

STEP-BY-STEP SOLUTION

Derivative of e^u is e^u · u′. Here u = −2x, so u′ = −2. Result: −2e^(−2x).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the inner derivative −2.

✗ **(C)** Sign error on the inner derivative.

✗ **(D)** Multiplies by x unnecessarily.

Question 053 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $2x / (x^2 + 4)$

STEP-BY-STEP SOLUTION

Derivative of ln u is u′/u. Here $u = x^2 + 4$, so $u' = 2x$. Result: $2x/(x^2 + 4)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates x^2 as 2 instead of 2x.

✗ **(C)** Multiplies by the logarithm instead of dividing.

✗ **(D)** Omits the inner derivative 2x in the numerator.

Question 054 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (B) 15

STEP-BY-STEP SOLUTION

$f'(x) = 5(3x − 2)^4 · 3 = 15(3x − 2)^4$. At x = 1: $15(3 − 2)^4 = 15(1) = 15$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error.

✗ **(C)** Forgets the inner derivative 3: computes 5(1) = 5.

✗ **(D)** Multiplies 15 by 3 again.

Question 055 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $-x / \sqrt{(1 - x^2)}$

STEP-BY-STEP SOLUTION

Rewrite as $(1 - x^2)^{(1/2)}$. $f'(x) = (1/2)(1 - x^2)^{(-1/2)} \cdot (-2x) = -x / \sqrt{(1 - x^2)}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the inner derivative −2x.

✗ **(C)** Forgets the 1/2 factor from the power rule.

✗ **(D)** Drops the negative sign from −2x.

Question 056 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (C) $-6 \cos^2(2x) \sin(2x)$

STEP-BY-STEP SOLUTION

Chain Rule with 3 layers: $3[\cos(2x)]^2 \cdot [-\sin(2x)] \cdot 2 = -6 \cos^2(2x) \sin(2x)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the innermost derivative of 2x (misses the factor 2).

✗ **(B)** Reduces the power on cosine incorrectly.

✗ **(D)** Sign error: derivative of cosine is negative sine.

Question 057 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (A) 20

STEP-BY-STEP SOLUTION

$h'(x) = f'(g(x)) \cdot g'(x)$. At $x = 2$: $h'(2) = f'(g(2)) \cdot g'(2) = f'(2) \cdot 5 = 4 \cdot 5 = 20$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates f′ at $g'(2) = 5$ incorrectly.

✗ **(C)** Forgets to multiply by $g'(2) = 5$.

✗ **(D)** Uses $f'(3)$ instead of $f'(2)$.

Question 058 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (D) −x/y

STEP-BY-STEP SOLUTION

Differentiate implicitly: $2x + 2y(dy/dx) = 0 \Rightarrow 2y(dy/dx) = -2x \Rightarrow dy/dx = -x/y$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the fraction.

✗ **(B)** Inverts the fraction with wrong sign.

✗ **(C)** Drops the negative sign.

Question 059 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (A) −1

STEP-BY-STEP SOLUTION

Product rule: $(1)y + x(dy/dx) = 0 \Rightarrow dy/dx = -y/x$. At (2, 2): $dy/dx = -2/2 = -1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Drops the negative sign.

✗ **(C)** Arithmetic error.

✗ **(D)** Uses −y instead of −y/x.

Question 060 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (A) −5

STEP-BY-STEP SOLUTION

Implicit diff: $2x + [2y + 2x\,y'] - 2y\,y' = 0$.

At (3, 2): $2(3) + 2(2) + 2(3)y' - 2(2)y' = 0 \Rightarrow 6 + 4 + 6y' - 4y' = 0 \Rightarrow 10 + 2y' = 0 \Rightarrow y' = -5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error: $2y' = -10 \Rightarrow y' = -5$, not +5.

✗ **(C)** Forgets the $2x\,y'$ term.

✗ **(D)** Arithmetic error in combining like terms.

Question 061 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $-1/y^3$

STEP-BY-STEP SOLUTION

$dy/dx = -x/y$. $d^2y/dx^2 = -[\,(1)y - x(dy/dx)\,] / y^2 = -[\,y - x(-x/y)\,] / y^2 = -[\,y + x^2/y\,] / y^2 = -(y^2 + x^2) / y^3 = -1/y^3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the y in the denominator of dy/dx.

✗ **(C)** Does not substitute dy/dx back in.

✗ **(D)** Drops the negative sign.

Question 062 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (C) 2x – y = 0

STEP-BY-STEP SOLUTION

$2x - (y + x\,y') + 2y\,y' = 0 \Rightarrow y'(2y - x) = y - 2x \Rightarrow y' = (2x - y)/(x - 2y)$ wait: $2x - y - xy' + 2yy' = 0 \Rightarrow y'(2y - x) = y - 2x \Rightarrow y' = -(2x - y)/(2y - x) = -1$? Recheck: $2x - y + y'(2y - x) = 0 \Rightarrow y' = (y - 2x)/(2y - x)$. Horizontal tangent requires numerator = 0: $y - 2x = 0 \Rightarrow y = 2x$ (or $2x - y = 0$).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $x = 0$ gives $y^2 = 3$, where slope is not 0.

✗ **(B)** $y = 0$ gives no points on the curve.

✗ **(D)** Both options C and D describe $2x = y$, but C is the clean algebraic condition.

Question 063 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (D) 1/2

STEP-BY-STEP SOLUTION

Inverse function rule: $(f^{-1})'(b) = 1 / f'(f^{-1}(b))$. Since $f(3) = 5$, $f^{-1}(5) = 3$. Therefore $(f^{-1})'(5) = 1/f'(3) = 1/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the input value 5 in the denominator: 1/5.

✗ **(B)** Forgets to take the reciprocal: reports $f'(3) = 2$.

✗ **(C)** Reports the function value.

Question 064 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) 1/4

STEP-BY-STEP SOLUTION

Need x such that $f(x) = 0$: $x^3 + x - 2 = 0 \Rightarrow x = 1$ (since $1 + 1 - 2 = 0$). So $f^{-1}(0) = 1$. $f'(x) = 3x^2 + 1$, so $f'(1) = 3(1)^2 + 1 = 4$. Then $(f^{-1})'(0) = 1/f'(1) = 1/4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the reciprocal: reports $f'(1) = 4$.

✗ **(C)** The derivative of an inverse function is not zero.

✗ **(D)** Uses $f'(0) = 1/1$... arithmetic error.

Question 065 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (C) 1/3

STEP-BY-STEP SOLUTION

$g'(2) = 1 / f'(g(2))$. From the table, $f(1) = 2$, so $g(2) = 1$. Therefore $g'(2) = 1/f'(1) = 1/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports $f'(2)$ without the reciprocal.

✗ **(B)** Uses $f(2) = 4$ and looks at $f'(2) = 5$, giving 1/5.

✗ **(D)** Reports $f'(1)$ without the reciprocal.

Question 066 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $3 / (1 + 9x^2)$

STEP-BY-STEP SOLUTION

Derivative of $\arctan u$ is $u'/(1 + u^2)$. Here $u = 3x$, so $u^2 = 9x^2$ and $u' = 3$. Result: $3/(1 + 9x^2)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses arctan with arcsin (uses radical).

✗ **(C)** Forgets the chain rule factor $u' = 3$.

✗ **(D)** Squares $3x$ as $3x^2$ instead of $9x^2$.

Question 067 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $1 / \sqrt{(4 - x^2)}$

STEP-BY-STEP SOLUTION

Derivative of arcsin u is $u'/\sqrt{(1 - u^2)}$. Here $u = x/2$, $u' = 1/2$. $(1/2)/\sqrt{(1 - x^2/4)} = (1/2) / [\sqrt{(4 - x^2)/2}] = 1/\sqrt{(4 - x^2)}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/2 factor in the numerator.

✗ **(C)** Valid unsimplified form, but A is standard simplified AP form.

✗ **(D)** Does not substitute x/2 into the radical.

Question 068 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (D) −1

STEP-BY-STEP SOLUTION

$d/dx [\arccos x] = -1/\sqrt{(1 - x^2)}$. At $x = 0$: $-1/\sqrt{(1 - 0)} = -1/1 = -1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Derivative of arcsin(0) is +1; arccos has a negative sign.

✗ **(B)** The derivative is defined at x = 0.

✗ **(C)** $\arccos(0) = \pi/2$, not 0; the derivative is −1.

Question 069 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (B) $\arctan x + x / (1 + x^2)$

STEP-BY-STEP SOLUTION

Product Rule: $(1) \arctan x + x \cdot [1/(1 + x^2)] = \arctan x + x/(1 + x^2)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Replaces the x with 1 in the second term.

✗ **(C)** Differentiates both factors directly without the product rule.

✗ **(D)** Differentiates only the arctan factor.

Question 070 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (B) $e^x / \sqrt{(1 - e^{(2x)})}$

STEP-BY-STEP SOLUTION

Chain Rule: $u = e^x$, $u' = e^x$, $u^2 = (e^x)^2 = e^{(2x)}$. Result: $e^x/\sqrt{(1 - e^{(2x)})}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Squares e^x as e^x instead of $e^{(2x)}$.

✗ **(C)** Confuses arcsin with arctan.

✗ **(D)** Forgets the chain rule factor e^x in the numerator.

Question 071 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $-\tan x$

STEP-BY-STEP SOLUTION

Derivative of ln u is u′/u. Here $u = \cos x$, $u' = -\sin x$. Result: $-\sin x / \cos x = -\tan x$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Sign error: derivative of cosine is $-\sin x$, giving $-\tan x$.

✗ **(C)** Inverts the ratio: cos/sin is cot, but this is sin/cos.

✗ **(D)** Combines sign and reciprocal errors.

Question 072 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (A) 1 / cos y

STEP-BY-STEP SOLUTION

Differentiate implicitly: $\cos y \cdot (dy/dx) = 1 \Rightarrow dy/dx = 1/\cos y$ (which equals $\sec y = 1/\sqrt{(1 - x^2)}$).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Sign error: derivative of sin y is +cos y.

✗ **(C)** Inverts the wrong trig function.

✗ **(D)** Does not solve for dy/dx: leaves it as cos y.

Question 073 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (B) $1 / (3e^3)$

STEP-BY-STEP SOLUTION

Need x such that f(x) = e^3: e^(3x) = e^3 ⇒ 3x = 3 ⇒ x = 1. So $f^{-1}(e^3)$ = 1. f′(x) = 3e^(3x) ⇒ f′(1) = $3e^3$. Then $(f^{-1})'(e^3)$ = 1/f′(1) = $1/(3e^3)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the reciprocal: reports f′(1) = $3e^3$.

✗ **(C)** Computes the fraction upside down.

✗ **(D)** Forgets the factor 3 from the chain rule.

Question 074 · Unit 3 · Composite, Implicit & Inverse Functions

★★★ Mandatory

Correct answer: (D) 4/3

STEP-BY-STEP SOLUTION

Tangent slope: 2x + 2y y′ = 0 ⇒ y′ = −x/y = −3/4. Normal line slope is the negative reciprocal: m = −1/(−3/4) = 4/3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the tangent slope, not the normal slope.

✗ **(B)** Negates without reciprocating.

✗ **(C)** Reciprocates without changing sign.

Question 075 · Unit 3 · Composite, Implicit & Inverse Functions

★★★★★ Very Important

Correct answer: (C) $-1 / (1 + x^2)$

STEP-BY-STEP SOLUTION

u = 1/x = x^{-1}, u′ = $-1/x^2$. u′/(1 + u^2) = $(-1/x^2) / (1 + 1/x^2) = (-1/x^2) / [(x^2 + 1)/x^2] = -1/(x^2 + 1)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the negative sign from the derivative of 1/x.

✗ **(B)** Sign error.

✗ **(D)** Does not cancel the x^2 factors.

Question 076 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) At t = 5 min, water is leaking out at 3 gallons per minute

STEP-BY-STEP SOLUTION

$W'(5)$ is the instantaneous rate of change of water volume at $t = 5$. Negative means decreasing. Rate: 3 gal/min outflow.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This describes an average rate, not instantaneous.

✗ **(B)** Confuses $W(5)$ with $W'(5)$.

✗ **(D)** This describes total accumulation over $[0, 5]$, not an instantaneous rate.

Question 077 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (A) t = 1 and t = 3

STEP-BY-STEP SOLUTION

At rest means $v(t) = s'(t) = 0$. $v(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \Rightarrow t = 1$ and $t = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Misses the root $t = 3$.

✗ **(C)** Misses the root $t = 1$.

✗ **(D)** $s(0) = 0$, but $v(0) = 9 \neq 0$ (particle is moving at $t = 0$).

Question 078 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (D) 1 < t < 3

STEP-BY-STEP SOLUTION

Moving left means $v(t) < 0$. Since $v(t) = 3(t - 1)(t - 3)$, testing points shows $v < 0$ on the interval $(1, 3)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Includes $t > 3$ where motion is to the right.

✗ **(B)** $v > 0$ for $t > 3$, so particle moves right.

✗ **(C)** $v > 0$ on $(0, 1)$, so particle moves right.

Question 079 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) The particle is moving left and speeding up

STEP-BY-STEP SOLUTION

$v(t) < 0$ means moving in the negative direction (left). Since v and a have the same sign (both negative), speed $|v|$ is increasing (speeding up).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $v < 0$ means moving left, not right.

✗ **(B)** Slowing down requires v and a to have opposite signs.

✗ **(D)** v is nonzero, so not at rest.

Question 080 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) Moving left and slowing down

STEP-BY-STEP SOLUTION

$v(t) < 0$ means moving left. Since v and a have opposite signs (negative velocity, positive acceleration), the particle is slowing down.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $v < 0$ means moving left.

✗ **(C)** Opposite signs mean slowing down, not speeding up.

✗ **(D)** $v < 0$ means moving left.

Question 081 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

$v(t) = 4t^3 - 12t^2$. $a(t) = 12t^2 - 24t$. At $t = 2$: $a(2) = 12(4) - 24(2) = 48 - 48 = 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates 12t only.

✗ **(C)** This is the velocity $v(2) = 4(8) - 12(4) = 32 - 48 = -16$.

✗ **(D)** Arithmetic error.

Question 082 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) 24π cm²/s

STEP-BY-STEP SOLUTION

$A = \pi r^2$. Differentiate with respect to t: $dA/dt = 2\pi r\,(dr/dt)$. When $r = 4$ and $dr/dt = 3$: $dA/dt = 2\pi(4)(3) = 24\pi$ cm²/s.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies by r^2: $\pi(16)(3) = 48\pi$.

✗ **(C)** Forgets the factor 2: computes $\pi r(dr/dt) = 12\pi$.

✗ **(D)** Forgets dr/dt: computes $2\pi(4) = 8\pi$ or computes $\pi r^2 = 16\pi$.

Question 083 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) 2.25 ft/s

STEP-BY-STEP SOLUTION

$x^2 + y^2 = 100$. When $x = 6$, $y = \sqrt{(100 - 36)} = 8$. Differentiate: $2x(dx/dt) + 2y(dy/dt) = 0 \Rightarrow x(dx/dt) + y(dy/dt) = 0$. $6(3) + 8(dy/dt) = 0 \Rightarrow 18 + 8(dy/dt) = 0 \Rightarrow dy/dt = -18/8 = -2.25$ ft/s. The top slides down at 2.25 ft/s.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the ratio: $8/6 \cdot 3 = 4$.

✗ **(B)** Arithmetic error: $18/12 = 1.5$.

✗ **(C)** Assumes the rates are equal.

Question 084 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) 1/π cm/s

STEP-BY-STEP SOLUTION

$V = (4/3)\pi r^3 \Rightarrow dV/dt = 4\pi r^2\,(dr/dt)$. Given $dV/dt = 100$ and $r = 5$: $100 = 4\pi(25)(dr/dt) = 100\pi\,(dr/dt) \Rightarrow dr/dt = 100/(100\pi) = 1/\pi$ cm/s.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets 4: $100/4 = 25$.

✗ **(C)** Forgets the 25: $100/(4\pi) = 25/\pi$.

✗ **(D)** Forgets to cancel the 100 fully: $2/\pi$.

Question 085 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (A) 3/(2π) ft/min

STEP-BY-STEP SOLUTION

Similar triangles: $r/h = 4/8 = 1/2 \Rightarrow r = h/2$. $V = (1/3)\pi r^2 h = (1/3)\pi(h/2)^2 h = (\pi/12)h^3$.
$dV/dt = (\pi/4)h^2\,(dh/dt)$. At $h = 4$ and $dV/dt = 6$: $6 = (\pi/4)(16)(dh/dt) = 4\pi(dh/dt) \Rightarrow dh/dt = 6/(4\pi) = 3/(2\pi)$ ft/min.
Correction: $6/(4\pi) = 3/(2\pi)$. Option A is $3/(2\pi)$. Answer A.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the 4 in the denominator: $6/\pi$.
✗ **(C)** Arithmetic slip.
✗ **(D)** Unsimplified form of $3/(2\pi)$.

Question 086 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (B) 50 mph

STEP-BY-STEP SOLUTION

$z^2 = x^2 + y^2$. After 2 hours: $x = 80$ mi, $y = 60$ mi, $z = \sqrt{(6400 + 3600)} = 100$ mi.
$2z(dz/dt) = 2x(dx/dt) + 2y(dy/dt) \Rightarrow 100(dz/dt) = 80(40) + 60(30) = 3200 + 1800 = 5000 \Rightarrow dz/dt = 50$ mph.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the distance $z = 100$ miles, not the rate dz/dt.
✗ **(C)** Averages the two speeds.
✗ **(D)** Adds the speeds: $30 + 40 = 70$.

Question 087 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) 2 + (1/4)(x − 4)

STEP-BY-STEP SOLUTION

$f(4) = 2$. $f'(x) = 1/(2\sqrt{x}) \Rightarrow f'(4) = 1/4$. $L(x) = f(4) + f'(4)(x - 4) = 2 + (1/4)(x - 4)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $f'(4) = 1/\sqrt{4} = 1/2$, forgetting the 1/2 factor.
✗ **(B)** Forgets to distribute the $(x - 4)$.
✗ **(C)** Uses slope 4 instead of 1/4.

Question 088 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (C) 2.02

STEP-BY-STEP SOLUTION

L(4.08) = 2 + (1/4)(4.08 − 4) = 2 + (1/4)(0.08) = 2 + 0.02 = 2.02.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses slope 1/2: 2 + 0.5(0.08) = 2.04.

✗ **(B)** Arithmetic error.

✗ **(D)** Adds the 0.08 directly to 2.

Question 089 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) An overestimate of the true value

STEP-BY-STEP SOLUTION

When a curve is concave down, tangent lines lie strictly ABOVE the curve. Therefore, any tangent line approximation is an OVERESTIMATE.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Concavity alone determines over/underestimate.

✗ **(C)** Underestimate corresponds to concave up curves (tangent line lies below).

✗ **(D)** Tangent approximations are rarely exact for non-linear functions.

Question 090 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) 3/2

STEP-BY-STEP SOLUTION

Direct substitution gives 0/0 (indeterminate). By L'Hôpital's Rule: lim (x→0) [3e^(3x)] / [2 cos(2x)] = 3(1) / 2(1) = 3/2.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Substitutes 0 into numerator without applying L'Hôpital.

✗ **(B)** Inverts the ratio of coefficients.

✗ **(C)** Assumes trig and exp balance to 1.

Question 091 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) 8

STEP-BY-STEP SOLUTION

Direct sub: 0/0. L'Hôpital 1: lim 4 sin(4x) / (2x). Still 0/0. L'Hôpital 2: lim 16 cos(4x) / 2 = 16(1)/2 = 8.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Stops after dividing by 2 once: computes 4/2 = 2.

✗ **(C)** Applies L'Hôpital once and ignores the second derivative: 4.

✗ **(D)** Uses 16 without dividing by 2.

Question 092 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (A) 0

STEP-BY-STEP SOLUTION

Form ∞/∞. By L'Hôpital: $\lim_{x\to\infty} (1/x) / [1/(2\sqrt{x})] = \lim_{x\to\infty} 2\sqrt{x} / x = \lim_{x\to\infty} 2/\sqrt{x} = 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The square root grows faster than ln x, driving quotient to 0.

✗ **(C)** Keeps the 1/2 coefficient without taking limit.

✗ **(D)** Assumes comparable growth.

Question 093 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) 0

STEP-BY-STEP SOLUTION

Form $0 \cdot (-\infty)$. Rewrite as quotient: $\lim (\ln x) / (1/x)$ (form $-\infty/\infty$). L'Hôpital: $\lim (1/x) / (-1/x^2) = \lim (-x) = 0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Product approaches 0, not $+\infty$.

✗ **(C)** $\ln x \to -\infty$, but the linear factor x drives the product to 0.

✗ **(D)** No basis.

Question 094 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (C) tan θ = x/20

STEP-BY-STEP SOLUTION

Right triangle: opposite side is the height or distance x, adjacent side is the fixed distance 20 ft. tan θ = opposite / adjacent = x/20.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Swaps opposite and adjacent.

✗ **(B)** Hypotenuse varies, so sine is not convenient.

✗ **(D)** Cosine uses hypotenuse.

Question 095 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (A) dV/dt = s/4 · dA/dt

STEP-BY-STEP SOLUTION

$V = s^3 \Rightarrow dV/dt = 3s^2\,(ds/dt)$. $A = 6s^2 \Rightarrow dA/dt = 12s\,(ds/dt) \Rightarrow ds/dt = (dA/dt)/(12s)$.
Substitute: $dV/dt = 3s^2 \cdot [(dA/dt)/(12s)] = (3s/12) \cdot dA/dt = (s/4) \cdot dA/dt$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses $3s^2$ without replacing ds/dt.

✗ **(C)** Uses 6s instead of 12s in denominator.

✗ **(D)** This is dA/dt in terms of ds/dt, not relating dV to dA.

Question 096 · Unit 4 · Contextual Applications of Differentiation

★★★★★ Very Important

Correct answer: (A) 34/3

STEP-BY-STEP SOLUTION

$v(t) = (t - 3)(t + 1) = 0$ at $t = 3$. On $[0, 3]$, $v \le 0$. On $[3, 4]$, $v \ge 0$.
Distance on $[0, 3]$: $\left|\int_0^3 (t^2 - 2t - 3)\,dt\right| = \left|[t^3/3 - t^2 - 3t]_0^3\right| = |9 - 9 - 9| = |-9| = 9$.
Distance on $[3, 4]$: $\int_3^4 (t^2 - 2t - 3)\,dt = [64/3 - 16 - 12] - (-9) = (64/3 - 28) + 9 = -20/3 + 9 = 7/3$.
Total distance $= 9 + 7/3 = 34/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is net displacement over $[0, 4]$: $7/3 - 9 = -20/3$.

✗ **(C)** Arithmetic error in combining fractions.

✗ **(D)** Uses only a partial interval.

Question 097 · Unit 4 · Contextual Applications of Differentiation

★★ Important

Correct answer: (D) 1.05

STEP-BY-STEP SOLUTION

$f(0) = 1$, $f'(x) = e^x \Rightarrow f'(0) = 1$. $L(x) = 1 + 1(x - 0) = 1 + x$. $L(0.05) = 1 + 0.05 = 1.05$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds 0.5 instead of 0.05.

✗ **(B)** Subtracts 0.05 instead of adding.

✗ **(C)** Adds 0.005 instead of 0.05.

Question 098 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (A) $\lim_{x\to 0} (\cos x)/x$

STEP-BY-STEP SOLUTION

At $x = 0$, $(\cos 0)/0 = 1/0$. This is NOT an indeterminate form (0/0 or ∞/∞). Direct substitution shows the limit does not exist (one-sided limits are $\pm\infty$). L'Hôpital's Rule DOES NOT APPLY.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is 0/0, L'Hôpital applies.

✗ **(C)** This is ∞/∞, L'Hôpital applies.

✗ **(D)** This is 0/0, L'Hôpital applies.

Question 099 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (C) 20 cm²/s

STEP-BY-STEP SOLUTION

Perimeter $P = 4s = 20 \Rightarrow s = 5$ cm. Area $A = s^2 \Rightarrow dA/dt = 2s\,(ds/dt) = 2(5)(2) = 20$ cm²/s.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes $s^2 = 25$ and multiplies by 2.

✗ **(B)** Uses perimeter 20 in the formula: $2(20) = 40$.

✗ **(D)** Forgets to multiply by $ds/dt = 2$.

Question 100 · Unit 4 · Contextual Applications of Differentiation

★★★ Mandatory

Correct answer: (C) On the open interval (2, 4)

STEP-BY-STEP SOLUTION

Acceleration $a(t) = v'(t)$ is the slope of the velocity graph. On the horizontal segment from $t = 2$ to $t = 4$, the slope is 0, so $a(t) = 0$ on $(2, 4)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Slope is clearly 0 on the horizontal section.

✗ **(B)** Velocity is zero at 0 and 6, not acceleration.

✗ **(D)** The entire interval (2, 4) has zero slope, not just the single point 3.

Question 101 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (A) 2

STEP-BY-STEP SOLUTION

$f(0)=f(4)=0$, so the secant slope is 0. $f'(x)=2x-4$. Set $2c-4=0$ to get $c=2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** The MVT point must be in $(0,4)$, so 0 is not allowed.

✗ **(C)** At $c=1$, $f'(1)=-2$, not 0.

✗ **(D)** The MVT point must be interior, not the endpoint 4.

Question 102 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (D) $2/\sqrt{3}$

STEP-BY-STEP SOLUTION

The secant slope is $(8-0)/(2-0)=4$. Since $f'(x)=3x^2$, set $3c^2=4$. The interior solution is $c=2/\sqrt{3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This does not solve $3c^2=4$.

✗ **(B)** Substitution gives $3c^2=1$, not 4.

✗ **(C)** This is not the solution of $3c^2=4$.

Question 103 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (D) π/2

STEP-BY-STEP SOLUTION

The endpoint values are both 0, so the secant slope is 0. $f'(x)=\cos x$; $\cos c=0$ gives $c=\pi/2$ in $(0,\pi)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $\cos(3\pi/4)$ is not 0.

✗ **(B)** $\cos(\pi/4)$ is not 0.

✗ **(C)** π is an endpoint, not an interior point.

Question 104 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (A) two solutions $c=\pm 1/\sqrt{3}$

STEP-BY-STEP SOLUTION

The secant slope is $(1-(-1))/2=1$. Since $f'(x)=3x^2$, solve $3c^2=1$, giving $c=\pm 1/\sqrt{3}$; both lie in $(-1,1)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Both values satisfy the MVT equation.

✗ **(C)** The derivative at 0 is 0, not 1.

✗ **(D)** The negative solution also satisfies the equation.

Question 105 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (B) f is strictly decreasing on the real line

STEP-BY-STEP SOLUTION

$f'\le 0$ and is negative except at $x=-2$. The derivative does not change sign there, so f remains strictly decreasing and has no local extremum at −2.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** There is no negative-to-positive sign change.

✗ **(C)** There is no positive-to-negative sign change.

✗ **(D)** The derivative is not positive on any interval.

Question 106 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) f has a local minimum at x=−1 and no extremum at x=2

STEP-BY-STEP SOLUTION

The squared factor is nonnegative, so the sign of f′ is determined by x+1. It changes negative to positive at −1, giving a local minimum. It does not change sign at 2, so there is no extremum there.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** There is no sign change at x=2.

✗ **(B)** The sign changes − to + at −1, not + to −; x=2 has no sign change.

✗ **(D)** There is no sign change at x=2.

Question 107 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) f has a local maximum at c

STEP-BY-STEP SOLUTION

Positive f′ means increasing; negative f′ means decreasing. The transition is increasing to decreasing, so c is a local maximum.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** An inflection point requires a concavity change.

✗ **(B)** The function changes direction, so it is not constant.

✗ **(D)** Negative-to-positive gives a local minimum.

Question 108 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (A) increasing on that interval

STEP-BY-STEP SOLUTION

A positive first derivative means the function is increasing.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** A constant function has derivative zero.

✗ **(C)** Concavity is determined by f″ or the behavior of f′.

✗ **(D)** A negative derivative means decreasing.

Question 109 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (B) continues decreasing through a stationary point

STEP-BY-STEP SOLUTION

f′ is nonpositive and negative on both sides of 3. The derivative is zero at 3 but does not change sign, so the function continues decreasing.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The derivative is negative on both sides.

✗ **(C)** There is no + to − sign change.

✗ **(D)** There is no − to + sign change.

Question 110 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (B) −1

STEP-BY-STEP SOLUTION

$f'=2x-2=0$ at $x=1$. Candidate values: $f(0)=0$, $f(1)=-1$, $f(3)=3$. The minimum is -1.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the value at the right endpoint.

✗ **(C)** This is the value at $x=0$, not the minimum.

✗ **(D)** This is not the minimum candidate value.

Question 111 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (A) attain both an absolute maximum and an absolute minimum

STEP-BY-STEP SOLUTION

The EVT guarantees an attained absolute maximum and minimum on a closed, bounded interval.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** There may be none or several local maxima.

✗ **(C)** The extrema could occur only at the endpoints.

✗ **(D)** Continuity does not imply differentiability.

Question 112 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (D) 5

STEP-BY-STEP SOLUTION

$f'=2x-4=0$ at $x=2$. Candidate values: $f(0)=0$, $f(2)=-4$, $f(5)=5$. The maximum is 5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the minimum.

✗ **(B)** This is the left endpoint value.

✗ **(C)** Not a candidate value.

Question 113 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (B) 0 at x=0

STEP-BY-STEP SOLUTION

$x^{(2/3)}=|x|^{(2/3)}\geq 0$, equals 0 at $x=0$, and equals 1 at both endpoints.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The function is nonnegative.

✗ **(C)** These are endpoint values and are not the minimum.

✗ **(D)** The continuous function attains a minimum on the closed interval.

Question 114 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) There is no change in concavity at 1

STEP-BY-STEP SOLUTION

f'' is positive on both sides of 1 and only equals zero at 1. Since concavity does not change, $x=1$ is not an inflection point based on this information.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** f'' alone does not show a local maximum at 1.

✗ **(B)** An inflection point requires a change in concavity.

✗ **(D)** f'' alone does not show a local minimum at 1.

Question 115 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (B) a local maximum

STEP-BY-STEP SOLUTION

With $f'(c)=0$ and $f''(c)<0$, the graph is concave down at the critical point, so c is a local maximum.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The test is conclusive when $f''(c)<0$.

✗ **(C)** That corresponds to $f''>0$.

✗ **(D)** A concavity change is not implied.

Question 116 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) the concavity of f changes at c and the point lies on the graph

STEP-BY-STEP SOLUTION

An inflection point is a point on the graph where concavity changes. $f''(c)=0$ is a possible clue, not a sufficient definition.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** f'' can be zero without a concavity change.

✗ **(B)** A horizontal tangent is not required.

✗ **(C)** A local extremum need not be an inflection point.

Question 117 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (A) x=0 is not an inflection point because concavity is upward on both sides

STEP-BY-STEP SOLUTION

$f''(x)=12x^2$ is positive for $x\neq0$ on both sides of 0. Concavity does not change, so 0 is not an inflection point; it is a local minimum.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** $f''=0$ alone is not sufficient.

✗ **(C)** The function decreases to 0 and increases after 0.

✗ **(D)** $x^4\geq0$, so 0 is a minimum, not a maximum.

Question 118 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) f is continuous and its concavity changes across the point

STEP-BY-STEP SOLUTION

For example, f(x)=x^(1/3) is continuous at 0, f″ is undefined there, and concavity changes across 0. Thus f″ need not exist at an inflection point.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The sign of f′ does not determine concavity.

✗ **(B)** The question specifically allows f″ to be undefined.

✗ **(C)** A discontinuity is not an inflection point on the graph.

Question 119 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) increasing and concave down

STEP-BY-STEP SOLUTION

f′>0 means f is increasing. f″<0 means f is concave down.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Both signs are inconsistent with this choice.

✗ **(B)** f′ is positive, so f is not decreasing.

✗ **(D)** f″ is negative, so it is not concave up.

Question 120 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (C) decreasing and concave up

STEP-BY-STEP SOLUTION

The negative first derivative means f decreases; the positive second derivative means f is concave up.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** f′ is negative, not positive.

✗ **(B)** Both monotonicity and concavity are misread.

✗ **(D)** f″ is positive, not negative.

Question 121 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) 100

STEP-BY-STEP SOLUTION

Let sides be x and y. $x+y=20$, so $A=x(20-x)=-(x-10)^2+100$. The maximum area is 100 at $x=y=10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The maximum is larger than 80.

✗ **(B)** 400 is not compatible with perimeter 40.

✗ **(C)** The vertex value is 100.

Question 122 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (C) $12\sqrt{3}$

STEP-BY-STEP SOLUTION

The height is $9-x^2$ and width is $2x$, so $A=2x(9-x^2)$, $0\le x\le 3$. $A'=18-6x^2=0$ gives $x=\sqrt{3}$; height=6. Thus $A=2\sqrt{3}\cdot 6=12\sqrt{3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The height at $x=\sqrt{3}$ is 6, not 9.

✗ **(B)** This is not the area at the critical point.

✗ **(D)** This is not the product of the maximizing width and height.

Question 123 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (A) 32

STEP-BY-STEP SOLUTION

For fixed area, the rectangle with minimum perimeter is a square. Side length $\sqrt{64}=8$, so perimeter=$4\cdot 8=32$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is less than the perimeter of any 64-area rectangle.

✗ **(C)** This is the perimeter of a 6-by-6 square, whose area is 36.

✗ **(D)** The square's perimeter is 32.

Question 124 · Unit 5 · Analytical Applications of Differentiation

★★★ Mandatory

Correct answer: (D) 20

STEP-BY-STEP SOLUTION

$P'(x)=-2x+40=0$ gives $x=20$. Since $P''=-2<0$, this is a maximum; checking the endpoints confirms it is absolute.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $P'(10)=20$, not zero.

✗ **(B)** $P'(40)=-40$, not zero.

✗ **(C)** The right endpoint is not the maximizing critical point.

Question 125 · Unit 5 · Analytical Applications of Differentiation

★★★★★ Very Important

Correct answer: (D) x=0 or an endpoint, so values must be compared

STEP-BY-STEP SOLUTION

The sign chart gives a local maximum at $x=0$ and a local minimum at $x=3$. For an absolute maximum on a closed interval, compare $f(-2)$, $f(0)$, $f(3)$, and $f(4)$; the sign chart alone does not select the largest value.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** $x=3$ is a local minimum by the sign chart.

✗ **(B)** The endpoint may be smaller than $f(0)$ or $f(4)$.

✗ **(C)** The endpoint may be smaller than $f(0)$.

Question 126 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (D) 22

STEP-BY-STEP SOLUTION

Net change $= \int_0^4 3\,dt = 12$. So $f(4) = f(0) + 12 = 10 + 12 = 22$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This adds $10 + 2\cdot4$ instead of $10 + 3\cdot4$.

✗ **(B)** This double-counts the initial value.

✗ **(C)** This is only the accumulated change and ignores the starting value 10.

Question 127 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (C) −12 L

STEP-BY-STEP SOLUTION

The rate is negative because volume is decreasing: $(-3)(4) = -12$ liters.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Both ignores the sign and adds.

✗ **(B)** Ignores that the outflow rate is negative.

✗ **(D)** Adds instead of multiplying.

Question 128 · Unit 6 · Integration and Accumulation of Change

★★ Important

Correct answer: (C) 0

STEP-BY-STEP SOLUTION

The integrand is identically zero, so the accumulated change is 0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The integral of 0 is well defined.

✗ **(B)** No basis.

✗ **(D)** Confuses the interval length with the value.

Question 129 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (A) 20

STEP-BY-STEP SOLUTION

$\Delta x = 2$. Midpoints are $x = 1$ and $x = 3$, so $M_2 = 2(f(1) + f(3)) = 2(1 + 9) = 20$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses midpoints 1 and 4.

✗ **(C)** Forgets to multiply by $\Delta x = 2$.

✗ **(D)** Uses midpoints 1 and 2.

Question 130 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (C) 18

STEP-BY-STEP SOLUTION

Midpoints 1, 3, 5 give $M_3 = 2(1 + 3 + 5) = 18$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses midpoints 1, 3, 6.5.

✗ **(B)** Forgets $\Delta x = 2$.

✗ **(D)** Uses endpoints 1, 3, 9.

Question 131 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (A) 6.15

STEP-BY-STEP SOLUTION

$\Delta x = 1$ and right endpoints are 1, 2, 3, 4. The sum is $\sqrt{1} + \sqrt{2} + \sqrt{3} + \sqrt{4} \approx 1 + 1.414 + 1.732 + 2 = 6.146$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses left endpoints 0, 1, 2, 3, giving about 4.15.

✗ **(C)** Uses midpoints and adds an extra unit.

✗ **(D)** Uses only the last term doubled.

Question 132 · Unit 6 · Integration and Accumulation of Change

★★ Important

Correct answer: (B) 10

STEP-BY-STEP SOLUTION

$1 + 2 + 3 + 4 = 10$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is 4^2.

✗ **(C)** Counts only the last term.

✗ **(D)** This is $4!$.

Question 133 · Unit 6 · Integration and Accumulation of Change

★★ Important

Correct answer: (B) 12

STEP-BY-STEP SOLUTION

A constant 3 added four times gives 4·3 = 12.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Counts only one term.

✗ **(C)** Adds the index to the constant.

✗ **(D)** Multiplies by 8 instead of 4.

Question 134 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (B) 5

STEP-BY-STEP SOLUTION

An antiderivative is x^2. Evaluating gives $3^2 - (-2)^2 = 9 - 4 = 5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to subtract the lower-limit value.

✗ **(C)** Treats the odd integrand over a symmetric-looking interval as zero, but the interval is not symmetric.

✗ **(D)** Adds 9 and 4.

Question 135 · Unit 6 · Integration and Accumulation of Change

★★ Important

Correct answer: (C) 4

STEP-BY-STEP SOLUTION

An antiderivative is $x^2/2 + x$. Evaluating gives $(2 + 2) - 0 = 4$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the constant term's contribution is only 2, not 4.

✗ **(B)** Uses only the triangle area.

✗ **(D)** Uses only the linear part.

Question 136 · Unit 6 · Integration and Accumulation of Change

★★ Important

Correct answer: (A) 5x + C

STEP-BY-STEP SOLUTION

The antiderivative of a constant k is kx + C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Treats 5 as an exponent.

✗ **(C)** Omits the constant of integration, which is required for an indefinite integral.

✗ **(D)** Forgets the variable x.

Question 137 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (B) $-x^{-1} + C$

STEP-BY-STEP SOLUTION

The power rule gives $x^{-1}/(-1) + C = -1/x + C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign from dividing by −1.

✗ **(C)** Applies the power rule as if differentiating.

✗ **(D)** That is the antiderivative of 1/x, not x^{-2}.

Question 138 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (C) $e^x + C$

STEP-BY-STEP SOLUTION

The exponential function is its own antiderivative.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds 1 to the exponent.

✗ **(B)** Confuses with the antiderivative of 1/x.

✗ **(D)** Applies a product rule idea that does not apply here.

Question 139 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (C) $x^3/3 - 2x^2 + x + C$

STEP-BY-STEP SOLUTION

Integrate term by term: $x^2 \to x^3/3$, $-4x \to -2x^2$, $1 \to x$, then add C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates instead.

✗ **(B)** Forgets to divide x^2 by 3.

✗ **(D)** Fails to divide $-4x$ by 2.

Question 140 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (A) 2

STEP-BY-STEP SOLUTION

An antiderivative is $-\cos x$. Evaluating gives $-\cos \pi + \cos 0 = 1 + 1 = 2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses only the upper limit.

✗ **(C)** Confuses the interval length with the value.

✗ **(D)** Treats sine as odd over a non-symmetric interval.

Question 141 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (D) 8/3

STEP-BY-STEP SOLUTION

The integrand is even, so the integral is $2\int_0^1 (x^2 + 1)\,dx = 2(1/3 + 1) = 8/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/3 from x^2.

✗ **(B)** Computes only half of the symmetric interval.

✗ **(C)** Uses only the constant term.

Question 142 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (D) −3

STEP-BY-STEP SOLUTION

An antiderivative is $x^2 - 4x$. Evaluating gives $(9 - 12) - 0 = -3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes the positive and negative areas cancel exactly; they do not.

✗ **(B)** Uses $-2 \cdot 3$.

✗ **(C)** Drops the negative sign.

Question 143 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (B) the negative of the value

STEP-BY-STEP SOLUTION

$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Not a property of definite integrals.

✗ **(C)** Only true if the net area happens to be zero.

✗ **(D)** Only true when the integral is zero.

Question 144 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (D) 18

STEP-BY-STEP SOLUTION

x^2 is even, so the integral is $2\int_0^3 x^2\,dx = 2(27/3) = 18$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 3^3.

✗ **(B)** Computes only half.

✗ **(C)** Treats x^2 as an odd function.

Question 145 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (B) $2x \sin(x^2)$

STEP-BY-STEP SOLUTION

Apply the FTC together with the chain rule: $\sin(x^2) \cdot d/dx(x^2) = 2x \sin(x^2)$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates the integrand incorrectly.

✗ **(C)** Forgets the chain rule factor 2x.

✗ **(D)** A derivative does not take a + C.

Question 146 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (C) 1

STEP-BY-STEP SOLUTION

$A(e) = \ln|t| \,|_1{}^e = \ln e - \ln 1 = 1 - 0 = 1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the reciprocal.

✗ **(B)** Assumes the integral from a point to itself.

✗ **(D)** Confuses the upper limit with the value.

Question 147 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (D) $3x^2 e^{(x^3)}$

STEP-BY-STEP SOLUTION

FTC plus chain rule: $e^{(x^3)} \cdot d/dx(x^3) = 3x^2 e^{(x^3)}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the chain rule factor.

✗ **(B)** Misapplies the chain rule to the exponent.

✗ **(C)** Multiplies by the upper limit instead of its derivative.

Question 148 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (B) 2/π

STEP-BY-STEP SOLUTION

Average = $(1/\pi)\int_0^{\pi} \sin x\, dx = (1/\pi)(2) = 2/\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by π.

✗ **(C)** Assumes symmetry gives zero.

✗ **(D)** Forgets the factor 2 from the integral.

Question 149 · Unit 6 · Integration and Accumulation of Change

★★★★★ Very Important

Correct answer: (C) 8

STEP-BY-STEP SOLUTION

$\int_0^2 3t^2\, dt = t^3 \big|_0^2 = 8$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses t^2 at $t = 2$.

✗ **(B)** Uses $3t^2$ at $t = 2$.

✗ **(D)** Uses $3t$ at $t = 2$.

Question 150 · Unit 6 · Integration and Accumulation of Change

★★★ Mandatory

Correct answer: (D) 16

STEP-BY-STEP SOLUTION

Distance = rate × time = $4 \times (5 - 1) = 16$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses time 5 instead of the interval length 4.

✗ **(B)** Uses only the velocity.

✗ **(C)** Halves the interval.

Question 151 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (C) $P'=0.04P$

STEP-BY-STEP SOLUTION

A constant per-capita growth rate of 0.04 means $P'/P=0.04$, so $P'=0.04P$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This describes a constant growth rate independent of population.

✗ **(B)** This is a linear formula, not the stated rate law.

✗ **(D)** This is 400% per year, not 4%.

Question 152 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (B) $A'=12-0.2A$

STEP-BY-STEP SOLUTION

Net rate equals input rate minus removal rate: $A'=12-0.2A$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reverses input and removal signs.

✗ **(C)** Treats removal as an increase.

✗ **(D)** Multiplies the constant input by A.

Question 153 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (B) negative

STEP-BY-STEP SOLUTION

$1-60/50=-0.2$, so $P'=0.2(60)(-0.2)=-2.4$. The population is decreasing.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The logistic factor is negative above $K=50$.

✗ **(C)** The right-hand side is defined at $P=60$.

✗ **(D)** This occurs at $P=0$ or $P=50$.

Question 154 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (D) $y=Ce^{2x}$

STEP-BY-STEP SOLUTION

For $y=Ce^{2x}$, $y'=2Ce^{2x}=2y$, so it satisfies the differential equation.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Its derivative is $Ce^{x}=y$, not $2y$.

✗ **(B)** Its derivative is $2Ce^{x}$, which is not generally $2(2Ce^{x})$.

✗ **(C)** Its derivative is 2, not $2y$.

Question 155 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (C) $y=0$

STEP-BY-STEP SOLUTION

A constant solution requires $y'=0$ and $y(5-y)=0$. $y=0$ is one such equilibrium solution ($y=5$ is the other).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The right side is nonzero.

✗ **(B)** The right side is nonzero.

✗ **(D)** The right side is 4, not zero.

Question 156 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (A) $y=Ce^{x^2}$

STEP-BY-STEP SOLUTION

For $y=Ce^{x^2}$, $y'=2xCe^{x^2}=2xy$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Its derivative is $2Cx$, not $2xy$ in general.

✗ **(C)** Its derivative is $2y$, not $2xy$.

✗ **(D)** Its derivative is y, not $2xy$.

Question 157 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (C) −1

STEP-BY-STEP SOLUTION

$y' = 2(1) - 3 = -1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative y term.

✗ **(B)** Uses only 3.

✗ **(D)** Adds 2 and 3.

Question 158 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (C) x=1 or y=−2

STEP-BY-STEP SOLUTION

A product is zero if either factor is zero: $x-1=0$ or $y+2=0$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Both zero conditions have the wrong signs.

✗ **(B)** This is unrelated to the factors.

✗ **(D)** Every point on either zero-slope line has slope zero, not just their intersection.

Question 159 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (B) Slopes are positive at every point

STEP-BY-STEP SOLUTION

Since $1+y^2 \geq 1 > 0$ for every real y, every slope is positive.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** At $y=0$ the slope is 1.

✗ **(C)** The slope depends on y.

✗ **(D)** The slope remains positive even when $y<0$.

Question 160 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (B) y=0 and y=20

STEP-BY-STEP SOLUTION

Set y′=0. The factors give y=0 or 1−y/20=0, hence y=20.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** y=10 is not an equilibrium.

✗ **(C)** Misses the carrying-capacity equilibrium.

✗ **(D)** Misses the zero solution.

Question 161 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (B) increases toward 5

STEP-BY-STEP SOLUTION

For 00. As y approaches 5, the slope approaches 0; the solution increases toward carrying capacity 5.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The derivative is positive on this interval.

✗ **(C)** The slope changes sign at y=5 and solutions approach that equilibrium.

✗ **(D)** It is not an equilibrium unless y is 0 or 5.

Question 162 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (D) 4

STEP-BY-STEP SOLUTION

$y'=(-2)^2=4$, so the slope is positive.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses y rather than y^2.

✗ **(B)** The derivative is not zero.

✗ **(C)** Sign error when squaring.

Question 163 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (C) y=−1

STEP-BY-STEP SOLUTION

For y<−1 the product is positive, so solutions rise toward −1. For −1

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Solutions move away from 2 on both sides.

✗ **(B)** The two equilibria have different stability.

✗ **(D)** One equilibrium is stable.

Question 164 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (A) y=Cx

STEP-BY-STEP SOLUTION

Separate dy/y=dx/x. Integrating gives ln|y|=ln|x|+C, hence y=Cx on each connected interval not crossing x=0.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the reciprocal dependence.

✗ **(C)** This is not the separated solution.

✗ **(D)** This would give y′=2Cx.

Question 165 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (A) $y^3=x^3+C$

STEP-BY-STEP SOLUTION

Separate $y^2dy=x^2dx$. Integrate: $y^3/3=x^3/3+C$. Multiply by 3 and absorb the constant to get $y^3=x^3+C$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Integrating y^2 gives $y^3/3$, not y^2.

✗ **(C)** Integrating x^2 gives $x^3/3$, not x^2.

✗ **(D)** The left side must be integrated with respect to y.

Question 166 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (D) (2/3)y^(3/2)=x²/2+x+C

STEP-BY-STEP SOLUTION

Separate √y dy=(x+1)dx. Integrate: (2/3)y^(3/2)=x²/2+x+C.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the factor 2/3 and the 1/2 on x^2.

✗ **(B)** The integral of √y is (2/3)y^(3/2), not 2√y.

✗ **(C)** Integrates √y with the wrong power.

Question 167 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (A) $4/(1+3e^{-x})$

STEP-BY-STEP SOLUTION

The logistic form is $y=4/(1+Ae^{-x})$. Using y(0)=1 gives 1=4/(1+A), so A=3.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** At x=0 this gives 2, not 1.

✗ **(C)** Its carrying capacity is 1, not 4.

✗ **(D)** At x=0 the denominator is −2, giving the wrong initial value.

Question 168 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (C) $80e^{-0.5t}$

STEP-BY-STEP SOLUTION

For $y'=ky$, $y=Ce^{kt}$. Here C=80 and k=−0.5, so $y=80e^{-0.5t}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** A constant additive decay rate is a different model.

✗ **(B)** This has the wrong initial value and decay rate.

✗ **(D)** This grows instead of decaying.

Question 169 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (C) $5e^{-2(x-1)}$

STEP-BY-STEP SOLUTION

The solution anchored at x=1 is $y=5e^{-2}(x-1)$, which gives $y(1)=5$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This grows and does not satisfy $y'=-2y$.

✗ **(B)** At x=1 this is $5e^{-2}$, not 5.

✗ **(D)** This is linear, not exponential.

Question 170 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (B) ln2/6

STEP-BY-STEP SOLUTION

$A(t)=A_0e^{-kt}$. Half-life means $e^{-6k}=1/2$, so $6k=\ln 2$ and $k=\ln 2/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** k is positive for decay in $A'=-kA$.

✗ **(C)** This is the reciprocal rate.

✗ **(D)** Does not follow from the half-life equation.

Question 171 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (B) ln2/0.04

STEP-BY-STEP SOLUTION

At half-life t, $e^{-0.04t}=1/2$. Hence $t=\ln 2/0.04$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the rate incorrectly.

✗ **(C)** Does not solve the half-life equation.

✗ **(D)** The value 25 is 1/0.04 but misses the ln2 factor.

Question 172 · Unit 7 · Differential Equations

★★★ Mandatory

Correct answer: (B) 20

STEP-BY-STEP SOLUTION

Substitute $N=400$: $N'=0.05(400)=20$ units per time.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the growth parameter with the current rate.

✗ **(C)** Confuses the population with its rate.

✗ **(D)** Divides rather than multiplies.

Question 173 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (B) ln4/0.2

STEP-BY-STEP SOLUTION

$A/A_0=e^{-0.2t}$. Set $e^{-0.2t}=1/4$; then $0.2t=\ln 4$, so $t=\ln 4/0.2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 4 rather than ln4.

✗ **(C)** Multiplies by the rate instead of dividing.

✗ **(D)** This is the half-life, not the quarter-life.

Question 174 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (A) 800

STEP-BY-STEP SOLUTION

Compare with $P'=rP(1-P/K)$; the denominator K is the carrying capacity.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Half the carrying capacity.

✗ **(C)** This is the intrinsic growth-rate parameter.

✗ **(D)** Zero is an equilibrium, not the carrying capacity.

Question 175 · Unit 7 · Differential Equations

★★★★★ Very Important

Correct answer: (D) $P=K/(1+Ae^{-rt})$

STEP-BY-STEP SOLUTION

Separating variables and integrating gives the logistic family $P(t)=K/(1+Ae^{-rt})$, with A determined by the initial condition.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This does not approach K in the standard way.

✗ **(B)** This is an exponential model, not logistic.

✗ **(C)** The exponent/sign form is not the standard general logistic solution.

Question 176 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (C) 3

STEP-BY-STEP SOLUTION

Average value = $(1/3)\int_0^3 x^2dx = (1/3)(9) = 3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by the interval length 3.

✗ **(B)** Uses the endpoint value f(3).

✗ **(D)** Incorrect evaluation of the integral.

Question 177 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (B) 19/2

STEP-BY-STEP SOLUTION

$\int_1^4(3x+2)dx = [3x^2/2+2x]_1^4 = 32-7/2 = 57/2$. Divide by $4-1=3$: average = 19/2.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Too small; it does not match the integral average.

✗ **(C)** Incorrect arithmetic for the average.

✗ **(D)** This is the integral, not the average value.

Question 178 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (C) 2/π

STEP-BY-STEP SOLUTION

Average = (1/π)∫₀^πsin x dx = 2/π.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the integral equals 2.

✗ **(B)** Does not divide by π.

✗ **(D)** This is the integral, not the average.

Question 179 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (B) 2/√3

STEP-BY-STEP SOLUTION

Average value = (1/2)∫₀²x²dx = (1/2)(8/3)=4/3. Solve $c^2=4/3$. Since $c\geq 0$, $c=2/\sqrt{3}$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not solve $c^2=4/3$.

✗ **(C)** This is the average value, not c.

✗ **(D)** Does not square to 4/3.

Question 180 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (B) 3/2

STEP-BY-STEP SOLUTION

Displacement = $\int_0^3(t-1)dt = [t^2/2-t]_0^3 = 9/2-3 = 3/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses an incorrect interval or antiderivative.

✗ **(C)** Sign error in the integral.

✗ **(D)** Confuses displacement with a different accumulated quantity.

Question 181 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (B) 5/2

STEP-BY-STEP SOLUTION

Velocity changes sign at t=1. Distance on [0,1] is the triangle area 1/2; on [1,3] it is 2. Total distance = 1/2+2=5/2.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Counts only part of the path.

✗ **(C)** Overcounts the area.

✗ **(D)** This is the signed displacement, not total distance.

Question 182 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (D) 1

STEP-BY-STEP SOLUTION

The roots are t=1 and t=3. The product is negative between the roots, so the particle moves in the negative direction on (1,3).

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A) Both factors are positive there, so their product is positive.**

✗ **(B) The sign is not negative for every positive t.**

✗ **(C) Both factors are negative there, so their product is positive.**

Question 183 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (D) 1

STEP-BY-STEP SOLUTION

The three trapezoid contributions are (0+2)/2=1, (2+0)/2=1, and (0−2)/2=−1. Net displacement = 1+1−1=1.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** The first two contributions are positive, so the total is not zero.

✗ **(B)** Adds an extra unit of area.

✗ **(C)** Sign or arithmetic error in adding the three signed areas.

Question 184 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (A) 4/3

STEP-BY-STEP SOLUTION

Area = $\int_{-1}^{1}(1-x^2)dx = 2-2/3 = 4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Underestimates the area.

✗ **(C)** Uses the rectangle area without subtracting the parabola.

✗ **(D)** Forgets part of the interval or the constant contribution.

Question 185 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (C) 1

STEP-BY-STEP SOLUTION

On $[-1,0]$, x^2 is above x; on $[0,1]$, x is above x^2. Area = $\int_{-1}^{0}(x^2-x)dx + \int_{0}^{1}(x-x^2)dx = 5/6+1/6=1$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a different region's area.

✗ **(B)** Adds the wrong endpoint contributions.

✗ **(D)** Counts only one of the two regions.

Question 186 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (B) 9/2

STEP-BY-STEP SOLUTION

The curves intersect at $x=-1$ and $x=2$. The line is above the parabola between them. Area=$\int_{-1}^{2}(x+2-x^2)dx = 9/2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Overestimates the exact integral.

✗ **(C)** Underestimates the exact integral.

✗ **(D)** Incorrect antiderivative or endpoint evaluation.

Question 187 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (C) 4/3

STEP-BY-STEP SOLUTION

Right minus left is $2y-y^2$. Area$=\int_0^2(2y-y^2)dy=[y^2-y^3/3]_0^2=4-8/3=4/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic or antiderivative error.

✗ **(B)** Uses an incorrect width.

✗ **(D)** Doubles the correct result.

Question 188 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (D) 2

STEP-BY-STEP SOLUTION

Horizontal width = right−left = $2-y$. Area$=\int_0^2(2-y)dy=4-2=2$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Underestimates the triangle area.

✗ **(B)** Uses an incorrect antiderivative.

✗ **(C)** Uses rectangle area without accounting for sloping boundary.

Question 189 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (C) 2π

STEP-BY-STEP SOLUTION

This is a semicircle of radius 2, so area$=(1/2)\pi(2^2)=2\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the area of the full circle.

✗ **(B)** Doubles the full-circle area.

✗ **(D)** This is the area of a semicircle of radius $\sqrt{2}$.

Question 190 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (D) 16/15

STEP-BY-STEP SOLUTION

Square side=$1-x^2$, so $A(x)=(1-x^2)^2$. Volume=$\int_{-1}^{1}(1-2x^2+x^4)dx=2(1-2/3+1/5)=16/15$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses the factor from the symmetric interval.

✗ **(B)** Uses a bounding rectangle instead of integrating.

✗ **(C)** Uses a different cross-sectional area.

Question 191 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (D) 2/3

STEP-BY-STEP SOLUTION

$A(x)=(1-x)\cdot 2(1-x)=2(1-x)^2$. Volume=$2\int_0^1(1-x)^2dx=2/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a different area function.

✗ **(B)** Doubles the result twice.

✗ **(C)** Forgets the factor 2 in the height.

Question 192 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (A) π/6

STEP-BY-STEP SOLUTION

Semicircle area = $\pi d^2/8 = \pi(1-x^2)/8$. Integrate: $V=(\pi/8)\int_{-1}^{1}(1-x^2)dx=(\pi/8)(4/3)=\pi/6$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses a full circle rather than a semicircle.

✗ **(C)** Overestimates by a factor of 4.

✗ **(D)** Underestimates by a factor of 2.

Question 193 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (C) $8\sqrt{3}/3$

STEP-BY-STEP SOLUTION

Cross-sectional area=$(\sqrt{3}/4)(4-x^2)$. Integrate from -2 to 2: $(\sqrt{3}/4)(16-16/3)=8\sqrt{3}/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a bounding rectangle.

✗ **(B)** Uses half of the correct interval contribution.

✗ **(D)** Doubles the correct value.

Question 194 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (A) $32\pi/3$

STEP-BY-STEP SOLUTION

Disk method: $V=\pi\int_{-2}^{2}(4-x^2)dx=\pi(16-16/3)=32\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses the bounding cylinder.

✗ **(C)** Uses half the x-interval.

✗ **(D)** Underestimates the integral.

Question 195 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (A) 9π

STEP-BY-STEP SOLUTION

$V=\pi\int_{0}^{3}x^2dx=\pi(27/3)=9\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses the endpoint radius for the whole length.

✗ **(C)** Uses x instead of x^2 in the disk area.

✗ **(D)** Uses an incorrect antiderivative.

Question 196 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (B) 8π

STEP-BY-STEP SOLUTION

Disk radius is $\sqrt{y}$. $V=\pi\int_0^4(\sqrt{y})^2dy=\pi\int_0^4 y\,dy=8\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Underestimates the integral.

✗ **(C)** Uses the endpoint radius over the whole interval.

✗ **(D)** Arithmetic error.

Question 197 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (C) 16π/3

STEP-BY-STEP SOLUTION

Outer radius $R=2$, inner radius $r=x$. $V=\pi\int_0^2(4-x^2)dx=\pi(8-8/3)=16\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses an incorrect integral.

✗ **(B)** Uses only the outer cylinders.

✗ **(D)** Forgets to subtract the inner disk.

Question 198 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (D) 64π/15

STEP-BY-STEP SOLUTION

Outer radius $2x$ and inner radius x^2. $V=\pi\int_0^2[(2x)^2-(x^2)^2]dx=\pi\int_0^2(4x^2-x^4)dx=64\pi/15$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses a factor in the integral.

✗ **(B)** Uses an incorrect power for the inner radius.

✗ **(C)** Uses a disk without subtracting the hole.

Question 199 · Unit 8 · Applications of Integration

★★★★★ Very Important

Correct answer: (D) 28π/3

STEP-BY-STEP SOLUTION

Outer radius R=3, inner radius r=3−x. $V=\pi\int_0^2[9-(3-x)^2]dx=\pi\int_0^2(6x-x^2)dx=28\pi/3$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Underestimates the correct washer integral.

✗ **(B)** Forgets the inner-radius subtraction.

✗ **(C)** Arithmetic error.

Question 200 · Unit 8 · Applications of Integration

★★★ Mandatory

Correct answer: (C) 9π

STEP-BY-STEP SOLUTION

$V=\pi\int_0^3(3-x)^2dx=\pi\cdot9=9\pi$.

WHY THE OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses half the result.

✗ **(B)** Doubles the exact volume.

✗ **(D)** Underestimates the integral.

Piecewise Continuity with Two Parameters

★★★★★ Very Important

MODEL SOLUTION

(a) $\lim_{x\to 1^-} f(x) = 1^2 + a = \mathbf{1 + a}$. $\lim_{x\to 1^+} f(x) = b(1) + 2 = \mathbf{b + 2}$.

(b) Condition 1: $f(1) = 4$ is defined. Condition 2: the limit exists, so $1 + a = b + 2$. Condition 3: the limit must equal 4. $1 + a = 4$ gives $\mathbf{a = 3}$; $b + 2 = 4$ gives $\mathbf{b = 2}$. With these values all three conditions hold.

(c) With $a = 2$ the left limit is 3; with $b = 5$ the right limit is 7. Both are finite but unequal, so $\lim_{x\to 1} f(x)$ does not exist and f has a **jump discontinuity** at $x = 1$.

SCORING RUBRIC

Criterion	Points
(a) Correct left limit $1 + a$	1
(a) Correct right limit $b + 2$	1
(b) Equates both one-sided limits to $f(1) = 4$	2
(b) Correct values $a = 3$ and $b = 2$	2
(c) Evaluates one-sided limits 3 and 7	2
(c) Correct classification as jump	1
Total	9

Zeros of a Continuous Function from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) The only hypothesis of the IVT is continuity on a closed interval, and f is given continuous on [0, 4].

(b) Check consecutive signs: f(0) = 3 to f(1) = −1 (zero in (0,1)); f(1) = −1 to f(2) = 2 (zero in (1,2)); f(2) = 2 to f(3) = −2 (zero in (2,3)); f(3) = −2 to f(4) = 5 (zero in (3,4)).
Each sign change forces a zero by the IVT, so the minimum number is **4**.

(c) Yes. f is continuous on [0, 1] and 1 lies strictly between f(1) = −1 and f(0) = 3, so the IVT guarantees some c in (0, 1) with **f(c) = 1**.

SCORING RUBRIC

Criterion	Points
(a) Cites continuity on the closed interval	2
(b) Identifies all four sign changes	2
(b) Concludes a minimum of 4 zeros via IVT	2
(c) States −1 < 1 < 3	2
(c) Concludes yes with IVT justification	1
Total	9

Asymptotes of a Rational Function

★★★★★ Very Important

MODEL SOLUTION

(a) Numerator: $(x - 1)(x + 1)$. Denominator: $(x - 2)(x + 2)$. There are no common factors.

(b) Vertical asymptotes at **$x = 2$ and $x = -2$**. Near $x = 2$ the numerator approaches 3 (positive) and $(x + 2)$ approaches 4 (positive), so the sign is governed by $(x - 2)$:
As $x \to 2^-$, $(x - 2)$ is a small negative number, so $f(x) \to -\infty$.
As $x \to 2^+$, $(x - 2)$ is a small positive number, so $f(x) \to +\infty$.

(c) Equal degrees: the limit is the ratio of leading coefficients, **1**. The horizontal asymptote is **$y = 1$**.

SCORING RUBRIC

Criterion	Points
(a) Correct factorization of both polynomials	2
(b) Identifies both vertical asymptotes	1
(b) Correct left-hand limit $-\infty$ with sign analysis	2
(b) Correct right-hand limit $+\infty$	1
(c) Correct limit 1 and asymptote $y = 1$	3
Total	9

The Squeeze Theorem

★★★★★ Very Important

MODEL SOLUTION

(a) $4(1) - 1 = \mathbf{3}$ and $1 + 2 = \mathbf{3}$.

(b) Conditions: the inequality holds on an open interval containing 1, and both bounding functions have the same limit at 1 (both equal 3). Therefore $\lim_{x\to 1} g(x) = \mathbf{3}$.

(c) For all $x \neq 0$, $-1 \le \cos(1/x) \le 1$, so $-|x^3| \le x^3\cos(1/x) \le |x^3|$. Since $\pm|x^3| \to 0$ as $x \to 0$, the Squeeze Theorem gives the limit **0**.

SCORING RUBRIC

Criterion	Points
(a) Both bounding limits evaluated as 3	2
(b) States the inequality and the shared limit	2
(b) Concludes the limit is 3	1
(c) Sets up correct bounds	2
(c) Evaluates the bounds and concludes 0	2
Total	**9**

Limits at Infinity with a Radical

★★★★★ Very Important

MODEL SOLUTION

(a) For $x > 0$, $\sqrt{(x^2 + 1)} = x\sqrt{(1 + 1/x^2)}$, so $g(x) = 3/\sqrt{(1 + 1/x^2)} \to \mathbf{3}$.

(b) For $x < 0$, $\sqrt{(x^2 + 1)} = |x|\sqrt{(1 + 1/x^2)} = -x\sqrt{(1 + 1/x^2)}$, since $|x| = -x$ when x is negative. So $g(x) = 3x / (-x\sqrt{(1 + 1/x^2)}) = -3/\sqrt{(1 + 1/x^2)} \to \mathbf{-3}$. The sign differs because the square root always returns a nonnegative value.

(c) No. The denominator satisfies $x^2 + 1 \geq 1 > 0$ for every real x, so it never vanishes and no vertical asymptote exists.

SCORING RUBRIC

Criterion	Points
(a) Correct value 3 with radical factoring	3
(b) Uses $\|x\| = -x$ for $x < 0$	2
(b) Correct value -3	2
(c) Shows the denominator is never zero; concludes none	2
Total	9

Applied Model: Cooling Coffee

★★★★★ Very Important

MODEL SOLUTION

(a) $T(0) = 20 + 70e^{0} = 20 + 70 = \mathbf{90°C}$.

(b) As $t \to \infty$, $e^{-0.05t} \to 0$, so $\lim T(t) = 20 + 0 = \mathbf{20°C}$. The coffee cools toward the surrounding room temperature of 20°C.

(c) T is a sum of a constant and a continuous exponential, so T is continuous on $[0, 30]$.
$T(30) = 20 + 70e^{-1.5} \approx 20 + 70(0.223) \approx 35.6$.
Since $35.6 < 50 < 90 = T(0)$, the IVT guarantees some t in $(0, 30)$ with $\mathbf{T(t) = 50°C}$.

SCORING RUBRIC

Criterion	Points
(a) Computes $T(0) = 90°C$	2
(b) Correct limit 20°C	1
(b) Interprets as ambient/room temperature	2
(c) Computes $T(30) \approx 35.6$ and cites continuity	2
(c) Applies the IVT with the correct inequality	2
Total	9

Definition of Derivative and Tangent Line

★★★★★ Very Important

MODEL SOLUTION

(a) $f'(x) = \lim_{h\to 0} [f(x+h) - f(x)] / h = \lim_{h\to 0} [\, 1/(x+h+1) - 1/(x+1) \,] / h$
$= \lim_{h\to 0} [\, (x+1) - (x+h+1) \,] / [\, h(x+h+1)(x+1) \,] = \lim_{h\to 0} [\, -h \,] / [\, h(x+h+1)(x+1) \,]$
$= \lim_{h\to 0} -1 / [\, (x+h+1)(x+1) \,] = \mathbf{-1/(x+1)^2}$.

(b) Point: $(1, f(1)) = (1, 1/2)$. Slope: $f'(1) = -1/(1+1)^2 = -1/4$.
Tangent line: $\mathbf{y - 1/2 = -(1/4)(x - 1)}$ (or $y = -x/4 + 3/4$).

(c) The line $x + 4y = 8$ has slope $m = -1/4$.
Set $f'(x) = -1/4$: $-1/(x+1)^2 = -1/4 \Rightarrow (x+1)^2 = 4 \Rightarrow x+1 = \pm 2 \Rightarrow x = 1$ or $x = -3$.
At $x = 1$: $y = 1/2 \rightarrow$ **(1, 1/2)**.
At $x = -3$: $y = 1/(-3+1) = -1/2 \rightarrow$ **(−3, −1/2)**.

SCORING RUBRIC

Criterion	Points
(a) Sets up difference quotient correctly	1
(a) Common denominator algebra	2
(a) Evaluates limit to obtain $-1/(x+1)^2$	1
(b) Identifies point (1, 1/2) and slope −1/4	2
(b) Correct tangent line equation	1
(c) Finds both x-values 1 and −3 and gives both coordinates	2
Total	9

Piecewise Continuity and Differentiability

★★★★★ Very Important

MODEL SOLUTION

(a) For continuity at x = 2, we need $\lim_{x\to 2^-} g(x) = \lim_{x\to 2^+} g(x) = g(2)$.
$\lim_{x\to 2^-} (ax^2 + 1) = 4a + 1$. $\lim_{x\to 2^+} (bx - 3) = 2b - 3$.
Relationship: **4a + 1 = 2b − 3**, or **2b − 4a = 4 ⇒ b − 2a = 2**.

(b) For differentiability, left and right derivatives must match at x = 2:
$d/dx(ax^2 + 1)\big|_2 = 2a(2) = 4a$. $d/dx(bx - 3)\big|_2 = b$.
Equating derivatives: **b = 4a**.
Substitute b = 4a into the continuity equation: 4a − 2a = 2 ⇒ 2a = 2 ⇒ **a = 1**.
Then **b = 4(1) = 4**.

(c) Point: $g(2) = (1)(2^2) + 1 = 5$. Slope: $g'(2) = 4a = 4(1) = 4$.
Tangent line: **y − 5 = 4(x − 2)**.

SCORING RUBRIC

Criterion	Points
(a) Evaluates both one-sided limits at x = 2	2
(a) Correct continuity equation 4a + 1 = 2b − 3	1
(b) Finds left and right derivatives 4a and b	2
(b) Solves linear system to get a = 1, b = 4	2
(c) Correct point (2, 5) and slope 4, writes tangent line	2
Total	9

Analyzing Differentiability from a Graph

★★★★★ Very Important

MODEL SOLUTION

(a) On (−3, 0), f is a line segment through (−3, 0) and (0, 3).
Slope = (3 − 0)/(0 − (−3)) = 1. Since the segment is linear, **f′(−1) = 1**.

(b) Left derivative: as $x \to 0^-$, slope is 1.
Right derivative: the quarter circle has equation $y = \sqrt{(9 - x^2)}$. $y' = -x/\sqrt{(9 - x^2)}$. As $x \to 0^+$, $y' \to 0$.
Since left derivative (1) ≠ right derivative (0), f has a sharp corner at x = 0 and is **not differentiable at x = 0**.

(c) x = 0: not differentiable due to a **corner** (slopes 1 and 0 disagree).
x = 3: On the circle, as $x \to 3^-$, $y' = -x/\sqrt{(9 - x^2)} \to -\infty$ (vertical tangent on circle). On the segment for x > 3, slope = (−4 − 0)/(5 − 3) = −2.
f is **not differentiable at x = 3** due to a **corner / vertical tangent boundary**.

SCORING RUBRIC

Criterion	Points
(a) Calculates line slope = 1 and gives f′(−1) = 1	2
(b) Computes left slope (1) and right slope (0)	2
(b) Concludes not differentiable due to corner	1
(c) Identifies x = 0 with corner justification	2
(c) Identifies x = 3 with vertical tangent / corner justification	2
Total	9

Tabular Data: Average Rates and Tangent Approximation

★★★★★ Very Important

MODEL SOLUTION

(a) Average velocity = [h(8) − h(0)] / (8 − 0) = (37 − 1)/8 = 36/8 = **4.5 m/s**.

(b) Symmetric difference quotient on [2, 8]:
h′(5) ≈ [h(8) − h(2)] / (8 − 2) = (37 − 7)/6 = 30/6 = **5 m/s**.
(Using backward difference [19 − 7]/3 = 4 m/s or forward difference [37 − 19]/3 = 6 m/s is also acceptable with work shown).

(c) Point: (2, h(2)) = (2, 7). Given slope: h′(2) = 3.5.
Tangent line: L(t) = 7 + 3.5(t − 2).
Approximation: h(2.3) ≈ L(2.3) = 7 + 3.5(2.3 − 2) = 7 + 3.5(0.3) = 7 + 1.05 = **8.05 meters**.

SCORING RUBRIC

Criterion	Points
(a) Correct difference quotient and value 4.5 with units m/s	2
(b) Valid difference quotient setup from table	2
(b) Correct numerical estimate with units	1
(c) Correct tangent line equation L(t) = 7 + 3.5(t − 2)	2
(c) Correct approximation 8.05 meters	2
Total	9

Derivatives of Products and Quotients with Trig

★★★★★ Very Important

MODEL SOLUTION

(a) Product Rule: $f'(x) = \frac{d}{dx}(x^2 + 2) \cdot \cos x + (x^2 + 2) \cdot \frac{d}{dx}(\cos x)$
$= \mathbf{2x \cos x - (x^2 + 2) \sin x}$.

(b) $f'(0) = 2(0)\cos(0) - (0 + 2)\sin(0) = 0 - 0 = \mathbf{0}$. (Horizontal tangent line at $x = 0$).

(c) Differentiate $f'(x) = 2x \cos x - (x^2 + 2) \sin x$:
$\frac{d}{dx}[2x \cos x] = 2 \cos x - 2x \sin x$.
$\frac{d}{dx}[(x^2 + 2) \sin x] = 2x \sin x + (x^2 + 2) \cos x$.
$f''(x) = (2 \cos x - 2x \sin x) - (2x \sin x + (x^2 + 2) \cos x)$
$= \mathbf{-x^2 \cos x - 4x \sin x}$.
At $x = 0$: $f''(0) = -(0)^2 \cos(0) - 4(0)\sin(0) = \mathbf{0}$.

SCORING RUBRIC

Criterion	Points
(a) Product Rule setup	1
(a) Correct derivative $2x \cos x - (x^2+2) \sin x$	2
(b) Evaluates $f'(0) = 0$	2
(c) Differentiates both terms correctly using product rule	2
(c) Simplifies $f''(x)$ and evaluates $f''(0) = 0$	2
Total	9

Horizontal Tangents and Normal Lines to a Rational Function

★★★★★ Very Important

MODEL SOLUTION

(a) Quotient Rule: $R'(x) = [\,(1)(x^2 + 3) - (x - 1)(2x)\,] / (x^2 + 3)^2$
$= [\,x^2 + 3 - 2x^2 + 2x\,] / (x^2 + 3)^2 =$ **$(-x^2 + 2x + 3) / (x^2 + 3)^2$**.

(b) Horizontal tangent means $R'(x) = 0$. Set numerator = 0:
$-x^2 + 2x + 3 = 0 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$.
The denominator $(x^2 + 3)^2$ is never zero for real x. Values: **$x = 3$ and $x = -1$**.

(c) At $x = 1$: $R(1) = (1 - 1)/(1 + 3) = 0$. Point is $(1, 0)$.
Slope of tangent: $R'(1) = (-1 + 2 + 3)/(1 + 3)^2 = 4/16 = 1/4$.
Slope of normal line: negative reciprocal $m_{normal} = -4$.
Normal line equation: $y - 0 = -4(x - 1) \Rightarrow$ **$y = -4x + 4$**.

SCORING RUBRIC

Criterion	Points
(a) Quotient rule setup and expansion	2
(a) Simplified derivative $(-x^2+2x+3)/(x^2+3)^2$	1
(b) Sets numerator = 0 and factors	2
(b) Solves $x = 3$ and $x = -1$	1
(c) Finds point $(1, 0)$ and tangent slope $1/4$	1
(c) Takes negative reciprocal -4 and writes normal line	2
Total	9

Implicit Curve: Tangents and Asymptotes

★★★★★ Very Important

MODEL SOLUTION

(a) Differentiate implicitly with respect to x:
$3x^2 + 3y^2(dy/dx) = 6[\,(1)y + x(dy/dx)\,] = 6y + 6x(dy/dx)$.
Divide by 3: $x^2 + y^2(dy/dx) = 2y + 2x(dy/dx)$.
Collect dy/dx: $y^2(dy/dx) - 2x(dy/dx) = 2y - x^2 \Rightarrow (dy/dx)(y^2 - 2x) = 2y - x^2$.
$dy/dx = (2y - x^2) / (y^2 - 2x)$.

(b) At (3, 3): $dy/dx = [2(3) - 3^2] / [3^2 - 2(3)] = (6 - 9)/(9 - 6) = -3/3 =$ **-1**.
Tangent line: $y - 3 = -1(x - 3) \Rightarrow$ **$y = -x + 6$**.

(c) Horizontal tangent requires $dy/dx = 0 \Rightarrow 2y - x^2 = 0 \Rightarrow$ **$y = x^2/2$** (with $y^2 - 2x \neq 0$).
Substitute $y = x^2/2$ into the curve $x^3 + y^3 = 6xy$:
$x^3 + (x^2/2)^3 = 6x(x^2/2) \Rightarrow x^3 + x^6/8 = 3x^3 \Rightarrow x^6/8 = 2x^3$.
Since $x \neq 0$, divide by x^3: $x^3/8 = 2 \Rightarrow x^3 = 16 \Rightarrow$ **x = 16^(1/3) = 2 · 2^(1/3)**.
Then $y = x^2/2 = (16^{(2/3)})/2 = (2^{(8/3)})/2 = 2^{(5/3)} =$ **4 · 2^(1/3) / 2 = 2^(5/3)**.
Coordinates: **(2 · 4^(1/3), 4^(2/3))** (or (16^(1/3), 16^(2/3)/2)).

SCORING RUBRIC

Criterion	Points
(a) Applies product rule to 6xy and implicit rule to y^3	2
(a) Correctly solves for dy/dx	1
(b) Evaluates slope = −1 at (3, 3)	2
(b) Writes tangent line $y - 3 = -(x - 3)$	1
(c) Sets $2y - x^2 = 0$ to get $y = x^2/2$	1
(c) Substitutes into curve and solves for x	1
(c) Gives complete coordinates (x, y)	1
Total	9

Derivative of an Inverse Function with Table

★★★★★ Very Important

MODEL SOLUTION

(a) From the table, f(2) = 4, so **g(4) = 2**.
By the inverse derivative formula: g′(4) = 1 / f′(g(4)) = 1 / f′(2) = **1/5**.

(b) The point on y = g(x) is (4, g(4)) = (4, 2). Slope is g′(4) = 1/5.
Tangent line: **y − 2 = (1/5)(x − 4)** (or y = x/5 + 6/5).

(c) Differentiate g′(x) = [f′(g(x))]$^{-1}$ with respect to x:
g″(x) = −[f′(g(x))]$^{-2}$ · f″(g(x)) · g′(x) = −f″(g(x)) / [f′(g(x))]^{3}.
At x = 4: g(4) = 2, f′(2) = 5, f″(2) = 6.
g″(4) = −(6) / (5)^{3} = **−6/125**.

SCORING RUBRIC

Criterion	Points
(a) Identifies g(4) = 2	1
(a) Uses g′(4) = 1/f′(2) to get 1/5	2
(b) Point (4, 2) and slope 1/5	2
(b) Correct tangent line equation	1
(c) Chain rule on g′(x) = 1/f′(g(x))	2
(c) Correct numerical evaluation −6/125	1
Total	9

Chain Rule with Tables for Composites

★★★★★ Very Important

MODEL SOLUTION

(a) Chain Rule: $h'(x) = f'(g(x)) \cdot g'(x)$.
$h'(1) = f'(g(1)) \cdot g'(1) = f'(2) \cdot 6 = (4)(6) = \mathbf{24}$.

(b) Chain Rule: $k'(x) = g'(f(x)) \cdot f'(x)$.
$k'(2) = g'(f(2)) \cdot f'(2) = g'(1) \cdot 4 = (6)(4) = \mathbf{24}$.

(c) Product Rule + Chain Rule: $p'(x) = 2f(x)f'(x) \cdot g(x) + [f(x)]^2 \cdot g'(x)$.
At $x = 3$: $f(3) = 2$, $f'(3) = 5$, $g(3) = 1$, $g'(3) = 3$.
$p'(3) = 2(2)(5)(1) + (2)^2(3) = 20 + (4)(3) = 20 + 12 = \mathbf{32}$.

SCORING RUBRIC

Criterion	Points
(a) Chain Rule setup $f'(g(1))\cdot g'(1)$	2
(a) Evaluates to 24	1
(b) Chain Rule setup $g'(f(2))\cdot f'(2)$	2
(b) Evaluates to 24	1
(c) Product Rule and chain rule setup	2
(c) Evaluates to 32	1
Total	9

Inverse Trigonometric Function and Tangent Lines

★★★★★ Very Important

MODEL SOLUTION

(a) Chain Rule: $f'(x) = 2 / (1 + (2x)^2) = \mathbf{2 / (1 + 4x^2)}$.

(b) Point: $x = 1/2 \Rightarrow f(1/2) = \arctan(2 \cdot 1/2) = \arctan(1) = \pi/4$. Point is $(1/2, \pi/4)$.
Slope: $f'(1/2) = 2 / (1 + 4(1/4)) = 2 / (1 + 1) = 2/2 = 1$.
Tangent line: $\mathbf{y - \pi/4 = 1(x - 1/2)}$ (or $y = x - 1/2 + \pi/4$).

(c) Set $f'(x) = 1$:
$2 / (1 + 4x^2) = 1 \Rightarrow 1 + 4x^2 = 2 \Rightarrow 4x^2 = 1 \Rightarrow x^2 = 1/4 \Rightarrow \mathbf{x = \pm 1/2}$.

SCORING RUBRIC

Criterion	Points
(a) Correct derivative $2/(1 + 4x^2)$	2
(b) Evaluates $f(1/2) = \pi/4$	2
(b) Evaluates slope = 1 and writes line	2
(c) Sets derivative = 1 and solves to get $x = \pm 1/2$	3
Total	9

Higher-Order Implicit Differentiation

★★★★★ Very Important

MODEL SOLUTION

(a) $2x + 8y(dy/dx) = 0 \Rightarrow 8y(dy/dx) = -2x \Rightarrow$ **$dy/dx = -x / (4y)$**.

(b) Quotient Rule on $dy/dx = -x / (4y)$:
$d^2y/dx^2 = -[\,(1)(4y) - x \cdot 4(dy/dx)\,] / (4y)^2 = -[\,4y - 4x(-x/(4y))\,] / (16y^2)$
$= -[\,4y + x^2/y\,] / (16y^2) = -[\,4y^2 + x^2\,] / (16y^3)$.
From the original curve, $x^2 + 4y^2 = 8$. Substitute into numerator:
$d^2y/dx^2 = -8 / (16y^3) =$ **$-1 / (2y^3)$**? Wait, check coefficients:
Curve is $x^2 + 4y^2 = 8$. If $4y^2 + x^2 = 8$, then $-8/(16y^3) = -1/(2y^3)$.
Adjusting prompt to: $d^2y/dx^2 = -1/(2y^3)$.

(c) For $y > 0$, $y^3 > 0$, so $-1/(2y^3) < 0$ everywhere.
Since $d^2y/dx^2 < 0$ for all $y > 0$, the curve is **concave down** everywhere in the upper half-plane.

SCORING RUBRIC

Criterion	Points
(a) Correct implicit derivative $-x/(4y)$	2
(b) Quotient rule applied to dy/dx	2
(b) Substitutes dy/dx and uses curve equation to simplify	2
(c) Notes $y^3 > 0$ for $y > 0$ and concludes concave down	3
Total	9

Multi-Layer Chain Rule Applied to Motion

★★★★★ Very Important

MODEL SOLUTION

(a) Chain Rule: $v(t) = x'(t) = \mathbf{2t / (t^2 + 1)}$.

(b) Quotient Rule: $a(t) = v'(t) = [\ 2(t^2 + 1) - 2t(2t)\ ] / (t^2 + 1)^2$
$= [\ 2t^2 + 2 - 4t^2\ ] / (t^2 + 1)^2 = \mathbf{(2 - 2t^2) / (t^2 + 1)^2 = 2(1 - t^2) / (t^2 + 1)^2}$.

(c) Velocity is maximized when $a(t) = v'(t) = 0$ and changes from positive to negative.
Set numerator = 0: $2(1 - t^2) = 0 \Rightarrow t^2 = 1 \Rightarrow \mathbf{t = 1}$ (since $t \geq 0$).
For $0 \leq t < 1$: $1 - t^2 > 0 \Rightarrow a(t) > 0$ (v increasing).
For $t > 1$: $1 - t^2 < 0 \Rightarrow a(t) < 0$ (v decreasing).
By the First Derivative Test for v, velocity is maximized at **t = 1** with maximum velocity $v(1) = 2/2 = 1$.

SCORING RUBRIC

Criterion	Points
(a) Correct velocity $v(t) = 2t/(t^2+1)$	2
(b) Correct acceleration $a(t) = 2(1-t^2)/(t^2+1)^2$	3
(c) Sets $a(t) = 0$ to get $t = 1$	2
(c) Justifies maximum with sign change / first derivative test	2
Total	9

Particle Motion Analysis

★★★★★ Very Important

MODEL SOLUTION

(a) $v(t) = s'(t) = \mathbf{6t^2 - 18t + 12 = 6(t-1)(t-2)\ m/s}$.
$a(t) = v'(t) = \mathbf{12t - 18\ m/s^2}$.

(b) The particle moves left when $v(t) < 0$.
$6(t-1)(t-2) < 0$ occurs when $(t-1)$ and $(t-2)$ have opposite signs: $\mathbf{1 < t < 2}$ (or interval $(1, 2)$).

(c) At $t = 1.5$:
$v(1.5) = 6(1.5-1)(1.5-2) = 6(0.5)(-0.5) = -1.5$ m/s < 0.
$a(1.5) = 12(1.5) - 18 = 18 - 18 = 0$ m/s^2.
Since $a(1.5) = 0$, the rate of change of speed is zero at that exact instant (transition point, **neither speeding up nor slowing down**).

(d) Particle changes direction at $t = 1$ and $t = 2$.
$s(0) = 0$.
$s(1) = 2(1) - 9(1) + 12(1) = 5$.
$s(2) = 2(8) - 9(4) + 12(2) = 16 - 36 + 24 = 4$.
$s(3) = 2(27) - 9(9) + 12(3) = 54 - 81 + 36 = 9$.
Total distance $= |s(1) - s(0)| + |s(2) - s(1)| + |s(3) - s(2)| = |5 - 0| + |4 - 5| + |9 - 4| = 5 + 1 + 5 =$ **11 meters**.

SCORING RUBRIC

Criterion	Points
(a) Correct velocity and acceleration expressions	2
(b) Sets $v(t) < 0$ and identifies interval $(1, 2)$	2
(c) Evaluates $v(1.5)$ and $a(1.5)$ and explains speed behavior	2
(d) Identifies direction changes and evaluates positions	2
(d) Correct total distance 11 meters	1
Total	9

Related Rates: Draining Conical Tank

★★★★★ Very Important

MODEL SOLUTION

(a) Full cone: R = 6, H = 12. Similar triangles: r/h = 6/12 = 1/2 ⇒ **r = h/2**.
$V = (1/3)\pi r^2 h = (1/3)\pi(h/2)^2 h = (1/3)\pi(h^2/4)h =$ **$(\pi/12)h^3$**.

(b) Differentiate with respect to t:
$dV/dt = (3\pi/12)h^2\,(dh/dt) = (\pi/4)h^2\,(dh/dt)$.
Water is leaking out, so $dV/dt = -5$ ft³/min.
When h = 4: $-5 = (\pi/4)(4)^2\,(dh/dt) = 4\pi\,(dh/dt) \Rightarrow dh/dt = -5/(4\pi)$ ft/min.
The water level is falling at a rate of **$5/(4\pi)$ ft/min** (≈ 0.398 ft/min).

(c) Speed of falling depth is $|dh/dt| = 5 / [(\pi/4)h^2] = 20 / (\pi h^2)$.
As h decreases (approaches 0), the denominator πh^2 decreases, so the quotient $20/(\pi h^2)$ increases.
Therefore, the water level falls **faster** as depth decreases.

SCORING RUBRIC

Criterion	Points
(a) Uses similar triangles r = h/2	2
(a) Obtains $V = (\pi/12)h^3$	1
(b) Differentiates V with respect to t using chain rule	2
(b) Correct numerical rate $dh/dt = -5/(4\pi)$ ft/min	2
(c) Analyzes $\|dh/dt\| = 20/(\pi h^2)$ and concludes faster as h decreases	2
Total	9

Related Rates: Sliding Ladder with Angle and Area

★★★★★ Very Important

MODEL SOLUTION

(a) $x^2 + y^2 = 25^2 = 625$. When $x = 7$: $y = \sqrt{(625 - 49)} = \sqrt{576} = 24$ ft.
$2x(dx/dt) + 2y(dy/dt) = 0 \Rightarrow x(dx/dt) + y(dy/dt) = 0$.
$7(2) + 24(dy/dt) = 0 \Rightarrow 14 + 24(dy/dt) = 0 \Rightarrow dy/dt = -14/24 =$ **$-7/12$ ft/s**.
Top slides down at **7/12 ft/s**.

(b) Area $A = (1/2)xy$.
$dA/dt = (1/2)[\,(dx/dt)y + x(dy/dt)\,] = (1/2)[\,(2)(24) + (7)(-7/12)\,]$
$= (1/2)[\,48 - 49/12\,] = (1/2)[\,(576 - 49)/12\,] = (1/2)(527/12) =$ **$527/24$ ft²/s** (≈ 21.96 ft²/s).

(c) $\cos\theta = x/25 \Rightarrow -\sin\theta\,(d\theta/dt) = (1/25)(dx/dt)$.
When $x = 7$ and $y = 24$: $\sin\theta = 24/25$.
$-(24/25)(d\theta/dt) = (1/25)(2) = 2/25$.
$d\theta/dt = -(2/25) / (24/25) = -2/24 =$ **$-1/12$ rad/s**.

SCORING RUBRIC

Criterion	Points
(a) Uses $x^2+y^2=625$ and finds $y = 24$	1
(a) Differentiates and finds $dy/dt = -7/12$ ft/s	2
(b) Product rule on area $A = (1/2)xy$	2
(b) Evaluates $dA/dt = 527/24$ ft²/s	1
(c) Trig relation $\cos\theta = x/25$ or $\tan\theta = y/x$	1
(c) Evaluates $d\theta/dt = -1/12$ rad/s	2
Total	9

Linearization, Error and Concavity from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) From the table: $f(5) = 14$ and $f'(5) = 2.5$.
Tangent line: $L(x) = 14 + 2.5(x - 5) \Rightarrow$ **$y - 14 = 2.5(x - 5)$**.

(b) $f(5.2) \approx L(5.2) = 14 + 2.5(5.2 - 5) = 14 + 2.5(0.2) = 14 + 0.5 =$ **14.5**.

(c) It is given that $f''(x) < 0$ on $[3, 7]$, which means the graph of f is **concave down**.
Because the graph is concave down, the tangent line lies **above** the curve, making 14.5 an **overestimate**.

(d) Symmetric difference quotient on $[3, 7]$:
$f''(5) \approx [f'(7) - f'(3)] / (7 - 3) = (1.5 - 3.5)/4 = -2/4 =$ **-0.5**.
(Using $[f'(5) - f'(3)]/2 = -0.5$ or $[f'(7) - f'(5)]/2 = -0.5$ yields the identical result).

SCORING RUBRIC

Criterion	Points
(a) Correct tangent line equation $y - 14 = 2.5(x - 5)$	2
(b) Correct approximation $f(5.2) \approx 14.5$	2
(c) Mentions $f'' < 0$ / concave down and concludes overestimate	2
(d) Sets up difference quotient for f'	2
(d) Correct estimate -0.5	1
Total	9

Velocity Graph: Acceleration, Direction and Distance

★★★★★ Very Important

MODEL SOLUTION

(a) a(t) = slope of v(t).
On (2, 4): slope = (−3 − 3)/(4 − 2) = −6/2 = −3 m/s² ⇒ **a(3) = −3 m/s²**.
On (4, 6): v is constant (−3), so slope is 0 ⇒ **a(5) = 0 m/s²**.

(b) The particle changes direction when v(t) changes sign.
On (2, 4), the line segment goes from (2, 3) to (4, −3) with equation v(t) = 3 − 3(t − 2) = 9 − 3t.
v(t) = 0 at **t = 3 seconds**. Since v changes from positive to negative at t = 3, the particle changes direction at t = 3.

(c) Speed is increasing when v(t) and a(t) have the SAME sign:
On (0, 2): v = 3 > 0, a = 0 (constant speed).
On (2, 3): v > 0, a = −3 < 0 (opposite signs: slowing down).
On (3, 4): v < 0, a = −3 < 0 (same signs: **speeding up on (3, 4)**).
On (4, 6): v = −3 < 0, a = 0 (constant speed).
On (6, 8): slope = (0 − (−3))/(8 − 6) = 1.5 > 0. v < 0, a > 0 (opposite signs: slowing down).
Conclusion: Speed is increasing only on **(3, 4)**.

(d) Total distance = area under |v(t)|:
On [0, 2]: rectangle area = 2 · 3 = 6.
On [2, 3]: triangle area = (1/2)(1)(3) = 1.5.
On [3, 4]: triangle area = (1/2)(1)(3) = 1.5.
On [4, 6]: rectangle area = 2 · 3 = 6.
On [6, 8]: triangle area = (1/2)(2)(3) = 3.
Total distance = 6 + 1.5 + 1.5 + 6 + 3 = **18 meters**.

SCORING RUBRIC

Criterion	Points
(a) Evaluates a(3) = −3 and a(5) = 0	2
(b) Finds t = 3 and notes sign change from positive to negative	2
(c) Analyzes signs of v and a on intervals	2
(c) Concludes speed increasing on (3, 4)	1
(d) Geometric area calculation for total distance = 18 m	2
Total	9

L'Hôpital's Rule with Parameters and Limits

★★★★★ Very Important

MODEL SOLUTION

(a) Direct substitution gives $(\cos 0 - 1)/(b \cdot 0) = 0/0$, an indeterminate form.
L'Hôpital's Rule 1: differentiate numerator and denominator with respect to x:
$\lim_{x\to 0} [-a \sin(ax)] / [2bx]$.
At x = 0, this is still 0/0. Apply L'Hôpital's Rule 2:
$\lim_{x\to 0} [-a^2 \cos(ax)] / [2b]$.
Now substitute x = 0: $[-a^2 \cos(0)] / [2b] = \mathbf{-a^2 / (2b)}$.

(b) We are given $-a^2/(2b) = -2$ and a = 4.
$-(4)^2 / (2b) = -2 \Rightarrow -16 / (2b) = -2 \Rightarrow -8/b = -2 \Rightarrow 2b = 8 \Rightarrow \mathbf{b = 4}$.

(c) Form 0/0. L'Hôpital 1:
Numerator derivative: $1 - \cos x$.
Denominator: $x - x \cos x \Rightarrow$ derivative: $1 - [\cos x - x \sin x] = 1 - \cos x + x \sin x$.
At x = 0: $(1 - 1)/(1 - 1 + 0) = 0/0$. L'Hôpital 2:
Numerator derivative: $\sin x$.
Denominator derivative: $\sin x + (\sin x + x \cos x) = 2 \sin x + x \cos x$.
At x = 0: 0/0. L'Hôpital 3:
Numerator derivative: $\cos x$.
Denominator derivative: $2 \cos x + \cos x - x \sin x = 3 \cos x - x \sin x$.
Substitute x = 0: $\cos(0) / [3 \cos(0) - 0] = 1 / 3 = \mathbf{1/3}$.

SCORING RUBRIC

Criterion	Points
(a) Verifies 0/0 and applies L'Hôpital first time	1
(a) Verifies second 0/0 and applies L'Hôpital second time	2
(a) Obtains $-a^2/(2b)$	1
(b) Sets $-16/(2b) = -2$ and solves b = 4	2
(c) Applies L'Hôpital systematically through 0/0 forms to get 1/3	3
Total	9

Analyzing f′ from a Graph

★★★★★ Very Important

MODEL SOLUTION

(a) The graph of f′ is above the axis for $0<t<3$ and $7<t<8$, so f increases on **(0,3) and (7,8)**. It is below the axis for $3<t<7$, so f decreases on **(3,7)**.

(b) f′ changes + to − at t=3, so f has a relative maximum at 3. It changes − to + at t=7, so f has a relative minimum at 7.

(c) Signed areas: [0,2]=+4; [2,3]=+1; [3,4]=−1; [4,6]=−4; [6,7]=−1; [7,8]=+1. Net change is 0, so $f(8)=f(0)+0=$**0**.

(d) $f(3)=0+4+1=5$. $f(7)=5-1-4-1=-1$. Compare $f(0)=0$, $f(3)=5$, $f(7)=-1$, $f(8)=0$. Absolute maximum is **5 at t=3**; absolute minimum is **−1 at t=7**.

SCORING RUBRIC

Criterion	Points
Correct increasing intervals	2
Correct decreasing interval	0
Correct relative maximum/minimum classifications	2
Correct signed-area setup and f(8)=0	3
Correct absolute max and min	2
Total	9

Mean Value Theorem

★★★★★ Very Important

MODEL SOLUTION

(a) f is a polynomial, so it is continuous on [0,4] and differentiable on (0,4). These are the MVT hypotheses.
(b) f(0)=1 and f(4)=1, so the average rate is (1−1)/(4−0)=**0**.
(c) f′(t)=2t−4. Set f′(c)=0, giving **c=2**, which lies in (0,4).
(d) The secant slope is 0 and f′(2)=0. Equal slopes imply the tangent and secant lines are parallel.

SCORING RUBRIC

Criterion	Points
Continuity and differentiability hypotheses	2
Average rate 0	2
Derivative equation and c=2	3
Correct parallel-slope explanation	2
Total	9

Absolute Extrema and the Candidates Test

★★★★★ Very Important

MODEL SOLUTION

(a) $g'(x)=3x^2-12x+9=3(x-1)(x-3)$, so the interior critical numbers are **1 and 3**.
(b) $g(0)=0$; $g(1)=1-6+9=4$; $g(3)=27-54+27=0$; $g(5)=125-150+45=20$.
(c) Comparing these candidates, the absolute maximum is **20 at x=5**. The absolute minimum is **0 at x=0 and x=3**.

SCORING RUBRIC

Criterion	Points
Correct derivative and critical numbers	2
All four candidate values correct	3
Absolute maximum, value and location	2
Absolute minimum, value and all locations	2
Total	9

Concavity and Points of Inflection

★★★★★ Very Important

MODEL SOLUTION

(a) $f'(x)=x^3-2x^2$. Then $f''(x)=3x^2-4x=\mathbf{x(3x-4)}$.
(b) The zeros of f'' are 0 and 4/3. Its sign is positive on $(-\infty,0)$, negative on $(0,4/3)$, and positive on $(4/3,\infty)$. Thus f is concave up on $\mathbf{(-\infty,0)\cup(4/3,\infty)}$ and concave down on $\mathbf{(0,4/3)}$.
(c) Concavity changes at both values. $f(0)=0$. $f(4/3)=64/81-128/81=-64/81$. Inflection points: **(0,0)** and **(4/3,−64/81)**.
(d) f'' can equal zero without changing sign; for example, x^4 has $f''(0)=0$ but is concave up on both sides. A concavity change must be verified.

SCORING RUBRIC

Criterion	Points
Correct first and second derivatives/factorization	2
Correct concavity intervals	2
Both inflection coordinates	3
Explains sign-change requirement	2
Total	9

Second Derivative Test

★★★★★ Very Important

MODEL SOLUTION

(a) $f'(x)=4x^3-8x=4x(x^2-2)$. Critical numbers: $\mathbf{-\sqrt{2}}$, $\mathbf{0}$, $\mathbf{\sqrt{2}}$. Also $f''(x)=12x^2-8$.
(b) $f''(0)=-8<0$, so $x=0$ is a relative maximum. At $x=\pm\sqrt{2}$, $f''=24-8=16>0$, so both are relative minima.
(c) $f(0)=0$ and $f(\pm\sqrt{2})=4-8=-4$. Relative maximum: $\mathbf{(0,0)}$. Relative minima: $\mathbf{(-\sqrt{2},-4)}$ and $\mathbf{(\sqrt{2},-4)}$.

SCORING RUBRIC

Criterion	Points
Correct derivatives and critical numbers	3
Correct Second Derivative Test classifications	3
Correct extremum coordinates and values	3
Total	9

Using a Sign Table for f′

★★★★★ Very Important

MODEL SOLUTION

(a) f′ is positive on (−2,−1) and (1,3), so f increases there. f′ is negative on (−1,1), so f decreases there.
(b) At x=−1, f′ changes + to −, so f has a relative maximum f(−1)=4. At x=1, f′ changes − to +, so f has a relative minimum f(1)=−2.
(c) The candidate values are 0, 4, −2, and 1. Absolute maximum is **4 at x=−1**; absolute minimum is **−2 at x=1**.

SCORING RUBRIC

Criterion	Points
All increasing/decreasing intervals	3
Both relative extrema and classifications	2
Candidate comparison and absolute maximum	2
Absolute minimum	2
Total	9

Riemann Sums from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) $\Delta x = 2$. Left endpoints $x = 0, 2, 4, 6$ give $L_4 = 2(3 + 5 + 4 + 2) = 2(14) = \mathbf{28}$.
(b) Right endpoints $x = 2, 4, 6, 8$ give $R_4 = 2(5 + 4 + 2 + 1) = 2(12) = \mathbf{24}$.
(c) Trapezoid rule: $T_4 = (2/2)[f(0) + 2f(2) + 2f(4) + 2f(6) + f(8)] = 1[3 + 10 + 8 + 4 + 1] = \mathbf{26}$.
This equals $(L_4 + R_4)/2 = (28 + 24)/2 = 26$, because each interior value is counted once in the left sum and once in the right sum, then halved.

SCORING RUBRIC

Criterion	Points
Correct Δx and left endpoints, value 28	3
Correct right endpoints, value 24	3
Correct trapezoid setup and value 26 with explanation	3
Total	9

Definite Integrals from a Graph

★★★★★ Very Important

MODEL SOLUTION

(a) Areas by region: [0,2] triangle above the axis, area 4, integral +4. The segment from (2,4) to (5,−2) crosses zero at t = 4: above-axis triangle on [2,4] has area 4, below-axis triangle on [4,5] has area 1, contributing −1. On [5,8] the triangle lies below the axis with area 3, contributing −3.
So $\int_0^8 f(t)\,dt = 4 + 4 - 1 - 3 = \mathbf{4}$.
(b) Total area uses absolute values: 4 + 4 + 1 + 3 = **12**.
(c) Average value = $(1/8)\cdot\int_0^8 f(t)\,dt = 4/8 = \mathbf{1/2}$.
(d) Part (a) is the *net signed* area, where regions below the axis subtract. Part (b) is the *total* area, where every region is counted positively.

SCORING RUBRIC

Criterion	Points
Correct region-by-region signed areas	1
Correct net integral value 4	2
Correct total area 12	3
Correct average value 1/2	2
Explains signed versus total area	1
Total	9

Analyzing an Accumulation Function

★★★★★ Very Important

MODEL SOLUTION

(a) By the Fundamental Theorem of Calculus, $\mathbf{F'(x) = x^2 - 4}$.

(b) $F(x) = t^3/3 - 4t\,|_0^x = \mathbf{x^3/3 - 4x}$.

(c) $F'(x) = (x - 2)(x + 2) = 0$ at $x = \pm 2$. F' is positive for $x < -2$, negative for $-2 < x < 2$, and positive for $x > 2$. So $x = -2$ is a **local maximum** and $x = 2$ is a **local minimum**.

(d) $F(2) = 8/3 - 8 = \mathbf{-16/3}$. $F(-2) = -8/3 + 8 = \mathbf{16/3}$.

SCORING RUBRIC

Criterion	Points
States $F'(x) = x^2 - 4$ with the FTC	2
Correct antiderivative formula	3
Both critical numbers found	1
Correct max/min classification	1
Both values $-16/3$ and $16/3$	2
Total	9

FTC and Average Value

★★★★★ Very Important

MODEL SOLUTION

(a) An antiderivative is $x^3 - x^2$. So $\int_1^4 f = (64 - 16) - (1 - 1) = 48 - 0 = \mathbf{48}$.

(b) Average value $= 48/(4 - 1) = \mathbf{16}$.

(c) Solve $3c^2 - 2c = 16$, that is $3c^2 - 2c - 16 = 0$.

$c = [2 \pm \sqrt{(4 + 192)}]/6 = [2 \pm 14]/6$, so $c = 8/3$ or $c = -2$.

Only $c = \mathbf{8/3}$ lies in $[1, 4]$.

SCORING RUBRIC

Criterion	Points
Correct antiderivative	1
Correct evaluation 48	2
Average value 16	2
Sets up $3c^2 - 2c = 16$	2
Selects $c = 8/3$ and rejects -2	2
Total	9

Motion: Displacement and Distance

★★★★★ Very Important

MODEL SOLUTION

(a) Displacement = $\int_0^5 (t^2 - 4t + 3)\,dt = [t^3/3 - 2t^2 + 3t]_0^5 = 125/3 - 50 + 15 = 125/3 - 35 = \mathbf{20/3\ m}$.
(b) $v(t) = (t - 1)(t - 3)$ is zero at $t = 1$ and $t = 3$.
$\int_0^1 v = 1/3 - 2 + 3 = 4/3$. $\int_1^3 v = 0 - 4/3 = -4/3$. $\int_3^5 v = 20/3 - 0 = 20/3$.
Total distance = $|4/3| + |-4/3| + |20/3| = 4/3 + 4/3 + 20/3 = \mathbf{28/3\ m}$.
(c) $v(t) > 0$ on **(0, 1) and (3, 5]**.

SCORING RUBRIC

Criterion	Points
Correct antiderivative and displacement 20/3	3
Identifies sign-change times t = 1 and t = 3	1
Splits the integral into three pieces	1
Correct total distance 28/3	2
Correct positive-direction intervals	2
Total	9

From a Riemann Sum to an Integral

★★★★★ Very Important

MODEL SOLUTION

(a) $\Delta x = 6/n$. The right endpoints are $x_i = a + i\Delta x = 6i/n$, so $a = 0$ and $b = 6$. The interval is **[0, 6]**.
(b) Since the sampled height is $(2 + x_i)^2$, the integrand is $f(x) = (2 + x)^2$ and the limit equals $\int_0^6 \mathbf{(2 + x)^2\,dx}$.
(c) $\int_0^6 (2 + x)^2\,dx = [(2 + x)^3/3]_0^6 = 512/3 - 8/3 = 504/3 = \mathbf{168}$.
Check by expanding: $\int_0^6 (x^2 + 4x + 4)\,dx = 72 + 72 + 24 = 168$.

SCORING RUBRIC

Criterion	Points
Identifies $\Delta x = 6/n$	1
Identifies $x_i = 6i/n$ and interval [0,6]	2
Writes the integral with $f(x) = (2+x)^2$	3
Antiderivative $(2+x)^3/3$	1
Exact value 168	2
Total	9

Logistic Slope Field and Long-Term Behavior

★★★★★ Very Important

MODEL SOLUTION

(a) Set $P'=0$: $P=0$ or $1-P/100=0$. Equilibria are **P=0 and P=100**.
(b) At $P=20$, $P'=0.5(20)(1-0.2)=10(0.8)=$**8 population units per time unit**. The positive sign means P is increasing at that instant.
(c) Starting below 100 and above 0, the solution increases and levels off toward the stable equilibrium **P=100**.
(d) The logistic solution is $P=100/(1+Ae^{-0.5t})$. $P(0)=20$ gives $A=4$. Thus $P(t)=100/(1+4e^{-0.5t})$. At $t=2$, $P(2)=100/(1+4/e)\approx$**40.5**.

SCORING RUBRIC

Criterion	Points
Both equilibria P=0 and P=100	2
Slope 8 and correct interpretation	2
Increasing toward 100 from P(0)=20	2
Correct logistic solution and A=4	2
Correct estimate P(2)≈40.5	1
Total	9

Verifying and Applying a Solution

★★★★★ Very Important

MODEL SOLUTION

(a) $y'=2xCe^{x^2}$. Since $y=Ce^{x^2}$, this is $2xy$, so the family satisfies $y'=2xy$.
(b) $y(0)=C=3$, so $\mathbf{y=3e^{x^2}}$.
(c) $y(1)=\mathbf{3e}$.
(d) If $y=0$ for every x, then $y'=0$ and the right side $2x(0)=0$. Therefore **y=0** is also a solution.

SCORING RUBRIC

Criterion	Points
Correct derivative using chain rule	2
Uses initial condition to get C=3	2
Correct value 3e	2
Checks both sides for y=0	3
Total	9

Reasoning from an Exponential-Decay Slope Field

★★★★★ Very Important

MODEL SOLUTION

(a) $y' = -0.5(8) = \mathbf{-4}$.

(b) $dy/y = -0.5dt$. Integrate: $\ln|y| = -0.5t + C$, so $y = Ce^{-0.5t}$. From $y(0) = 8$, $C = 8$. Thus $\mathbf{y = 8e^{-0.5t}}$.

(c) $y(2) = 8e^{-1} = \mathbf{8/e}$.

(d) The equilibrium is $\mathbf{y = 0}$; the given positive solution decreases toward 0, so $\lim y(t) = \mathbf{0}$.

SCORING RUBRIC

Criterion	Points
Slope -4	2
Correct separation/integration	2
Applies initial condition to obtain $8e^{-0.5t}$	1
$y(2) = 8/e$	2
Equilibrium and limit 0	2
Total	9

Separation of Variables with an Initial Condition

★★★★★ Very Important

MODEL SOLUTION

(a) Multiply both sides by $(y+1)dx$: $\mathbf{(y+1)dy=(x+1)dx}$.
(b) Integrating gives $y+y^2/2=x+x^2/2+C$.
(c) At (0,1), $C=1+1/2=\mathbf{3/2}$. The particular relation is $y+y^2/2=x+x^2/2+3/2$.
(d) At $x=1$, $y+y^2/2=3$. Multiply by 2: $y^2+2y-6=0$. Thus $y=-1\pm\sqrt{7}$; $y>-1$ selects $\mathbf{y(1)=\sqrt{7}-1}$.

SCORING RUBRIC

Criterion	Points
Correct separation	2
Integrates both sides correctly	2
Includes integration constant	1
Finds $C=3/2$	2
Solves and selects $\sqrt{7}-1$ using $y>-1$	2
Total	9

Exponential Population Model from a Table

★★★★★ Very Important

MODEL SOLUTION

(a) $P(t)=P_0e^{kt}$. Using $400=200e^{4k}$ gives $e^{4k}=2$, so **$k=\ln 2/4$** per hour.
(b) **$P(t)=200e^{((\ln 2)t/4)}$**.
(c) $P(6)=200e^{(3\ln 2/2)}=200\cdot 2\sqrt{2}=$**$400\sqrt{2}$**.
(d) $P'(4)=kP(4)=(\ln 2/4)(400)=$**$100\ln 2$ population units per hour**.

SCORING RUBRIC

Criterion	Points
Uses exponential model and solves $k=\ln 2/4$	3
Correct particular solution	2
Correct $P(6)=400\sqrt{2}$	2
Correct $P'(4)=100\ln 2$ with units	2
Total	9

Radioactive Decay and Half-Life

★★★★★ Very Important

MODEL SOLUTION

(a) Half-life gives $e^{-8k}=1/2$, hence **k=ln2/8** per hour.
(b) **A(t)=640e^(-(ln2)t/8)**.
(c) Twenty-four hours is three half-lives, so $A(24)=640(1/2)^3=$**80 g**.
(d) $40/640=1/16=(1/2)^4$, so four half-lives are needed. Time=**32 hours**.

SCORING RUBRIC

Criterion	Points
Correct half-life equation and k	2
Correct exponential model	2
Correct mass 80 g	2
Recognizes four half-lives and finds 32 hours	3
Total	9

Determining a Logistic Model from Rates

★★★★★ Very Important

MODEL SOLUTION

(a) The nonzero population where the rate is zero is the carrying capacity, so **K=200**.
(b) At P=50, 30=r(50)(1−50/200)=37.5r. Thus **r=0.8** per time unit.
(c) Logistic growth is maximal at P=K/2=100. From the table the maximum rate is **40 population units per time unit at P=100**.
(d) P=0 is unstable for positive populations because rates are positive just above 0. P=200 is stable because rates are positive below it and negative above it.

SCORING RUBRIC

Criterion	Points
Identifies K=200 from nonzero zero-rate equilibrium	2
Solves for r=0.8	2
Maximum rate 40 at P=100	3
Correct stability classification of both equilibria	2
Total	9

Average Value and the Integral Mean Value Theorem

★★★★★ Very Important

MODEL SOLUTION

(a) $f_{avg}=(1/4)\int_0^4 x^2dx=(1/4)(64/3)=\mathbf{16/3}$.
(b) Solve $c^2=16/3$. Since $c\in[0,4]$, $\mathbf{c=4/\sqrt{3}}$.
(c) f is continuous on [0,4]. The Integral Mean Value Theorem guarantees a point c in the interval where f(c) equals the average value.

SCORING RUBRIC

Criterion	Points
Correct average-value formula and integral	2
Average value 16/3	1
Solves $c^2=16/3$	2
Selects $c=4/\sqrt{3}$ in the interval	1
States continuity and applies the theorem	3
Total	9

Motion from a Velocity Graph

★★★★★ Very Important

MODEL SOLUTION

(a) Acceleration is the slope of v(t). Slopes are 0, −2, 0, and 1 m/s² on the four intervals.
(b) The graph crosses v=0 at t=3 and stays below the axis until t=8, so the particle moves left on **(3,8)**.
(c) Signed areas: +4, +1, −1, −4, −2, totaling displacement **−2 m**. Total distance=4+1+1+4+2=**12 m**.
(d) s(8)=s(0)+displacement=5+(−2)=**3 m**.

SCORING RUBRIC

Criterion	Points
Correct four acceleration slopes	2
Correct negative-direction interval	2
Displacement −2 m	2
Distance 12 m	1
Final position 3 m	2
Total	9

Area Between Curves and Average Vertical Distance

★★★★★ Very Important

MODEL SOLUTION

(a) Solve $x+2=x^2$: $(x-2)(x+1)=0$. Intersections occur at **$x=-1$ and $x=2$**.
(b) At $x=0$, line value 2 exceeds parabola value 0. Thus $y=x+2$ is above $y=x^2$ between the intersections.
(c) $A=\int_{-1}^{2}(x+2-x^2)dx=9/2$ square units.
(d) Average vertical distance = area / interval length = $(9/2)/3=$**$3/2$**.

SCORING RUBRIC

Criterion	Points
Both intersection x-values	2
Correct upper/lower curve identification	2
Correct integral and area 9/2	3
Correct average vertical distance 3/2	2
Total	9

Volume with Square Cross Sections

★★★★★ Very Important

MODEL SOLUTION

(a) The vertical base segment has length $\mathbf{4-x^2}$.

(b) A square with side $4-x^2$ has cross-sectional area $\mathbf{A(x)=(4-x^2)^2}$.

(c) $V=\int_{-2}^{2}(4-x^2)^2dx=\int_{-2}^{2}(16-8x^2+x^4)dx=$ **512/15 cubic units**.

(d) Volume integrates cross-sectional area, not length. Square area is side×side, hence $(4-x^2)^2$.

SCORING RUBRIC

Criterion	Points
Correct side length $4-x^2$	2
Correct area $(4-x^2)^2$	2
Correct integral setup and evaluation 512/15	3
Explains length versus area	2
Total	9

Disk Method: A Semicircle Revolved

★★★★★ Very Important

MODEL SOLUTION

(a) Disk radius is the height of the curve: $\mathbf{R(x)=\sqrt{4-x^2}}$.

(b) $V=\pi\int_{-2}^{2}R(x)^2dx=\pi\int_{-2}^{2}(4-x^2)dx$.

(c) $V=\pi[4x-x^3/3]_{-2}^{2}=\pi(32/3)=$**32π/3 cubic units**. This is a sphere of radius 2.

SCORING RUBRIC

Criterion	Points
Correct radius $\sqrt{(4-x^2)}$	2
Correct disk setup	3
Correct antiderivative/evaluation	2
Exact volume 32π/3	2
Total	9

Washer Method About the x-Axis

★★★★★ Very Important

MODEL SOLUTION

(a) Outer radius $R=2$ and inner radius $r=x$.

(b) $V=\pi\int_0^2(R^2-r^2)dx=\pi\int_0^2(4-x^2)dx$.

(c) $V=\pi[4x-x^3/3]_0^2=\pi(8-8/3)=$**$16\pi/3$ cubic units**.

SCORING RUBRIC

Criterion	Points
Correct outer and inner radii	2
Correct washer setup	3
Correct evaluation $16\pi/3$	4
Total	9

Horizontal Slices and Rotation About the y-Axis

★★★★★ Very Important

MODEL SOLUTION

(a) Solve $y^2=2y$: $y(y-2)=0$, so intersections occur at **y=0 and y=2**.
(b) Right boundary $x=2y$, left boundary $x=y^2$. Area$=\int_0^2(2y-y^2)dy=$**4/3**.
(c) Washer radii are $R=2y$ and $r=y^2$. $V=\pi\int_0^2[(2y)^2-(y^2)^2]dy=\pi\int_0^2(4y^2-y^4)dy=$**64π/15 cubic units**.

SCORING RUBRIC

Criterion	Points
Correct intersection y-values	2
Correct horizontal-slice setup and area 4/3	3
Correct washer radii, integral, and volume 64π/15	4
Total	9

End of Master Set

This resource contains original AP Calculus AB–aligned practice questions for Units 1–8. It is not affiliated with, endorsed by, or approved by the College Board. “AP” and “Advanced Placement” are registered trademarks of the College Board.

> **Print check:** Before exporting, preview the entire document. In Chrome print settings choose A4, enable Background graphics, and review any validation report.